



DEPARTMENT OF THE NAVY
OFFICE OF THE CHIEF OF NAVAL OPERATIONS
2000 NAVY PENTAGON
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OPNAVINST 1500.76E
N13M
14 May 2025

OPNAV INSTRUCTION 1500.76E

From: Chief of Naval Operations

Subj: NAVY SYSTEMS, TRAINING REQUIREMENTS, AND MANAGEMENT

Ref: See Enclosure (1)

Encl: (1) References
(2) Manpower and Training Requirements Planning
(3) Training Estimate and Navy Training System Plan
(4) Assessment, Analysis and Evaluation
(5) Training Installation and Transfer Agreement
(6) Rating Continuum
(7) Glossary and Acronyms
(8) Examples of Products

1. Purpose. To establish policy for planning, determining and documenting manpower, personnel and training (PMT) requirements and developing of professional, technical and leadership training for Navy and Navy and Marine Corps integrated weapons and warfare systems across the continuum of Naval training (apprentice, journeyman, and master level) throughout the system's life cycle. This revision aligns with changes in policy, procedures, responsibilities and reporting identified in reference (a), missions, functions and tasks published in reference (b) and additional guidelines established in the references identified in enclosure (1). This instruction is a complete revision and should be reviewed in its entirety.

2. Cancellation. OPNAVINST 1500.76D.

3. Scope and Applicability. This instruction applies to Navy and Marine Corps Defense System programs, as identified in references (c), (d), (e) and (f). Specific Department of Defense (DoD) instructions or directives for guidance and management can be found under references (c), (g), (h), (i) and (j) for major capability programs, middle tier and software programs. Additionally, this instruction applies to modernization programs. Joint and Inter-Service training programs will follow the procedures outlined in references (d) and (f). This instruction facilitates the alignment of the system, manpower and training requirements throughout the lifecycle of the system and rating continua. This instruction also serves as guidance for the Department of the Navy (DON) in building training effectiveness assessment models. Compliance with this policy will ensure PMT requirements traceability in support of new or modernized (or both) Navy training capabilities via the enclosures contained in this instruction.

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4. Background. This instruction requires the identification of total force manpower and training requirements and the development of a training sustainment plan for support of Navy and integrated Navy and Marine Corps or modernization programs. Navy training system plans (NTSP) are regulatory documents identified by reference (e). References (e) and (f) provide procedural guidance for the Joint Capabilities Integration and Development System (JCIDS) as the primary means for the Joint Requirements Oversight Council (JROC) to fulfill its statutory responsibilities. This instruction includes assessing joint military capabilities and identifying, approving and prioritizing gaps in these capabilities to meet requirements in the National Defense Strategy. References (k) and (l) address the requirements development for human systems integration.

5. Policy

a. Manpower and Training Requirements Planning (MTRP). The MTRP process identifies the strategy and tasks associated with developing manpower and training requirements and integrating them into warfare system design, development, testing, fielding and operations. Front end analyses (FEA) are a primary component of MTRP and are comprised of the baseline comparison system (BCS) analysis, the task analysis, the training systems requirements analysis (TSRA) and the rating domain analysis (RDA) and career flow map (CFM). The TSRA is a structured process used to examine training requirements and identify alternative approaches to training job tasks. The TSRA is comprised of the training situation analysis as documented in the training situation document (TSD), the Task and Learning Analysis as documented in the instructional performance requirements document (IPRD), the media selection as documented in the instructional media requirements document (IMRD), and the output of the Navy's Assessment, Analysis and Evaluation (AAE). A technical assessment of potential recommended training systems, a complete description of the alternatives (including costing data for development, implementation and sustainment) and a recommended solution with supporting rationale is documented in the training decision coordinating paper (TDCP). The military characteristics document (MCD) discusses how the training device (TD) integrates with the training system to fulfill the training requirements outlined in the TSRA and TDCP. TSRA results inform Program Office, Naval Education and Training Command (NETC) and Type Commander (TYCOM) training plans in alignment with ready for training (RFT) and follow-on training execution in the Fleet. Revisions to the TSRA associated documents will be included in a revised NTSP or in a revised FEA as applicable at the program manager's (PM) discretion. The MTRP analysis should utilize references (f), (m) and (n) to post findings in official repositories and be completed as identified in enclosure (2).

b. Training Estimate (TE). The TE describes the PMT requirements and funding across the Future Years Defense Program (FYDP). The acquiring organization will provide the identified training and produce a TE and total life-cycle cost estimate (TLCE) to support the production and development phase requirements. Urgent or joint emergent needs, modernization and rapid or abbreviated programs, including non-developmental items, commercial-off-the-shelf (COTS), government-off-the-shelf (GOTS) rapid deployment capability recommended for the transition into a program of record, will be incorporated into a TE and TLCE and a follow-on NTSP. The TE must be completed prior to capability design review. The completion of the TE and TLCE is the entrance criteria for the FEA documents that provide content for the full development of an

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NTSP, unless a modernized or updated system NTSP is in progress. The TE must be completed and approved for business systems identified in reference (m) and as identified in enclosure (2).

c. NTSP. The NTSP is required for integrated weapons and warfare systems and functional areas (e.g., habitability, damage control, etc.). An NTSP applies to equipment, systems and subsystems (ESS) or to a larger scale such as a ship platform. The NTSP is the output of MTRP. The MTRP consists of the items in Exhibit 2-1 of enclosure (2) with the applicable product support analysis data, life-cycle management data, maintenance task analysis, credential transparency description language and personnel qualifications standards (PQS). All NTSPs must state whether the NTSP capability is an acquisition category (ACAT) I through IV program of record, a non-ACAT capability or rapid acquisition program. The PM must complete an annual review of the NTSP to ensure the operational and sustainment requirements are met as identified in enclosures (2) and (3).

d. AAE. The AAE describes the strategy, goals and objectives to conduct assessment, analysis and evaluation. The AAE uses AAE Business Rule 1 for NETC training and Business Rule 2 for all other solutions that identify methodologies to collect and evaluate data of ESS training systems and other requirements that improve readiness as identified in enclosure (4). Updates will be developed in line with MTRP triggers (see Exhibit 2-1 of enclosure (2)). For further guidance refer to the AAE Strategy, Process, Standard Operating Procedures and Business Rules located at: https://flankspeed.sharepoint-mil.us/sites/MyNavyHR_Communities/OPNAV_N13M5.

e. Training Installation and Transfer Agreement (TITA). The TITA describes training products, equipment, facilities and infrastructure requirements that need to be in place to ensure sustainment of the training system throughout the life of the requirement. Approval of the TITA shifts responsibility for complete training systems delivered by the Training Support Agency (TSA) to the identified training agency (TA). The TITA is required for all ACAT programs, Fleet and NETC training solutions and may be tailored to the needs of the PM. The TITA must be completed and approved before the RFT date and as identified in enclosure (5).

f. Rating Continuum. Office of the Chief of Naval Operations (OPNAV), Director, Military Personnel Plans and Policy (OPNAV N13) ensures completion of RDA with the output of a Rating Continuum or CFM. All Navy enlisted classifications (NEC) are organized according to their career fields. The foundation for a rating continuum is the technical, professional and leadership requirements as plotted on each ratings' CFM and supported by tasks identified in the rating career domain continuum (RCDC) and the training and education needed throughout a career. The Rating Continuum must be completed and approved as identified in reference (o) and enclosure (6). For the listing of ratings and associated stakeholders, refer to the Ready Relevant Learning (RRL) Matrix located at: https://flankspeed.sharepoint-mil.us/sites/MyNavyHR_Communities/OPNAV_N13M5.

g. DON Requirements Processes. This process provides a timeline for Manpower and Training requirements documents described in reference (e), and aligned to NAVEDTRA 142 (Series) Navy training process (NTP) and the following NETC Link: <https://flankspeed.sharepoint->

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mil.us/sites/MYNAVYHR_NETC/N7/N71Prod/Forms/GroupedShort.aspx. A rapid acquisition program will follow a more constrained milestone (MS) schedule, aligned to meet warfighter readiness requirements until formal lifecycle sustainment training is in place and will be transitioned into a program of record. Reference (m) describes the business support cycle with authority-to-proceed.

h. FEA. The FEA is conducted using various analyses to create MTRP documents using references (e), (f), (m) and (n) and enclosures (2), (3), (4), (5) and (6). The resulting analysis data and documents are posted to official repositories. The entry point for the analysis is dependent on receipt of the configuration management data and need.

(1) Technology Development and Engineering Manufacturing Development

(a) Trigger: Plan. PMT Advisory Board, alignment see enclosure (2)

(2) Requirements: Analyze and Design

(a) BCS analysis see enclosure (2)

(b) Task Analysis see enclosure (2)

(c) TSRA comprises

1. Training Situation Analysis see enclosure (2)

2. Training Task Analysis see enclosure (2)

3. Learning Analysis see enclosure (2)

4. Media Selection or media fidelity analysis (MFA) see enclosure (2)

5. MCD see enclosure (2)

a. TE or TLCE see enclosure (3)

b. AAE see enclosure (4)

c. Rating Continuum analysis see enclosure (6)

(d) Production and Deployment. Course development and modernization: design and develop; course fielding implementation; assessment and feedback.

1. NTSP see enclosure (3)

2. TITA see enclosure (5)

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6. Roles and Responsibilities. Exhibit 1 provides the product roles and responsibilities.

*Note: NETC, U.S. Marine Corps Combat Development Command, Commander Naval Installations Command (CNIC) and Naval Facilities Engineering Systems Command (COMNAVFACSYSCOM) are included as required.

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Product	Framework Timeline Completion By	Deputy Chief of Naval Operations for Manpower, Personnel and Training (N1)	Resource Sponsor (RS) CNO (N1, N2/N6, N4,N9, N095)	Systems Command (SYSCOM)	PM and Program Executive Offices (PEO)	TSA	Training Systems Program Office (TSPO) and Training Systems Program Manager (TSPM)	TA	NETC*	SYSCOM Tech Authority	U. S. Fleet Forces Command (USFLTFORCOM) Commander, Pacific Fleet (COMPACFLT) and TYCOM(s)	USMC Combat and Integration and HQ Aviation*	Commander Navy Installations Command (CNIC)*	NAVFAC*
Technology Development-Engineering and Manufacturing Development														
Requirements – ANALYZE AND DESIGN														
FEA	Gate 5	C	A		L,V	C	C	C	C	V,C	C	V,C		C
TSRA TSD	Prior to First Configuration Steering Board	C	A		L	C	C	C	C	V,C	C			C
Task Analysis, Learning Analysis IPRD	Prior to First Configuration Steering Board	C	A		L		C			V,C	C			
Media Selection IMRD and TDCP	MS C	C	A		L	C	C	C	C	V,C	C		C	C
MCD	MS C	C	A		L		C			V,C	C			
RDA and CFM	Post MS C	A	C		C	C	L*	C	C	C	V,C		C	C
TE and TLCE	Prior to or at MS C	C, V	A		L,V	C	C	C	C	V, C	C	V, C		
AAE	MS C	A	C		L,C	V,C	C	C	C	V,C	V,C			C
Manpower Estimate Report	MS C (Update at full rate production as needed)	A	C		L		C	C	C	V,C	C	C		C
Production and Deployment														
Development, Modernization, Testing – DESIGN AND DEVELOP & PHASE IV Fielding – IMPLEMENT														
NTSP	MS C	C	A		L,V	C	C	C	C	V,C	V,C	V,C		C
TITA	Full rate production	A	C		L,V	C	C	C	C	V,C	V,C	V,C		C
Curriculum, Training, Equipment Devices	Prior to RFT	C	C	C	A,L	V,C	V,C	C	C	V,C	C		C	C
Assessment and Feedback – EVALUATE														
AAE	First Course Convening	A	A,C		L	C	C	C	C	V,C	V,C	V,C		C
Operations and Support – Sustainment – Disposal														
Life-Cycle Sustainment – MAINTAIN														

Crew Scheduling and Phasing Plan	Low-rate initial production and Full rate production	C	A		L		C	C	C		C	C		
Life-Cycle Maintenance	After RFT	A	A	C	L	L	L	L	C	C	C	C	C	C
Continual Training Surveillance, Formal Course Review, Training Requirements Review	After RFT	A	A			C	L	L	L	C	C	C		C
Disposal		C	A	C	L	C	C	C	C		C	C		C

A – Approval C – Concurrence L – Lead L* - Lead (Enterprise Required) V - Validation In Black - Not applicable

Exhibit 1: Product Roles and Responsibility Chart

a. Deputy Chief of Naval Operations, Manpower, Training, and Education (CNO N1)

- (1) Serves as a member of the PMT Advisory Board.
- (2) Provides oversight of process, policy and procedures for this instruction.
- (3) Acts as the executive agent for:

(a) Naval Total Force (N13N5) - Single assessment sponsor for military personnel and the executive agent for Navy manpower and training integration. Oversees and directs recruiting, organization, training of military personnel for assignment to combatant commands. Exercises centralized supervision and coordination of Navy's manpower and training requirements and coordinates their allocation among Resource Sponsors (RS) in line with reference (a).

(b) OPNAV N13. Prepare for employment of the Navy by developing plans for recruiting, training, educating, mobilizing, demobilizing and administrating. Coordinate with all OPNAV RSs for alignment of systems, PQS, occupational standards (OCCSTD) and performance support requirements. Manage and coordinate manpower requirements, policy and infrastructure with Secretary of the Navy (SECNAV), Joint Chiefs Staff (JSC), Office of the Secretary of Defense (OSD) and Congress per reference (a).

1. PMT integration and the Chief of Naval Operations (CNO) principal advisor on all manpower and training issues impacting the cost and executability of personnel policy decisions and Fleet manning levels. This includes ensuring SYSCOMs and Training Systems Program Office (TSPO) have an auditable process reviewable by the Assistant Secretary of the Navy Research, Development and Acquisition (ASN (RD&A)) and manpower stakeholders per DoD Directive 5000.1.

2. Assessing the impact and equities of manpower and training requirements, modernization configuration management on DON resources. Assessing resourcing and programming of training plans in collaboration with RS via the established Program Objective Memorandum (POM) process.

3. In collaboration with appropriate stakeholders and per references (o) and (p), is assigned duties and responsibilities in support of the RRL program.

4. Leads the validation and prioritization of functional and technical information technology (IT) requirements to support the hosting, transport and delivery of RRL modernized training content and other data.

(4) Funds

(a) Development and update of the RCDC, and RDA for each rating continuum and Maintain Rating Continuums in a common data source or authoritative data environment.

(b) Funds and transfers sustainment of modernized training in coordination with the program office, TSPO, NETC and TYCOM in a TITA. CNO N1 funds RRL curricula, manpower requirements and learning resource centers and base operating support cost.

(c) ESS operational and maintenance workload analysis in coordination with Navy Manpower Analysis Center (NAVMAC), TYCOMs and RSs. Provides comprehensive workload analysis determined during production reviews by NAVMAC.

(d) Development of OCCSTD and on-the-job training requirements per reference (q).

(e) As a RS, CNO N1 is the RS for RRL. CNO N1 will work with Deputy Chiefs of Naval Operations (DCNO), Deputy Assistant Secretary of the Navy, Financial Management and Budget (FMBE), OSD and Congress to obtain appropriate funding for RRL, and respond to queries regarding program, budget and execution issues.

(f) Coordinates overall RRL POM resourcing requests across RSs to align with Sailor 2025 and RRL implementation.

(g) Coordinates the overall budget request and execution of RRL funding related to course delivery with other RSs and the Specialized Skills Training Budget Submitting Office.

(h) Prioritizes and assesses across enterprises systems, PQS, performance support and OCCSTD training requirements with recommendations from USFLTFORCOM and COMPACFLT, in coordination with the TYCOMs.

(5) Ensures manpower equities, requirements and analyses are appropriately addressed and vetted as applicable, to include NAVMAC coordination of workload with TSPOs and TYCOMs and coordination with the respective Marine Corps occupational field sponsor.

(6) Validates and approves FEA and training products aligned to the TE, Task Analysis, MTRP and RDA established by the TSA, TA and Fleet before RFT with USFLTFORCOM, COMPACFLT and TYCOM(s).

(7) Concurs with NTSP after validation of the planned, executable requirements per stakeholder review and adjudication.

(8) Validates Preliminary Ship Manpower Document (PSMD) or Squadron Manpower Document (PSQMD), Navy Officer Occupational Classification System and Navy Enlisted Occupational Classification System (NEOCS) packages.

(9) Validates and concurs when Crew Scheduling and Phasing Plans (as outlined in reference (r)) manpower is programmed and aligned to PSMDs and NTSPs.

(10) Validates enlisted and officer distribution, training quota management, temporary duty under instruction and permanent change of station move requirements.

(11) Approves TITAs prior to training system's transition to NETC after PM, TYCOM and RS concurrence.

(12) Approves AAE and ensures it provides an assessment based on the criteria identified in subparagraph 5d for a training product and Rating Continuum.

(13) Ensures AAE results are validated and conducted by the PM in conjunction with the TYCOM. Coordinate with NETC and other stakeholders to include future training requirements for system(s) and the Rating Continuum. As a RS for programs, systems or non-programs of record under CNO N1 purview, approves the AAE in coordination with PM, NETC and TYCOM.

(14) Ensures RFT is met prior to loading courses in Corporate Enterprise Training Activity Resource Systems (CeTARS).

(15) Validates, that development, procurements and operational and support costs are programmed by the RS, including Contractor Operated Maintenance Support before concurring on the NTSP.

(16) Validates, that identifying credentialing requirements are programmed by the RS supported by reference (s).

(17) See Exhibit 1 for the specific product Approval, Concurrence, Lead, Validation responsibilities.

b. Deputy Chief of Naval Operations for Fleet Readiness and Logistics (CNO N4)

(1) Integrates training facility construction requirements into overall Navy project programming and prioritization processes utilizing DD Form 1391, Fiscal Year (FY) ____ Military Construction Project Data, per reference (t) and enclosure (5).

(2) Completes TITA facility and infrastructure planning, development and implementation content, leading to the final acceptance of the training system from the TSA.

(3) Coordinates with the TAs and other RSs to submit resource and sustainment requirements for programmatic funding, complying with RFT timelines, updated as applicable.

- (4) Implements TITA facility and infrastructure portion.
- (5) See Exhibit 1 for the specific product Approval, Concurrence, Lead, Validation responsibilities.

c. RSs [CNO N1, CNO N4, Deputy Chief of Naval Operations for Information Warfare and Director of Naval Intelligence (CNO N2N6), Deputy Chief of Naval Operations for Warfighting Requirements and Capabilities (CNO N9) and Chief of Navy Reserve (CNO N095)].

- (1) Chairs PMT Advisory Boards, as required for designated ESS.
- (2) Identifies and validates Fleet training readiness requirements in support of Fleet Live Virtual Constructive training, Distributed Mission Training and Navy Maritime Operations Centers. Validates identification of Fleet training requirements for maintenance, basic, advanced, integrated and sustainment phase training. Complies with Inter-Service Training procedures identified in reference (d).
- (3) Coordinates with the TAs and other RSs to program for sustainment and resource requirements, including resolution of seams issues outside the POM process and before RFT.
- (4) Coordinates actions with the Marine Corps Resource Advocate when applicable.
- (5) Programs infrastructure support requirements for all warfare systems, weapons or Command, Control, Communications, Computers, Combat Systems and Intelligence (C5IS) systems per reference (u), aligned with major delivery MSs across the FYDP, including outfitting of facilities.
- (6) Establishes and validates requirements and capabilities identified in required operational capability and projected operational environment, Concept of Operations (CONOPS), JCIDS documents, KPPs and KSAs. Plans, programs and funds requirements for design, development, procurement, engineering changes, alterations, modifications, modernization and sustainment for the lifecycle of the system(s). Integrate and coordinate cross-RS PMT requirements before the budget submission. Prioritize requirements and wholeness issues in the planning, programming, budgeting and execution process.
- (7) Identifies the appropriate online repository for MTRP, NTSP and TITA.
- (8) If a community requires training on a system resourced by a different sponsor, the sponsor responsible for the community with the new requirement will be the RS for the training. In the case of training not associated with a system, the RS will be the sponsor responsible for the applicable rating.
- (9) Surface training efforts should include Surface Training Advanced Virtual Environment (STAVE) and program of record requirements or standards identified in Director, Surface Warfare (OPNAV N96) STAVE Interoperability Standard of February 2021.

(10) Funds

(a) Military Personnel Navy, Reserve Personnel Navy, temporary duty under instruction, students' account and Operation and Maintenance, Navy (O&MN) and Operation and Maintenance, Navy Reserve (O&MNR) sustainment funding required to sustain training both to meet RFT date throughout the lifecycle of the system and to identify excess shortfalls in students' accounts or temporary duty under instruction funding (or both). The RS that funds the system(s) and training funds technical training equipment (TTE), TD, Contractor Operated Maintenance Support, equipment upgrades, ESS, instructors and their sustainment.

(b) Requirements necessary for NTSP development and execution including MTRP, PQS, training system development, Rating Continuum, curricula, studies, operations refresh, contractor maintenance services (CMS) and sustainment of approved training systems.

(c) Associated training equipment, devices and materials based on the same precedence as the system the training supports. Assesses manpower and training requirements, modernization and configuration management processes to ensure training systems support the requirements identified in the MTRP and updated FEA documents.

(d) Development and approval of MCDs before the procurement of TDs, per enclosure (2).

(e) USFLTFORCOM and COMPACFLT, in coordination with the TYCOMs, identify and prioritize operational needs for new training solutions; the RSs validate the needs as formal requirements and provide program funding; and the SYSCOM and TSPO develop training solutions in consultation with NETC Learning Centers in alignment with the MTRP and TITA. In the case of training not directly tied to a system, the responsible RS funds the affected Learning Center. An exception to this policy is when two or more RSs or TYCOMs require the same training. When this occurs, the sponsor and TYCOM that require the training will fund the new training requirement.

(f) Initial or interim training (original equipment manufacturer (OEM) or vendor-provided) in a digital format or a methodology prescribed at applicable Resource Requirements Review Boards until RFT.

(g) Vendor training (including modifications), operational equipment and technical manuals for training purposes (other than training aircraft and expendable ordnance), technical documentation and courseware.

(h) Additional duty travel for initial training, NEC and Military Occupational Specialty (MOS) producing courses and non-NEC and non-MOS producing courses.

(i) Initial production equipment, training material, training systems (including

curriculum, job aids, performance support, TDs, etc.) and technical manuals for the new system's delivery. The installation schedule must align with the training schedules to ensure availability of adequately trained personnel for the first operational unit.

(j) Initial training for System Maintainers at the regional maintenance centers done by OEM or In-Service Engineering Agent (ISEA) to support their roles as subject matter experts (SME), Fleet Technical Assistance and associated Fleet support personnel (e.g., Operational Test and Evaluation Force (OPTEVFOR), TYCOMs, Board of Inspection and Survey, Navy shipyards, etc.).

(k) Development or revision, or both, to Navy PQS per reference (s) or equivalent Marine Corps Training and Readiness Program products (or both) for new and modernized aircraft and ship classes, systems and equipment.

(l) Credentialing requirements supported by reference (s).

(11) Identifies facility and infrastructure project requirements for submission to the appropriate property managers for their concurrence and submission of the DD Form 1391, per enclosure (5) and reference (t).

(12) Approves TSRA, NTSP and TITA.

Note: TITAs that transition training solutions to NETC require OPNAV N13 approval via the approved Navy Tasking system (currently Enterprise Task Management Software Solution (ETMS2)).

(13) Provides PMT supportability of all manpower and training requirements and modernization programs. Appropriate action to align PMT issues identified by USFLTFORCOM and COMPACFLT, Fleet commanders, TYCOMs, program offices, Resource Managers and NETC Learning Centers is required. USFLTFORCOM and COMPACFLT must be notified if Fleet training requirements cannot be supported (resourced).

(14) Approves all portions of the FEA documents in coordination with the PM.

(15) Approves the Task Analysis in coordination with the applicable TYCOM(s).

(16) Ensures NTSP:

(a) Requirements are clearly articulated and addressed in coordination with the Fleet.

(b) Is programmed and approved with a Course Identification Number (CIN) and course data processing training system to meet the RFT date.

(c) Additional process steps for programs are designated with JROC or Joint Capabilities Board-interest and major defense acquisition programs are performed (ACAT ID and ACAT IC), per reference (e).

(d) Numbering is provided per Enclosure (3).

(e) Issue resolution with key stakeholders before NTSP is approved.

(17) Ensures TITA:

(a) Is planned for development and implementation, leading to the final acceptance of the training system from the TSA and final acceptance by OPNAV N13 to meet the established RFT date, per enclosure (5).

(b) Formal liaison with CNO N1; USFLTFORCOM; COMPACFLT; TYCOM(s); Fleet commanders; CNO N095; applicable TSAs or TAs; and inter-Service agencies or services in the case of joint programs occurs to achieve satisfactory final acceptance of the training system at RFT is approved.

(c) Is signed via the approved Navy Tasking system (currently ETMS2) for OPNAV N13 approval of the TITA.

(18) Assesses and funds an AAE, including Executing Organization designation (in coordination with PM or TYCOMs and TA) and execution of the AAE per enclosure (4).

(19) Approves the AAE submitted by the Executing Organization after coordination with PM, NETC and TYCOM.

(20) Provides feedback to the Fleet when time or resources preclude completion of Fleet PMT requirements ahead of initial operational capability (IOC).

(21) See Exhibit 1 for the specific product Approval, Concurrence, Lead, Validation responsibilities.

d. SYSCOM

(1) Serves as a member of PMT Advisory Board, which may include triggering or requesting these boards.

(2) In addition to the roles and responsibilities listed in references (e) and (v):

(a) Complies with reference (w) during the development of systems and before approval of onboard installation or fielding of system equipment or software. Complies with Navy

Modernization Process Management and Operations Manual (NMP-MOM) (SL720-AA-MAN-030 Revision 4), before approval of onboard installation. Coordinates training and training requirements with Naval Information Forces Command (COMNAVIFOR), Naval Information Warfare Systems Command (COMNAVWARSYSCOM) and RSs.

(b) Coordinates with the appropriate SYSCOM TSPO, Training Systems Program Manager (TSPM), TSA, NAVMAC and TA on manpower analysis (i.e., manpower estimate) and operational and maintenance task analysis results prior to program office commencing development of the training system.

(c) Uses an auditable process and digital medium to record and report manpower analysis and task analysis certification results and MTRP documentation and NTSP certification results as objective quality evidence (OQE) to support RS and stakeholder requests.

(3) Conducts or participates in any engineering site surveys.

(4) Ensures facility and infrastructure readiness by:

(a) Coordinating with COMNAVFACSYSCOM to execute military construction (MILCON) as directed and funded by the RS.

(b) COMNAVFACSYSCOM executing Special Projects following the TITA and reference (x) as directed and funded by CNIC.

(c) COMNAVFACSYSCOM funding MILCON training facility maintenance, renovation and sustainment.

(d) COMNAVFACSYSCOM coordinating with Fleet Technical Assistance providers on training and manpower.

(e) COMNAVFACSYSCOM collaborating with the appropriate PM to schedule NAVFAC Fleet technical support training.

(f) COMNAVFACSYSCOM providing consolidated feedback to appropriate program offices for Fleet technical support training conducted to help improve the training process.

(5) Identifies TSPOs as applicable.

(6) Software

(a) Refer to reference (g) for specifics on Software Acquisition Pathway programs.

(b) Pathway programs are not subject to two-pass seven-gate process per reference

(e).

(c) Programs developed in an agile framework to support Development, Security and Operations for software applications are executed within a software development cycle and budget.

(7) See Exhibit 1 for the specific product Approval, Concurrence, Lead, Validation responsibilities.

e. PEOs and PMs

(1) Serve as a member of the PMT Advisory Board. Announce, host and provide administrative support for PMT Advisory Board, ensuring compliance with DoD and DON conference and travel guidance.

(2) PEO for Manpower, Logistics, and Business Solutions (MLB) has been assigned roles and responsibilities by ASN (RD&A) in order to improve RRL development of IT and training, integration and sustainment, as discussed in references (o), (p), (y), (z), (aa) and (ab). PEO MLB will:

(a) Be responsible for overall management of the RRL IT and training solution.

(b) Assign an RRL Portfolio Manager and designate appropriate PM support.

(3) For manpower and training requirements and modernization system(s) including updating PSMD or PSQMD requirements, identify training requirements, including training infrastructure, instructors (as required), contracted maintenance and Fleet Technical Support Training. The training requirements should also include interim training, TDs, TTE, capability upgrades, course curriculum, courseware, significant revisions (e.g., enabling and terminal learning objectives) and emerging C5IS. Ensure completion of a cost analysis for the transition of specific interim training courses to formal Navy schoolhouses in time for RS submission of POM. Provide life-cycle support until training system(s) or curriculum reach RFT with concurrence from the TA and approval by the RS. Inter-Service Training Organization programs will follow procedures identified in reference (d).

(4) When a program requires integration with multiple program offices, the PM coordinates between PEO to determine requirements and resources.

(5) Support the systems' modernization process.

(6) Refer to reference (g) for development and execution on Software Acquisition Pathway programs.

(7) Identify, plan and submit to RS all system and resource requirements, including the development of the NTSP per enclosures (3) and (5) and coordinate current and future FY cost

estimates and priorities for training systems with RS(s). Document the current and future FY cost estimates in the program's cost analysis requirements document and the program life-cycle cost estimates. New or modified training may require program funding in coordination with applicable RS and TYCOMs.

(8) Provide a list of TE or NTSPs to be developed, updated or recommended for cancellation to the RSs and OPNAV N13 as needed in order to support budget planning as well as update TE or NTSPs.

(9) Perform MTRP as described in enclosure (2).

(10) Ensure TE and TLCE is validated as program changes dictate and, at a minimum, annually review the TE to determine if updates are required. Report results to the applicable RSs and post the results on the appropriate SYSCOM website. Minimum required data is:

(a) Title and number of the TE and TLCE.

(b) Date of completed validation.

(c) PM point of contact (POC).

(d) TE and TLCE updates required or not required.

(e) If an update is required, provide justification and impact and indicate the FY the TE and TLCE will be updated.

(11) Develop the MTRP that includes the Training Situation Analysis, Task Analysis, Learning Analysis, Media Selection or MFA, MCD and AAE in coordination and validation with the TYCOM for approval by the RS.

(12) Coordinate with NETC and TYCOMs to ensure training requirements for new and modernized training systems are identified in CeTARS and Fleet Training Management and Planning System (FLT MPS).

(13) Validate measures of effectiveness for KPP or KSA associated with the critical tasks identified in enclosures (3) and (5). For ESS that do not have a system training KPP or KSA, the results of the FEA will be used to establish measures of effectiveness for the logistics products and included in the NTSP.

(14) Determine the need for an MCD and develop it per enclosure (2).

(15) Ensure NTSP is:

(a) Developed and submitted to the RS to meet the schedules described in enclosure

(3).

(b) Coordinated between the RS and CNO N1 to determine if and when an NTSP is needed.

(c) Validated as program changes dictate and, at a minimum, annually review the NTSPs to determine if updates are required. Report results to the applicable RSs and post the results on the appropriate SYSCOM website. Minimum required data is:

1. Title and number of the NTSP.

2. Date of completed validation.

3. PM POC.

4. NTSP updates required or not required.

5. If an update is required, provide justification and impact and indicate the FY the NTSP will be updated.

(16) Develop and concur on the TITA to ensure TITA documents the applicable formal security Risk Management Framework for curriculum materials and the transition of all Fleet training requirements and resourcing from the RS(s) and program office(s) to the TSA and TA.

(17) Develop training solutions for applicable initial, interim and follow-on training. Ensure System Maintainers at the regional maintenance centers are provided with initial training by the OEM or ISEA to support their roles as SMEs.

(18) Serve as initial PQS model manager and transfer responsibilities to the TA per reference (ab), except submarine qualification programs, which are managed by the TYCOM.

(19) Identify the TA and host installation public works department (PWD) facility requirements (e.g., air, water, power, space) to support the installation of the training system and training system changes and execution of site surveys.

(20) Provide the TA

(a) Updated curriculum materials, technical manuals, Maintenance Index Pages and Maintenance Assist Modules for training equipment, PQS or equivalent Marine Corps Training and Readiness Program (or both) as described in enclosure (5).

(b) Initial outfitting of new or modified training equipment before RFT.

(c) Resources for conversion and restoration of TA facilities when installing and removing TTE, per references (v) and (x).

(21) Coordinate with the TA responsible for follow-on training. Arrange inter-service training support, per reference (d).

(22) Develop, procure, deliver, install and modernize TTE, TDs and other training material requirements identified in the TE and TLCE, NTSP and NTSP updates throughout the lifecycle of the system, including unique systems assigned to a PEO or PM.

(23) Develop technical manual(s), documentation and updates for use in initial and follow-on training. Develop technical manual(s) concurrently with the development of the training system. Distribute technical manual(s) and subsequent updates to the TSA, TA and Learning Centers.

(24) Manage TTE and TD disposal of excess.

(25) Notify the RS and PEO by e-mail, letter or Navy message traffic to allow appropriate risk mitigation action for training systems' resourcing discrepancies.

(26) Provide initial operational equipment, alternative media, technical manuals and logistics products to the training commands in a digital format for those items required to train personnel in the operation, maintenance, employment and equipment support.

(27) Develop and coordinate an AAE with applicable Program Office(s), NETC, TYCOM(s) and other stakeholders.

(28) Conduct AAE in coordination with CNO N1 and TYCOM(s) validating the effectiveness of training products and processes as prescribed in the TE and TLCE or NTSP. NETC Training Requirement Review messages and other TA, Fleet or SYSCOM technical training audits need to be considered. Training effectiveness will be evaluated according to the NETC and TYCOM recommendations based on an approved FEA analyses. RSs will provide feedback for action items associated with the TE and TLCE or NTSPs and address action items during annual validations and future programming actions.

(29) Plan and execute CMS contracts for training systems under TSA custody and responsibility. Submit CMS changes to the RS before the end of the warranty period.

(30) Provide the IMRD and TDCP for RS approval in coordination with TYCOM and NETC.

(31) See Exhibit 1 for the specific product Approval, Concurrence, Lead, Validation responsibilities.

f. The TSA as assigned or as designated:

(1) Serves as a member of the PMT Advisory Board.

(2) Funds

(a) As required, plans and provides budget submissions with the PM for development of task analysis, FEA, PQS, media analysis, studies, curriculum, operations refresh, CMS, NTSPs, approved training systems and train the trainer for modernization of systems.

(b) Reimbursable funding as appropriate for coordination, development and contracts.

(c) Along with the PM, coordinates with the TA to determine funding information for maintaining the training. The TSA is responsible for funding the maintenance of the training until RFT, along with the TA concurrence. The TSA must ensure the PM and TSPO develop training aligned with completed sustainment engineering analyses (e.g., operational, organizational, intermediate and depot level tasks) and fund training until transition.

(3) Conducts training system audits and identifies overhaul candidates.

(4) Develops approved PMT products as directed.

(5) Develops and delivers the OPNAV 4790/CK, Ship's Configuration Change Form, and the OPNAV 4790/CK(c), Ship's Configuration Change Form (Continuation), to the TA in order to support configuration management of the installed and delivered TTE, as described in reference (ac). Develop and deliver Requisition and Invoice/Shipping Documents (DD Form 1149) for all delivered TTE and TDs.

(6) Transfers Plant Property to CNIC, Navy Region or their designated representatives per references (d) and (t).

(7) Executes the transition of responsibility for the training system(s), TDs and support equipment, including acceptance testing and delivery.

(8) Coordinates with the PM or TA for scheduling of TD maintenance and overhauls, as well as depot-level calibrations.

(9) Maintains configuration of learning facility drawings

(10) Develops, procures, delivers, installs and modernizes TTE, TDs and other training material requirements identified in the TE and TLCE, NTSP and NTSP updates throughout the lifecycle of the system.

(11) See Exhibit 1 for the specific product Approval, Concurrence, Lead, Validation responsibilities.

g. TSPO and TSPM

(1) Surface Training Systems Program Office (PMS-339) represents the Naval Surface and Expeditionary Enterprise Training within Naval Sea Systems Command (COMNAVSEASYSKOM), per OPNAVINST 1500.57 Series. Serves as the Director, Expeditionary Warfare (OPNAV N95) and OPNAV N96 single POC for strategic program planning, policy, life-cycle management, research and development and technical insertion into all existing and future Navy surface and expeditionary warfare training systems. Acts as lead for COMNAVSEASYSKOM managed common systems and models.

(2) Naval Aviation Training Systems and Ranges Program Office (PMA-205) represents the Naval Aviation Training and Naval Air Systems Command (COMNAVVAIRSYSKOM). Ensures the Naval Air Warfare Center Training Systems Division (NAWCTSD) is informed and consulted regarding TSPO plans and decisions. Acts as the lead for COMNAVVAIRSYSKOM managed common systems and models for TSPOs.

(3) COMNAVWARSYSKOM Department (Code 4.4) represents the information warfare enterprise PEO Command, Control, Communications, Computers and Intelligence (PEO C4I) and Space Systems as SMEs who provide people, processes and tools for PMT Analysis, NTSP development, training integration, training assessment and training transition. Additionally, promotes consistency of products through the implementation of OPNAV instruction to drive success in effectively training the Fleet on new technologies. Furthermore, provides a holistic view of the COMNAVWARSYSKOM enterprise and maintains close connections with external stakeholders such as CNO N2N6; COMNAVIFOR; and the NETC. Finally, initiates acts in a unified collaborative approach for the management, design, delivery and sustainment of quality and cost-effective training across the Information Warfare enterprise fielded by COMNAVWARSYSKOM. In lieu of an overarching Life-Cycle Sustainment Plan (LCSP), specific program of record LCSP, User's Logistic Support Summary and legacy Integrated Logistics Support Plans are in effect to cover various functional areas not mentioned therein for Shore, Surface and Subsurface sustainment requirements.

(4) PEO Undersea Warfare Systems Training and Readiness (UWS TR) performs as the Undersea Warfare Enterprise TSA for Submarine Training per OPNAVINST 3502.5. PEO UWR TR serves as the Director, Undersea Warfare (OPNAV N97) single POC for all submarine training systems strategic program planning, policy, life-cycle management, research and development and technical insertion into all existing and future Navy submarine training systems. Except for training material or training systems equipment under the cognizance of Naval Reactors (NAVSEA 08) or the strategic weapon system under the cognizance of Director, Strategic Systems Programs (SSP), acts as lead for COMNAVSEASYSKOM submarine managed common systems and models. As such, COMNAVSEASYSKOM PEO UWS TR will be responsible for configuration management and lifecycle support for all trainers and training support materials under their cognizance.

(5) COMNAVFACSYSKOM and Expeditionary Warfare Center (COMNAVFACSYSKOM EXWC) is designated as a science and technology reinvention

laboratory and represents COMNAVFACSYSCOM as an active member of Naval Laboratory and Centers Coordinating Group and as a member of the Naval Research and Development Establishment. COMNAVFACSYSCOM EXWC provides research, development, testing and evaluation for all areas under COMNAVFACSYSCOM cognizance, including facility engineering and construction, capital improvements, maintenance and facility services, environmental remediation and conservation, real estate, public works, utilities (including fuels management for self-performed functions), transportation and weight handling equipment, antiterrorism force protection ashore, contingency engineering, ocean facilities and infrastructure and expeditionary support. COMNAVFACSYSCOM EXWC exercise the SYSCOM technical and management authorities throughout the total life cycle of assigned Navy Expeditionary Forces material including research, development and sustainment from concept development to demilitarization and disposal. COMNAVFACSYSCOM EXWC Expeditionary Programs Office executes COMNAVFACSYSCOM's assigned SYSCOM responsibilities for expeditionary and construction training systems and equipment through COMNAVFACSYSCOM TSA or TSPO.

(6) PEO Carriers (PMS 312/PMS 378/PMS 379) represents PEO Carriers enterprise training within COMNAVSEASYSYSCOM. Represents the COMNAVVAIRSYSCOM Carriers platforms enterprise as SMEs who provide people, processes and tools for PMT Analysis, NTSP Development, Training Integration, Training Assessment and Training Transition. Serves as the Director, Air Warfare (OPNAV N98) single POC for strategic program planning, policy, life-cycle management, research and development and technical insertion into all existing and future Navy Carrier training systems except for training material or trainer systems equipment under the cognizance of NAVSEA 08. Acts as lead for COMNAVSEASYSYSCOM Carrier managed common systems and models. Responsible for configuration management and lifecycle support for all trainers and training support materials under their cognizance.

(7) PEO MLB RRL Portfolio Management Office (PfMO) functions as the TSPM and end-to-end integrator for the RRL initiative. In that role, the PfMO will work with TSPOs to ensure RRL processes and processes laid out in this instruction interface and complement each other to align system training into the appropriate portions of the RRL rating Career-Long Learning Continuum effort. The PfMO functions as TSA for content modernization efforts through TITA approval and is a stakeholder on the PMT Advisory Board as well as other assigned responsibilities per references (o) and (aa).

(8) See Exhibit 1 for the specific Approval, Concurrence, Lead, Validation responsibilities.

h. The TA

(1) Provides support for PMT advisory boards.

(2) Assigns appropriate Learning Center, who will identify POCs to coordinate with the SYSCOMs, TSPOs and TSAs during the TITA process.

(3) Concurs with the FEA Analyses as a key stakeholder in the FEA and NTSP process on:

(a) Training analysis in coordination with the program office using authoritative task-based data derived from the OEM and electronic computer-aided design data configured to the approved task analysis and product supportability.

(b) Coordination of TSRA results of the Training Situation Analysis documented in the TSD, Task Analysis and Learning Analysis documented in the IPRD, Media Selection or MFA documented in the IMRD and TDCP, MCD; and the AAE.

(c) Training Situation Analysis, Task Analysis and Learning Analysis, Media Selection or MFA, MCD and AAE with the Learning Center to align and sequence the new or revised work a Sailor is expected to perform before approval by CNO N1. Conduct learning performance evaluation and AAE planning with measures of effectiveness that align learning objectives in the Navy Task Classification system and common database with Fleet performance indicators.

(4) For NTSP and TITA

(a) Participates in the NTSPs and TITAs planning, development, implementation and validation leading to TA final acceptance of the training system from the TSA to meet the established RFT date.

(b) Coordinates with TSA to validate the delivery of the TITA requirement during installation. Coordinates with the RS for programming sustainment resource requirements in the POM before RFT if required.

(c) Provides the PM or TSA with input for the update of TITAs through final acceptance.

(d) Compiles TITA data for identification and inclusion in the Real Property Inventory.

(5) Validates training curriculum materials to ensure initial training and follow-on training meets requirements. Validates curriculum updates and develops plans in coordination with the TSA prior to transition of updated materials (excludes classified materials). Initial and follow-on training will be validated for effectiveness every 3 years to ensure it adequately supports readiness and continues to meet the needs of the Fleet.

(6) Executes approved training system sustainment requirements at RFT date. The TA may expressly provide for the delegation of TA responsibilities within its chain of command.

(7) Executes funded follow-on training requirements of the approved training system.
(8) Executes responsibilities of PQS model manager after Fleet introduction, per reference (ab).

(9) For AAE execution

(a) Provides support as required for AAE planning and execution.

(b) Provides the PM, TYCOM(s) and subordinate commander staff with training effectiveness data per enclosure (4).

(10) Participates in the facility and infrastructure requirements planning, development, implementation and validation of Installation Design Plans (IDP) to:

(a) Provide early identification of facilities (e.g., buildings, spaces, utilities) available to accommodate the establishment of a training system.

(b) Coordinate facility impacts with the host command.

(c) Monitor and support MILCON and Special Projects requirements identification.

(d) Deliver TSA-provided facility drawings and updates to local COMNAVFACSYSCOM or CNIC representatives (e.g., Public Works).

(e) Training System

1. Participates in the acceptance testing of the new, modified or updated training system.

2. Accepts the transfer of responsibility for the training system from the PM, TSPO and TSPM and TSA. Upon agreement with PM, TSPO and TSPM and TSA that installation deficiencies exist, monitor deficiencies until all deficiencies are resolved.

(11) Provides OPNAV 4790/CKs (hardcopy or electronic) to the Configuration Data Manager (CDM) for incorporation in the Configuration Data Managers Database - Open Architecture. The OPNAV 4790/CK ensures appropriate configuration management of new and modified training capabilities as well as signs for and provides records management for Requisition and Invoice or Shipping Documents (DD Form 1149s).

(12) See Exhibit 1 for the specific product Approval, Concurrence, Lead, Validation responsibilities.

i. NETC

- (1) Serves as a member of the PMT Advisory Board.
- (2) When applicable, serves as the TA for training within the NETC enterprise.
- (3) Validates MTRP provided by the PM or CNO N1 and identifies how the learning gaps between baseline comparison PMT requirements and new ESS will be integrated for follow-on sustainment training solutions.
- (4) Plans and executes, in coordination with the RS, applicable CMS contracts for training systems under NETC custody and responsibility. Submits CMS changes to the RS before the end of the warranty period.
- (5) Coordinates with the respective TYCOM to determine the adequacy of skills training as it affects mission capability and Fleet and Fleet Marine Force readiness per enclosure (4).
- (6) Designates the lead Learning Center for those programs that cross multiple Learning Centers.
- (7) Complies with Inter-Service Training Review Organization training policies and procedures identified in reference (d).
- (8) Provides a standardized equipment list to PM or TSA for learning resource centers.
- (9) For the TSRA:
 - (a) Validates TSRA Training Situation Analysis TSD, Task Analysis and Learning Analysis IPRD, media selection IMRD and TDCP, MCD and AAE (e.g., performance assessment) results and recommendations in coordination with TYCOM (e.g., Fleet performance assessment) based on the performance of task identified in RS approved task analysis.
 - (b) Maintains training within Rating Continuums in alignment with Navy Task Classifications and OCCSTD.
- (10) Coordinates and provides input into the NTSP
 - (a) Resources for subordinate activities and Learning Centers for participation supporting the NTSP process. Computation on instructor, throughput, average of board, annual training input requirements and an individual's account requirements for the development of NTSPs within 30 days of the request. Additionally, provide the process (e.g., instructor computation model) to determine requirements.
 - (b) Provides TE and TLCE and requirements documented in the NTSP providing concurrence recommendations.
 - (c) Provides annual training input requirements calculation.

- (11) Concurs on the TITA in coordination with the RS, PM, TYCOMs and CNO N1.
 - (12) Assists the TYCOM(s) in executing the evaluation by providing assessment data as outlined in the AAE.
 - (13) Coordinates inputs to include (e.g., convening dates, quotas, curriculum, TTE and RFT) in CeTARS with the applicable TSPO manager.
 - (14) Manages course delivery, supply chain execution and supporting shore infrastructure to meet Fleet capacity and performance needs.
 - (15) Maintains configuration of learning facility drawings.
 - (16) Coordinates, delivers, installs and fields TTE, TDs and other training material requirements identified in the TE and TLCE, NTSP and NTSP updates throughout the lifecycle of the system.
 - (17) See Exhibit 1 for the specific product Approval, Concurrence, Lead, Validation responsibilities.
- j. USFLTFORCOM and COMPACFLT, in coordination with TYCOMs
- (1) Serves as a member of the PMT Advisory Board.
 - (2) Provides policy, guidance and oversight for Fleet training, per reference (ad).
 - (3) Provides notification to RS and PM for all ship availabilities.
 - (4) Validates and concurs to NTSPs, TSRA Rating Continuum Training Situation Analysis TSD (Scoping), Task and Learning Analysis IPRD (RDA), media selection IMRD and TDCP (MFA), MCD and AAE prior to development of training solution.
 - (5) See Exhibit 1 for the specific product Approval, Concurrence, Lead, Validation responsibilities.
- k. Deputy Commandant, Combat Development and Integration and Deputy Commandant for Aviation, Headquarters Marine Corps
- (1) Serves as lead for Marine Corps for JCIDS, and as the primary conduit for all aviation related issues.
 - (2) For NTSP:

(a) Provides guidance to subordinate activities to prioritize and fund Capability Development Documents and participates in the NTSP process.

(b) Validates NTSPs for integrated Navy and Marine Corps training programs.

(3) See Exhibit 1 for the specific product Approval, Concurrence, Lead, Validation responsibilities.

1. CNIC

(1) Participates in site surveys as described in enclosure (5) and maintains liaison with offices and commands involved in the engineering process.

(2) Reviews TITAs to ensure facility impacts are addressed and that potential facility risks are adequately mitigated.

(3) Coordinates facility and infrastructure requirements with other RSs or SYSCOMs for the planning, programming and budgeting of MILCON and Special Projects, as described in reference (x).

(4) Ensures the recording and maintenance of real property.

(5) See Exhibit 1 for the specific product Approval, Concurrence, Lead, Validation responsibilities.

7. Action. OPNAV activities will ensure that NTSPs and training planning documents comply with requirements set forth within this instruction.

8. Glossary and Acronyms and Product Examples. For a glossary and acronyms, see enclosure (7). For PMT product examples, see enclosure (8).

9. Records Management

a. Records created as a result of this instruction, regardless of format or media, must be maintained and dispositioned per the records disposition schedules found on Directives and Records Management Division portal page at <https://portal.secnav.navy.mil/orgs/DUSNM/DONAA/DRM/Records-and-Information-Management/Approved%20Record%20Schedules/Forms/AllItems.aspx>.

b. For questions concerning the management of records related to this instruction or the records disposition schedules, please contact the local records manager or the OPNAV Records Management Program (DNS-16).

10. Forms

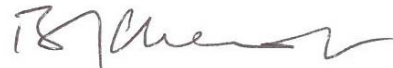
a. Forms are available for download at Navy Forms Online:
<https://forms.documentservices.dla.mil/order/>:

- (1) OPNAV 4790/60, Work Center Register Control Processing Copy
- (2) OPNAV 4790/2K, Ship's Maintenance Action Form (2-Kilo)
- (3) OPNAV 4790/CK, Ship's Configuration Change Form
- (4) OPNAV 4790/CK(c), Ship's Configuration Change Form (Continuation)

b. Forms are available for download at the DoD Forms website:
<https://www.esd.whs.mil/directives/forms/>:

- (1) DD Form 1149, Requisitions and Invoice/Shipping Document
- (2) DD Form 1391, FY ____ Military Construction Project Data

11. Review and Effective Date. Per OPNAVINST 5215.17A, Total Force Manpower, Training and Education Requirements (OPNAV N13M) will review this instruction annually around the anniversary of its issuance date to ensure applicability, currency and consistency with Federal, DoD, SECNAV and Navy policy and statutory authority using OPNAV 5215/40 Review of Instruction. This instruction will be in effect for 10 years, unless revised or cancelled in the interim and will be reissued by the 10-year anniversary date if it is still required, unless it meets one of the exceptions in OPNAVINST 5215.17A, paragraph 9. Otherwise, if the instruction is no longer required, it will be processed for cancellation as soon as the need for cancellation is known following the guidance in OPNAV Manual 5215.1 of May 2016.



R. J. CHEESEMAN, JR
Deputy Chief of Naval Operations
For Personnel, Manpower, and Training, N1

Releasability and distribution:

This instruction is cleared for public release and is available electronically only via DON Issuances website, <https://www.secnav.navy.mil/doni/default.aspx>.

REFERENCES

- (a) OPNAVINST 5420.117A
- (b) OPNAVINST 5450.352B
- (c) DoD Instruction 5000.80 of 30 December 2019, Operation of the Middle Tier of Acquisition
- (d) OPNAVINST 1500.27G
- (e) SECNAVINST 5000.2G
- (f) CJCSI 5123.01I
- (g) DoD Instruction 5000.87 of 2 October 2020, Operation of the Software Acquisition Pathway
- (h) DoD Directive 5000.01 of 28 July 2022, The Defense Acquisition System
- (i) DoD Instruction 5000.02 of 8 June 2022, Operation of the Adaptive Acquisition Framework
- (j) DoD Instruction 5000.85 of 4 Nov 2021, Major Capability Acquisition
- (k) OPNAVINST 5310.23A
- (l) DoD Instruction 5000.95 of 1 April 2022, Human Systems Integration in Defense Acquisition
- (m) DoD Instruction 5000.75 of 24 January 2020, Business Systems Requirements and Acquisition
- (n) NAVEDTRA 138
- (o) Memorandum of Understanding between ASN (RD&A) and VCNO: Ready Relevant Learning Charter, Ser N09/22U100539 of 26 May 2022
- (p) United States Fleet Forces Command Ready Relevant Learning Career Long Learning Continuum Manual of 18 November 2021
- (q) NAVPERS 18068F
- (r) OPNAVINST 3500.23D
- (s) OPNAVINST 1540.56B
- (t) SECNAVINST 11011.47D
- (u) NAVSO P-1000 of November 2019
- (v) SECNAVINST 5000.36A
- (w) DoD Directive 5000.59 of 15 October 2018, DoD Modeling and Simulation Management
- (x) OPNAVINST 11010.20J
- (y) United States Fleet Forces Command Ready Relevant Learning Process Manual Sailor 2025 of 20 Aug 2020
- (z) United States Fleet Forces Command Ready Relevant Learning Training Effectiveness and Assessment Manual of 3 Jan 2023
- (aa) United States Fleet Forces Command Acquisition Decision Memorandum to Direct Acquisition Management of Ready Relevant Learning Capability of 18 July 2021
- (ab) OPNAVINST 3500.34H
- (ac) OPNAVINST 4790.4F

OPNAVINST 1500.76E
14 May 2025

(ad) COMUSFLTFORCOMINST/COMPACFLTINST 3501.3E

MANPOWER AND TRAINING REQUIREMENTS PLANNING

1. Introduction. MTRP communicates the “who,” “what,” “where,” “when,” and “how” strategy needed to develop requirements and integrate sustainment requirements into system design, development, testing, fielding and operations and support. The task-based approach to manpower and training requirements determination provides PMs with the flexibility to tailor the level of PMT analysis to their program. In support of this effort, this enclosure (2) guides the production and sustainment of the MTRP to meet the user’s operational needs, lifecycle, performance support and the estimation of total ownership costs. The focus is to maintain the readiness of a system and describe the role the human plays to make the system mission capable. This enclosure (2) supports RS planning and programming of requirements and resources necessary to deliver effective integrated logistics products supporting integrated weapons and warfare system readiness. The analyses consider all interventions to improve the system and human performance to satisfy training requirements and align to reference (n) NTP.
2. Purpose. To document the MTRP conducted analyses that identify the requirements, proposed training system solution(s) and manpower required to operate and maintain ESS.
3. Process. The MTRP is task-based to ensure production and sustainment meet the user’s operational needs, life cycle, performance support and the estimation of total ownership costs.
 - a. Although a significant amount of data types are common between ESS, there are distinct differences in detail and scope. There is also a significant difference in detail and scope between ESS and platforms. Additionally, there is a significant amount of ESS Government Furnished Information required to track, monitor and ensure deliveries to support alignment with requirements.
 - b. During the Material Solution Analysis phase, the BCS Analysis specifies assessment of mission effectiveness, operational suitability and estimated lifecycle costs of alternative solutions based on system design specifications. During engagements with stakeholders, logistics supportability, configuration management, materiel options and limitations of future training for Abbreviated Acquisition Programs and Rapid Acquisition Program will be assessed to establish any triggers for execution. The BCS Analysis establishes the foundational starting point for follow-on PMT requirements analysis, development and assessment efforts. The BCS Analysis also ensures that PMT stakeholders use the same set of baseline data when subsequently considering and evaluating new, modified or updated system(s) training requirements and alternatives.
 - c. Each PM should tailor their process to ensure that training is included in the development of the intellectual property strategy. The process should include the development and management of the Crew Scheduling and Phasing Plan.

d. Stakeholders will participate in preliminary design reviews to advocate for requirements before completing a Cost Analysis Requirements Description, Program Life-cycle Cost Estimate and an approved LCSP.

Note 1: Before completing this enclosure (2), specific to technical training for engineered systems, Logistics Supportability Analysis and Product Supportability Analysis data (i.e., physical system design, maintenance significant items and maintenance task analysis – may also be referred to as LCSP) developed during the Engineering and Manufacturing Development EMD will be used to identify the tasks that need to be trained or require performance support (or both). A similar process for identifying tasks associated with operating tactical systems defined in MIL-STD-46855A, Human Engineering Requirements for Military System, Equipment and Facilities, will identify authoritative operator tasks. Whether for tactical operations or in support of maintenance, Logistics Supportability Analysis and Product Support Analysis (or LCSP now) data provides input to the MTRP process by describing the work, as well as the conditions and standards to enable readiness.

Note 2: For tasks, determine the requisite knowledge, skills and abilities to perform the task and align OCCSTD and career continuums for each rating using the RDA. As an output, the Rating Continuum will be developed, managed and sustained by CNO N1, in coordination with USFLTFORCOM and TYCOMs. For more on Rating Continuum see enclosure (6).

Note 3: PMs may proceed directly to Section Two - Requirements Analyze and Design and complete any applicable portions of the BCS Analysis planning for those data elements not previously addressed in the FEA.

Note 4: The processes may be tailored to meet KPP or KSA items.

e. The RSs will use the inputs and MTRP process steps to validate the PMT requirements shown in Exhibit 2-1 that follows. Exhibit 2-1 below describes the MTRP process. The enclosure consists of two sections. Section I - Trigger: Plan and Section II - Requirements: Analyze and Design.

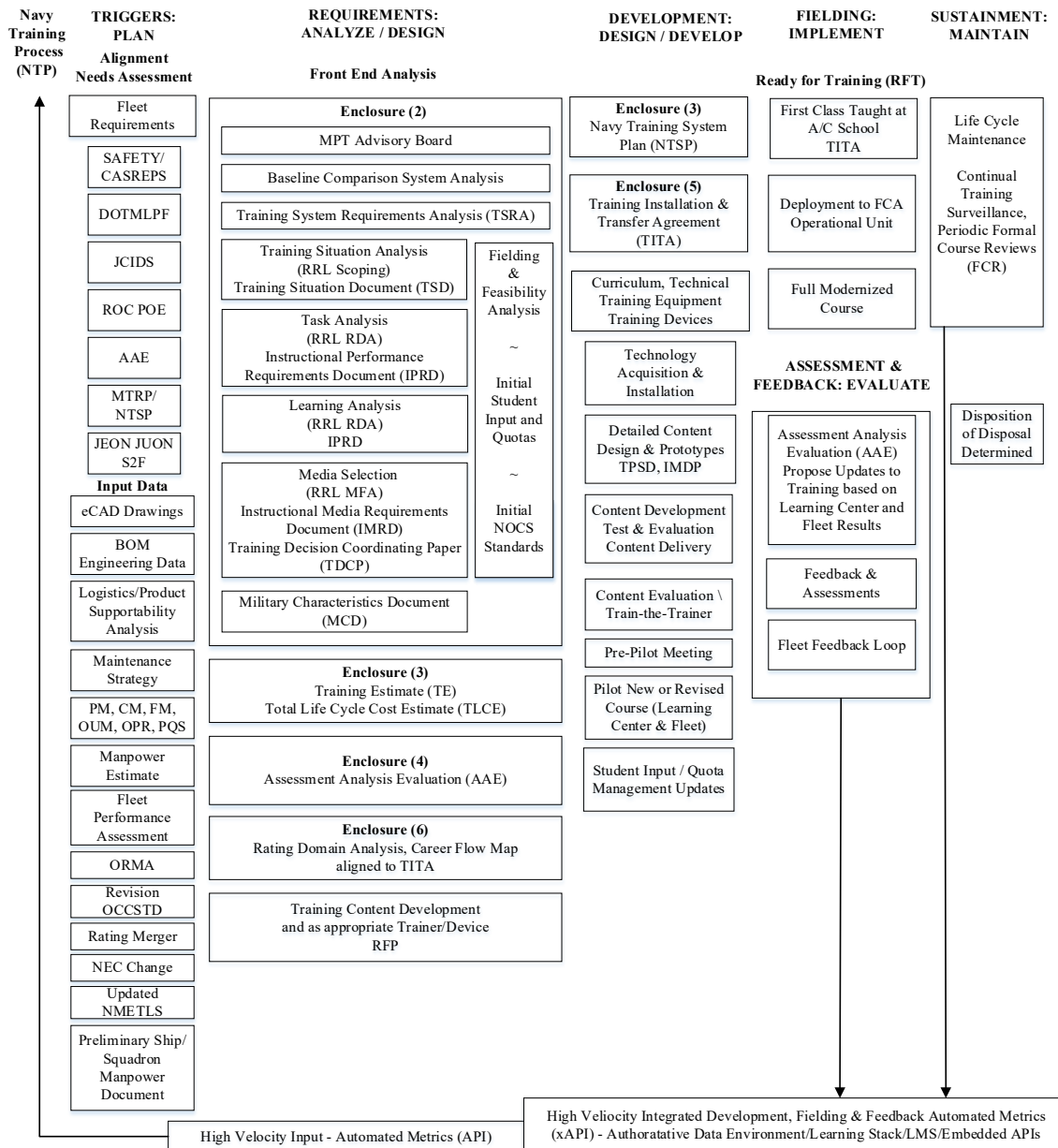


Exhibit 2-1: Manpower and Training Requirements Planning Process

(1) Trigger Plan

(a) PMT Advisory Board

1. Function. The PMT Advisory Board ensures effective communications are in place between the respective PMs, OPNAV, the PMT community and the Fleet. In the NTP, an Alignment Meeting occurs to kick off the PMT Advisory Board.

2. Establishment. The establishment of an PMT Advisory Board is required at the start of the Trigger Plan and will continue throughout the ESS lifecycle program of record. An Alignment Meeting is held to identify participating organizations' POCs, ensure engagement by participating organizations, review roles and responsibilities, communicate the plan and scope of the effort. The PMT Advisory Board is required early in the program, preferably after approval of the system design specifications.

Note: During the Trigger: Plan activities it is critical to validate Fleet issues. If an issue has multiple causes a Needs Assessment is conducted to determine the issue's root cause. If the root cause is training, the needs assessment will determine the appropriate intervention. To conduct a Needs Assessment, use the NETC NTP – NTP Workbook and Needs Assessment Out brief templates for speed to the Fleet.

3. Membership. The RS chairs the PMT Advisory Board. Participating organizations will designate representatives who are knowledgeable of their organizations' involvement in programs and empowered to make decisions on behalf of their organization. Membership of the PMT Advisory Board will include, but not be limited to, representation from:

- a. RS
- b. PM
- c. OPNAV N13M
- d. SYSCOMs, TSPO and TSA
- e. USFLTFORCOM including COMPACFLT and TYCOM
- f. TA, NETC and Learning Center(s)
- g. COMNAVFACSYSCOM and CNIC, Joint and other Service Representatives, as required
- h. PEOs and PEO MLB

4. Minimum Requirements. PMT Advisory Boards provide a stakeholder participation forum to facilitate the MTRP validation to consist of:

- a. BCS Analysis
- b. FEA for the analyses completed
 - (1) Task Analysis

(2) TSRA

- (a) Training Situation Analysis – TSD
- (b) Training Task Analysis – IPRD
- (c) Learning Analysis – IPRD
- (d) Media selection or MFA – IMRD and TDCP
- (e) MCD

c. AAE

d. TE and TLCE

e. NTSP

5. Other Considerations. The RS may convene additional PMT Advisory Boards when necessary.

(2) Requirements: Analyze and Design (FEA)

(a) Introduction. FEA is comprised of multi-phased analyses. These analyses identify the proposed training system solution(s) and manpower required to operate and maintain ESS. The analyses consider all interventions to improve the system and human performance to satisfy training requirements. This section provides overview of all the analyses that can be conducted. Entry into the analyses will vary based on former analyses, existing data or need.

(b) Purpose. To document the analyses that are conducted that provide a comprehensive analysis of the manpower, personnel, training, development and sustainment requirements for new or modified ESS and platforms (including all systems and subsystems). Analysis results enable stakeholders to project new or revised ESS training and manpower requirements throughout the lifecycle of the ESS or platform (including all systems and subsystems).

(c) Process. To conduct these analyses use the format templates posted on SharePoint as well as identified in this enclosure (2). The BCS Analysis, Task Analysis and the TSRA determines the “what, when and how” to deliver training. The TSRA includes the TSD. The Training Task and Learning Analyses are documented in an IPRD. The Media Selection is documented in an IMRD and is aligned to the TDCP. Functional requirements are documented in the MCD. The results of the analyses inform the Program Office, NETC and TYCOM training plans and delivery in alignment with RFT and follow-on performance in the Fleet.

Note 1: Revisions to the TSRA Task and Learning Analyses IPRD, media selection IMRD and TDCP and MCD, technical publications (e.g. interactive electronic technical manuals, portable document formats, etc.), PQS, preventative maintenance, and new or updated course content as required, will be included in the revised NTSP or in a revised FEA TSRA document. Complete the BCS Analysis to determine if a new or modified analysis is required with the introduction of new or modified warfare system. Procedures may vary by SYSCOM. Then proceed to the appropriate entry analysis in the subparagraphs of the TSRA.

Note 2: The analyses discussed in enclosure (2) using the NETC NTP incorporates RDA to include Career Progression Timing, Rating and Path considerations. Resultant analyses are documented in the NTP Requirements Workbook and Models, TSD, IPRD, IMRD as Appendices. For further discussion see enclosure (6).

(d) BCS Analysis

1. Introduction. The BCS Analysis establishes the foundational starting point for follow-on PMT requirements analysis, development and assessment efforts. The BCS Analysis also ensures PMT Stakeholders use the same set of baseline data when evaluating new or modified system(s) training requirements and alternatives. The BCS Analysis utilizes an existing system to define a quantitative baseline for training and manpower projections for the new or modified warfare system as a starting point for subsequent analysis. The BCS Analysis is a method for estimating manpower and training impacts and resource requirements early in a system's lifecycle, with the goal of establishing TDs, equipment and facility installation requirements and integrated product data. Additionally, the BCS Analysis provides a task analysis of an existing system for use in the examination of alternative technologies and engineering solutions. The application of this process ensures integration of PMT into the design effort of the warfare system to improve performance and reduce lifecycle costs.

2. Purpose. To document the baseline data to evaluate new or modified system(s) training requirements and alternatives.

3. Process. Follow the process as described in this section. Complete the actions in the subparagraphs for BCS to determine if a new or modified task is required with the introduction of new or modified ESS. Procedures may vary by SYSCOM.

a. Deliverable: BCS Analysis data is documented as discussed in this enclosure (2).

4. BCS Analysis Format: There is no prescribed format for the BCS Analysis.

a. The PM selects or develops the new or modified warfare system and identifies the BCS to generate the training requirements and cost estimates. The TE will reflect this data after alignment with the Analysis of Alternatives results and Lifecycle Sustainment

Plan. The data informs the Cost Analysis Requirements Description (when available) and consists of a short descriptive paragraph(s) of the early known operation and maintenance (O&M) functional characteristics. The PM identifies the “as is” training and cost.

b. BCS Analysis is based on an operational warfare system or a composite derived from multiple operational warfare systems. In selecting a BCS, the goal is to choose the system(s) that most closely represent the design, operational and support characteristics of the warfare system(s) under development. Initially and as a minimum, the estimate for new PMT requirements should be based on the BCS Analysis when no other integrated engineering product data, engineering changes or manpower documentation is available for projecting out-year requirements.

c. The TE will reflect the BCS Analysis results (as applicable) and is used to inform the TLCE.

d. The SYSCOM will validate the BCS Analysis. The PMT Advisory Board, as applicable, will endorse prior to RS approval of the BCS Analysis.

e. The BCS analysis is divided into steps as shown in Exhibit 2-2 (last two are informational) with platform-specific ESS completion requirements.

(1) For ESS and complex ESS, in the Task Analysis section complete steps 2.a, b and e.

(2) For ship platforms, in Task Analysis, complete steps 2.a, b, d and e.

(3) For aircraft platforms, in Task Analysis complete steps 2.a, b, c and e.

Note: An analyst knowledgeable and adept in the PMT disciplines should perform each step.

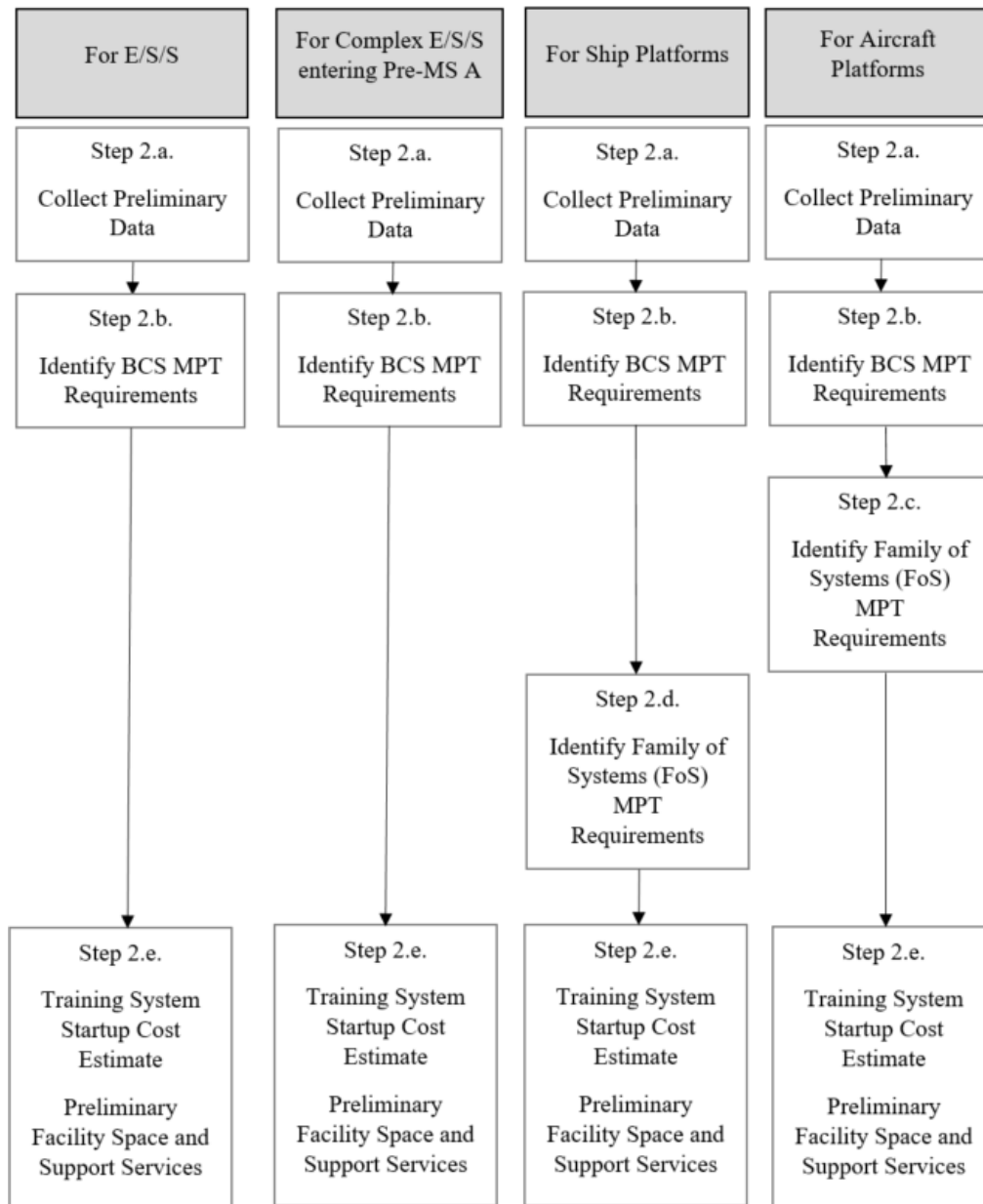


Exhibit 2-2: BCS Analysis Flow Chart

f. When the BCS is not the operational platform or ESS being replaced, then a hybrid baseline system is used. The hybrid baseline is a composite of ESS from several different end items which, when taken together, closely resemble the new ESS. The PMT Advisory Board concurrence is required with the hybrid baseline system.

g. All BCS analysis data can be tailored for projected requirements based on known data associated with the new platform and ESS, noting where there are differences to help defend or modify the projections as necessary.

h. The PMT requirements for a total platform can be divided into two categories; equipment driven, and non-equipment driven. Maintenance personnel, equipment operators and underway and in port watch stations are examples of equipment driven manpower. Supervisors, division and department heads, damage control personnel and administrative personnel are examples of non-equipment driven manpower.

i. In determining the total platform PMT requirements, the analyst should first identify Joint service requirements and then select a comparable existing platform. Once this is completed, the ship's department (ship application) or aircraft squadron's department and work center (aircraft applications) composition must be determined.

j. The non-equipment driven manpower requirements are based on performance tasks. The workload associated with completing the performance tasks is identified. NAVMAC will incorporate associated workload into the Navy Manpower Requirements System Model to determine billet requirements to support the workload.

5. BCS Procedure

a. Collect preliminary data for the corresponding Part, Section and Element of the NTSP format for the new system. Refer to enclosure (3) for more detail of data elements and potential sources.

b. POCs. List the principals, stakeholders, PM, TSPO key personnel, their activity, code and office contact information, including program-specific contacts and PMT Advisory Board members.

c. New or modified program documentation. Provide an unclassified narrative summary of the operational uses of the new weapon system, its functional description, known operational and maintenance concepts.

d. Provide a list of the components and configurations of the new or modernized weapon system, its mission, performance goals, constraints and interfaces.

e. Replacement(s). List each platform or ESS (or both) replaced by the new or modified system.

6. Identify manpower and training requirements

a. Collect manpower data. Consistent with skill identifiers utilized in Total Force Structure Management System (TFSMS) and Total Force Manpower Management System (TFMMS) (e.g., quality), identify and record the type of current manpower resources required for the operation, maintenance, facilities maintenance and own unit support of the BCS.

b. The Ship or Squadron Manpower Document and activity manpower document for shore establishments are excellent sources for this information. Ensure the Ship or Squadron Manpower Document activity manpower document used as the source reflects current manpower requirements. Any changes to manpower requirements should be submitted to NAVMAC per reference (r). The current manpower requirements are organized by manpower type under their associated department (ship applications) or department and work center (aircraft applications). Manpower type should be identified by using the descriptors:

- (1) Operator
- (2) Organizational maintainer
- (3) Intermediate level maintainer
- (4) Depot level maintainer
- (5) Watch station - Readiness Conditions I to V
- (6) Other manpower (e.g., OPTEVFOR staff, NETC instructor staff)
- (7) Supervisor
- (8) Own unit support

c. At shore activities, the BCS may be manned by multi-shifts (e.g., shift 1, shift 2 or shift 3). For airborne operators, indicate the operator title and the number of operators per aircraft. The current manpower quality and quantity should be organized by manpower type and categorized as:

(1) Officer (rank, designator, Navy Officer Billet Classification code or USMC equivalent).

(2) Enlisted (rating, pay grade, NEC, or USMC MOS).

(3) Civil Service (type, grade, series).

(4) Contractor (enter type as CTR, the salary range in thousands).

(5) List of the various manpower types by department or department and work center, including quality and quantity of the current manpower requirements.

d. Manpower quality refers to the skill and experience level of personnel. Describe any special mission requirements or constraints (e.g., directed manning, two-person rule, etc.) that affect the determination of operator manning.

- e. List of total manpower minus instructors.

7. Collect training data

a. Identify and record the operator titles associated with each operator type of manpower. Identify the frequency of involvement with the BCS for each operator. The frequency of involvement is the number of man-hours during peak usage conditions that the operator is involved in interacting with the BCS. If the BCS does not require a dedicated watch station, describe other manning scenarios under which the equipment is operated. For example, electrical watchstanders powering-on or powering-off equipment; communications center watchstanders periodically changing frequency; bridge personnel operating the equipment; or roving patrols checking and adjusting equipment.

b. Identify and record the specific performance tasks each operator of the BCS must perform and the component on which it is performed. The BCS components will be identified by work unit code (aircraft application), equipment identification code (shipboard applications) or other as specified by SYSCOM. Operator tasks are normally specified at a lower level of detail than maintenance tasks (e.g., identify aircraft, maintain aircraft tracks, enter periodic frequency change, start the engine, shift fuel pumps, etc.). Identify and record the operator's minimum skill set necessary to perform each task or group of tasks. The operator task skill set requirements are best verified by an existing operator. At a minimum, indicate rate and rating for enlisted requirements and rank and designator for officer operators. Document the NEC for enlisted requirements, Navy Officer Billet Classification for officers and series for civil service. Indicate any special skill requirements not identified by NEC, Navy Officer Billet Classification or MOS.

8. Identify BCS maintainer tasks

a. Identify and record the maintainer titles associated with each maintainer type of manpower (e.g., electrical, mechanical, etc.).

b. Identify and record the specific organizational level corrective maintenance tasks performed by each maintainer on the BCS. Identify the components of the BCS by work unit code (aviation applications) or equipment identification code (shipboard applications). The tasks should be grouped and recorded by work unit code for aviation applications and equipment identification code for shipboard applications. The tasks should be of a high level and correspond to the coding system of the maintenance data collection system used to record maintenance actions on the BCS. In the aviation community, the action taken codes used on the OPNAV 4790/60, Work Center Register Control Processing Copy, and Naval Aviation Logistics Command Management Information System (NALCOMIS) will be used as tasks. In the surface and undersea communities, the action taken codes used on the OPNAV 4790/2K, Ship's Maintenance Action Form (2-Kilo), will suffice. In this manner, historical and OEM data can be used based on inputs from TYCOMs or ISEAs prior to first deployment to

determine the amount of time expended on these tasks or project corrective maintenance based on preventative maintenance data.

c. Identify and record, by work unit code or equipment identification code, the specific organizational level planned maintenance tasks associated with the BCS. These tasks should be of a high level, identified as planned maintenance tasks and correspond to the coding system of the maintenance data collection system used to record maintenance actions on the BCS or its component under analysis. The Maintenance Requirement Card and Maintenance Index Page are a good source of planned maintenance tasks. Equipment guide lists or some other method used for documenting the total equipment quantities can be used for determining the total planned maintenance workload hours.

d. Collect and record the historical organizational level corrective and planned maintenance hours for each work unit code or equipment identification code of the BCS. In most cases, a considerable number of ships or aircraft use the BCS being investigated. Since most equipment receives more frequent usage at sea or on deployment, corrective and planned maintenance data collected from operational ships or aircraft squadrons should be collected to ensure data reflects peak usage conditions. For aviation applications, total flight hours for the period of the historical data must also be determined. The hours expended on corrective maintenance and some planned maintenance is expressed as man-hours per week for shipboard applications and maintenance man-hours per flight hour for aviation applications. Other kinds of planned maintenance actions may be more appropriately expressed in terms of man-hours per day, man-hours per sortie or other metrics. Those components of the BCS with the highest amount of maintenance hours should be flagged as high maintenance drivers. The Decision Knowledge Programming for Logistics Analysis and Technical Evaluation is the authoritative source for aviation data. COMNAVSEASYSCOM is a suggested source for maintenance data trend analysis and verification of organizational maintenance.

e. Identify and record the specific intermediate level corrective maintenance tasks performed by each maintainer of the BCS. Identify the components of the BCS by work unit code (aviation applications) or equipment identification code (shipboard applications) using the appropriate maintenance data collection system. The tasks should be grouped and recorded by work unit code for aviation applications and equipment identification code for shipboard applications.

f. List the depot level corrective maintenance workload by work unit code or equipment identification code.

9. Identify BCS operator training requirements.

____ a. List the training courses required for each different operator by new or modified course title, course length, source rating, NEC or MOS awarded and course location.

- b. List the characteristics of each operator course, e.g., the training type, presentation environment, technique, media and training path.
- c. List the current annual student throughput and TD requirements, including facility footprint, hotel services and costs for each operator course.
- d. List the in-service training required by each operator.

Note: PQS, on-the-job training handbooks and training system utilization handbooks are examples of in-service training.

10. Identify BCS maintainer training requirements

- a. List the training courses required for each different maintainer by new or modified course title, length, source ratings, NEC and MOS awarded and course location.
- b. List the characteristics of each maintainer course, e.g., the training type, presentation environment, technique, media and training path.
- c. List the current annual throughput and training equipment and device requirements, including facility footprint, hotel services and costs for each maintainer course.
- d. List the in-service training required by each maintainer.

11. For aircraft programs, identify family or system of systems manpower and training requirements.

- a. Identify family or system of systems manpower requirements.
- b. Examine the predecessor and various Navy comparable squadrons that closely match the mission and operational scenario of the new squadron and select the type of squadron (e.g., patrol, etc.) that most closely matches. The chosen squadron will be the basis against which the new squadron will be compared to estimate the new squadron's baseline manpower data. Using comparability analysis, identify the baseline data required to determine the non- equipment and aircrew manpower requirements for the aircraft squadron. Non-equipment and aircrew manpower requirements are determined by formulas developed by NAVMAC or an individual or activity qualified in this discipline. Once the baseline data for these formulas are identified, the equipment related, non-equipment related and aircrew manpower requirements should be developed by COMNAVAIRSYSCOM, NAVMAC or an individual or activity qualified in this discipline. List the baseline data for manpower formulas, algorithms, matrices, etc. The baseline data for the manpower formulas consist of:

(1) Number of aircraft in the squadron

- (2) Monthly utilization rate
- (3) Length of flying day
- (4) Length of a maintenance day
- (5) Average sortie length
- (6) Workweek
- (7) Number of shifts
- (8) Seat factor
- (9) Crew ratio
- (10) Equipment workload by work center

c. Develop, with the assistance of NAVMAC or an individual or activity qualified in this discipline, a Squadron Manpower Document for the aircraft squadron as it will ultimately be. Ensure the Squadron Manpower Document contains the equipment-related manpower requirements as well as the non-equipment and aircrew requirements. Provide a PSQMD per OPNAVINST 1000.16L.

d. Provide an aircraft and squadron phase-in schedule.

12. Identify predecessor aircraft non-equipment and aircrew training requirements.

a. For each different non-equipment (e.g., plane captain, safety personnel, etc.) or aircrew (pilot, naval flight officer, etc.) billet title of the predecessor squadron, determine all the formal follow-on training courses that are required. Class "A" school for enlisted personnel school will be assumed unless it is the only training received. For officers, a designation will be assigned at the time of commissioning. Training received to obtain a designator or qualification will be warfare dependent. Determine new or modified course titles, course lengths, source ratings, source designators (officers), NECs, MOSs or NOBCs awarded (if applicable), prerequisites and course locations with unit identification codes. Provide a list of follow-on training courses associated with the aircraft.

b. Describe the presentation environment, technique and media and training path(s) associated with the follow-on training courses of the aircraft.

c. List the follow-on courses, annual student throughput, TTE, TDs and test equipment, including facility footprint, hotel services and costs associated with the aircraft.

d. List aircraft in-service training.

13. Identify a new family or system of systems training requirements.

a. For each different non-equipment or aircrew billet title requiring training, estimate the initial training requirements. List the initial training requirements for the family or system of systems.

b. For each different non-equipment or aircrew billet title, determine the formal follow-on training courses that will be required. Class "A" school for enlisted personnel school will be assumed unless it is the only training received. For officers, a designation will be assigned at the time of commissioning. Training received to obtain a designator or qualification will be warfare dependent. For officers and enlisted operators, new or modified course titles, course lengths, source ratings, NECs, MOS or Navy Officer Billet Classification awarded (if applicable), prerequisites and logical course locations (with unit identification codes). List the new or modified follow-on training courses associated with the family of systems or system of systems.

c. Provide the presentation environment, technique and media and training path(s) associated with the follow-on training courses for the family or system of systems.

d. List the follow-on courses annual student throughput, TTE, TD, test equipment, including facility footprint, hotel services and costs associated with the family or system of systems.

e. Provide the family or system of systems for in-service training.

f. Identify any alternative concepts or strategies that, with equipment design changes, could result in more efficient use of Navy and Marine Corps training resources. The goal of this task is to standardize training requirements and optimize the selection of training alternatives. As a part of this task, predecessor-training resources will be assessed to determine their ability to support either non-equipment or aircrew training needs. Develop a narrative explanation of either all feasible non-equipment or aircrew training requirements for the family or system of systems developed during the performance of this task. The statement will detail the advantages and disadvantages of each training scenario as well as the potential source of training resources (e.g., new, existing, etc.). Compile the equipment related training requirements and the non-equipment and aircrew training requirements.

g. For family or system of systems programs, either new construction, modernization or conversion, identify family or system of systems manpower and training requirements.

14. Identify family or system of systems manpower requirements.

a. Examine the predecessor and various Navy comparable ship classes that closely match the mission and operational scenario of the new ship and select the ship class that most closely matches. The chosen ship will be the basis against which the new ship will be compared to estimate the new ship's baseline manpower data. Using comparability analysis, NAVMAC will assist by identifying the baseline data required to determine the non-equipment manpower requirements for the ship. Non-equipment manpower requirements are determined by various formulas developed by NAVMAC. Once the baseline data for these formulas are identified, the equipment related, and non-equipment related manpower requirements should also be developed by NAVMAC. List the baseline data for manpower formulas. The baseline data for the manpower formulas consist of:

- (1) Staffing standards
- (2) Directed functions
- (3) Workweek
- (4) Watchstations
- (5) Battle bill watchstation titles
- (6) Equipment planned maintenance values
- (7) Equipment corrective maintenance ratios
- (8) Facility maintenance manpower factors (e.g., production delay, support actions, etc.)

b. Develop, with the assistance of NAVMAC or an individual or activity qualified in this discipline, a PSMD for the new ship as configured. Ensure the PSMD contains the equipment and the non-equipment manpower requirements developed earlier.

c. Once the manpower requirements are developed, determine the source (e.g., new, existing, phased-out ship, etc.) of the required manpower resources. Typically, the majority of the manpower resources will come from the predecessor (replaced) ship. However, in some cases, a new ship will have different resource requirements than the predecessor. During the performance of this task, ensure any excess, additional or differing quality of manpower resources is identified. Provide the changes in manpower from a predecessor to the new ship and rationale for the differences.

15. Identify predecessor ship non-equipment training requirements.

a. For each different non-equipment billet title of the predecessor ship, determine all the formal follow-on training courses that are required. Class "A" school for enlisted personnel school will be assumed unless it is the only training received. For officers, a designation will be assigned at the time of commissioning. Training received to obtain a designator or qualification will be warfare dependent. Determine CIN (new and modified or all) course titles, course lengths, source ratings, source designators (officers), NECs or NOBCs awarded, prerequisites and course locations (with unit identification codes). List the follow-on training courses associated with the ship.

b. Provide the presentation environment, technique, media and training path(s) associated with the follow-on training courses.

c. Provide the annual student throughput, TTE, TD and test equipment, including facility footprint, hotel services and costs associated with the follow-on training courses.

d. List the in-service training associated with the ship.

16. Identify a new ESS family of systems or system of systems training requirements.

a. For each different new or modified platform and ESS non-equipment billet title requiring training, estimate the initial training requirements. List the initial training requirements for the family or system of systems.

b. For each different new platform and ESS non-equipment billet title, determine all the formal follow-on training courses that will be required. Class "A" school for enlisted personnel school will be assumed unless it is the only training received. For officers, a designation will be assigned at the time of commissioning. Training received to obtain a designator or qualification will be warfare dependent. For officers and enlisted operators, new or modified course titles, course lengths, source ratings, NECs or Navy Officer Billet Classification awarded (if applicable), prerequisites and logical course locations (with unit identification codes). List the follow-on training courses associated with the family of systems or system of systems.

c. Provide the presentation environment, technique, media and training path(s) associated with the follow-on training courses for the family or system of systems.

d. Provide the annual student throughput, TTE, TD and test equipment, including facility footprint, hotel services and costs associated with the follow-on training courses for the family or system of systems.

e. List the in-service training associated with the family of systems or system of systems.

f. Review the predecessor non-equipment training requirements and identify any alternative concepts or strategies that, with equipment design changes, could result in more efficient use of Navy training resources. The goal of this task is to optimize both training requirements and the selection of training alternatives. As a part of this task, predecessor-training resources should be assessed to determine their ability to support non-equipment training needs. Develop a narrative explanation of all feasible non-equipment training requirements for the new ESS (family or system of systems) developed during the performance of this task. The explanation should detail the advantages and disadvantages of each training scenario as well as the potential source of training resources (e.g., new, existing, etc.). Compile the ESS related training requirements and the non-equipment training requirements. Develop a narrative explanation of all possible ESS and non-equipment training requirements for the family or system of systems.

g. Two preliminary critical actions occur. The first action is the development of the initial training system cost estimate for future funding requirements. The second is identifying facility and infrastructure requirements for facility modification or construction to support new training system requirements.

h. For weapon systems entering before MS A and B, programmatic and technical data should be available to assist in developing a TE utilizing the results of the BCS Analysis and what is known of the new weapon system.

(3) TSRA

(a) Introduction. TSRA is a sequenced set of analyses that identifies the basic purpose of the training ESS, type of training provided, location and performance requirements, recommends appropriate learning objectives with recommended training strategies for new or modified training requirement(s). The TSRA determines the “what, when and how” to deliver training. The TSRA defines training requirements (e.g., training tasks, learning objectives, instructional strategies, instructional methods, media selection, throughput and facilities) for new or modified training, provides rationale for developing TDs and identifies training development and implementation risks. The TSRA includes the Training Situation Analysis documented in a TSD, Task and Learning Analyses documented in an IPRD and Media Selection documented in an IMRD. A technical assessment of potential training systems recommended and a complete description of the alternatives (including costing data for development, implementation and sustainment) and a recommended solution with supporting rationale is documented the TDCP. These documents depict the delta between the “as-is” and “to-be” comparison in order to conduct a TSRA. The results of the TSRA analyses inform the Program Office, NETC and TYCOM training plans and delivery in alignment with RFT and follow-on performance in the Fleet. For legacy systems where an NTSP already exists, revisions to the TSRA Task and Learning Analyses, Media Selection and MCD will be included in the revised NTSP or in a revised FEA TSRA document. The determination of which analyses should be the starting point is based on prior analyses, existing data and need.

1. Purpose. To document the conducted analyses that may include Training Situation Analysis, Task Analysis, Learning Analysis, Media Selection, MCD in the respective analysis documents.

2. Process. For all the analyses conducted, the process applies.

a. The PM will select the entry point for the TSRA analyses.

b. The PMT Advisory Board will validate the IMRD before RS approval. The IMRD (inclusive of the TDCP data) provides the PM a technical assessment of potential training system solution(s) based on the ROC and environment using the results of the completed analyses, media selection process and an analysis of training alternatives.

c. Upon IMRD review and approval, the PM will update the TITA to support the RFT date including facility modification, trainer design, procurement and delivery, as required.

d. Once the TSRA is baselined, the PM will submit the TSRA (completed analyses) to the SYSCOM and RS who will approve the analyses for systems requirements. The PM in coordination with the RS (and the PMT Advisory Board as applicable) will determine when to proceed to the next phase of the requirements process, as predicated on the outcome of the approved analyses and any new training requirement(s) identified.

e. The PM will submit the TSRA (completed analyses documents) to OPNAV N13, USFLTFORCOM, COMPACFLT and TYCOM(s) to evaluate the impacts on the applicable OCCSTDs, NECs, credentialing, rating continuums, development costs and analytics for future POM submission.

(1) Identify new ESS training requirements.

(2) For each ESS billet, determine the formal training course annual student throughput, TTE, TD and test equipment. For enlisted personnel, class "A" school or "block learning" will be used to establish a rate, "C" school will be used to establish an NEC. For officers, a designation will be assigned at the time of commissioning. Follow-on training will be warfare dependent. For officers and enlisted operators, course title, length, source rating, awarded NEC or Navy Officer Billet Classification, prerequisites and logical course locations will be included.

(3) For each ESS equipment training requirement, determine the appropriate training level (apprentice, journeyman or master).

(4) For each ESS non-equipment billet course, identify training type, environment, media and training path.

(5) For each ESS provide a list of all required in-service training.

Note 1: The TSRA section is broken into the following analyses: Training Situation Analysis, Task Analysis, Learning Analysis and Media Selection or MFA that follow.

Note 2: The Task Analysis can occur prior to the TSRA or in combination to the Training Situation Analysis. Regardless of whether the Task Analysis occurs before or in parallel to the Training Situation Analysis, a complete, approved Task Analysis is required before follow-on analysis Training Task Analysis can commence. The approved Task Analysis must be included (either in full or by reference) within the IPRD.

Note 3: A Training Situation Analysis may or may not be completed dependent on available data, prior analyses or need. For the purpose of process flow, the Task Analysis is described following the Training Situation Analysis along with the Training Task Analysis, as it is the entry point for the Training Task Analysis.

3. Training Situation Analysis

a. Introduction. The Training Situation Analysis documents the “What” is currently trained the “As Is” and identifies "Gaps" for new or modified training requirement(s). The Training Situation Analysis is used to verify the efficiency of a training system to meet existing training needs and to survey training programs and technologies for applicability to new training needs. The following Exhibit 2-3 shows the Product Roles and Responsibility Chart for the TSRA.

Product	Acquisition Timeline Completi n By	CNO N1	RS CNO (N1, N2/N6 , N4, N9, N095)	SYSCO MS	P M / P E O	T S A	TSPO/ TSPM	T A	NETC	SYSCO M Tech Authority	USFLTFO RCOM , COMPAC FLT and TYCOM(s)	USMC Combat and Integration and HQ Aviation	CNI C	N A V F A C
TSRA - TSD	Prior to First Configurati on Steerage Board	C	A		L	C	C	C	C	V,C	C			C

A - Approval C - Concurrence L - Lead V - Validation In Black - Not applicable

Exhibit 2-3: Product Roles and Responsibility Chart for Training Situation Requirements Analysis

b. Purpose. To document the Training Situation Analysis in the TSD.

c. Deliverable: Training Situation Analysis is documented in the tailored TSD Data Item Description (DID), DI-SESS-81517C template and as identified in this enclosure (2).

d. Training Situation Analysis TSD Format: The template for the TSD is located at: https://flankspeed.sharepoint-mil.us/sites/MyNavyHR_Communities/OPNAV_N13M5.

Note: The TSD format contains the recommended sections of the document. Table of Content items can be adjusted based on prior analyses or need.

4. Task Analysis

a. Introduction. Task Analysis may be conducted prior to the Training Situation Analysis or in conjunction with the Training Situation Analysis. A Task Analysis must be conducted as the entry point for Learning Analysis. The Task Analysis provides mission and task information. Task Analysis documents the tasks required to operate (watch station requirements) and maintain the new ESS RCDC. The maintenance will include Organizational (O) level, Intermediate (I) level and Depot (D) level maintenance tasks (including planned maintenance, corrective maintenance and troubleshooting). Task Analysis includes tasks, conditions, performance standards, task attributes, task references and sources, task criterion and required tools. During a Task Analysis training tasks are identified for “To Be” training as well as “When” the training is required. Also, cybersecurity-cyber hygiene maintenance tasks must be identified for all shipboard and maritime operations center network systems and hosted applications, including non-network systems with information assurance vulnerability concerns. The Task Analysis provides mission, collective and individual training task information. Training Task Analysis also includes tasks, conditions, performance standards, task attributes, task references and sources, task criterion and required tools for the selected training tasks. The Task Analysis contains data necessary to support the design of a training program. The following Exhibit 2-4 shows the Product Roles and Responsibility Chart for the Task Analysis.

Product	Acquisition Timeline Completion By	CNO N1	RS CNO (N1, N2/N6, N4,N9, N095)	SYSOMS	P M / P E O	T S A	TSP0/ TSPM	T A	NETC	SYSOM Tech Authority	USFLTFOR COM, COMPACF LT and TYCOM(s)	USMC Combat and Integration and HQ Aviation	CNIC	N A V F A C
Task Analysis , IPRD	Prior to First Configurati on Steering Board	C	A		L		C			V,C	C			

A - Approval C - Concurrence L - Lead V - Validation In Black - Not applicable

Exhibit 2-4. Product Roles and Responsibility Chart for Task Analysis

b. Purpose. To document the Task Analysis in the IPRD.

c. Deliverable: Task List with identified Training Tasks is documented in the tailored IPRD DID, DI-SESS-81518D template and as identified in this enclosure (2).

d. Task Analysis IPRD Format: The template for IPRD is located at:
https://flankspeed.sharepoint-mil.us/sites/MyNavyHR_Communities/OPNAV_N13M5.

Note: The IPRD format contains the recommended sections for the document. Table of Content items can be adjusted based on prior analyses or need.

5. Learning Analysis

a. Introduction. Learning Analysis documents “What” training is required for the new or modified ESS to the “To Be” training requirement(s). The Learning Analysis provides learning objectives, instructional strategies and instructional methods for the tasks selected for training. The Learning Analysis contains data necessary to support the design of a training program. The following Exhibit 2-5 shows the Product Roles and Responsibility Chart for the Learning Analysis.

Product	Acquisition Timeline Completion By	CNO NI	RS CNO (N1, N2/N6, N4, N9, N095)	SYSCOMS	P M / P E O	T S A	TSPO/ TSPM	T A	NETC	SYSKOM Tech Authority	USFLTFOR COM, COMPACF LT and TYCOM(s)	USMC Combat and Integration and HQ Aviation	CNIC	N A V F A C
Learning Analysis IPRD	Prior to First Configurati on Steering Board	C	A		L		C			V,C	C			

A - Approval C - Concurrence L - Lead V - Validation In Black - Not applicable

Exhibit 2-5: Product Roles and Responsibility Chart for Learning Analysis

b. Purpose. To document the Learning Analysis in the IPRD.

c. Deliverable: Tailored IPRD DID, DI-SESS-81518D.

d. Learning Analysis IPRD Format: The template for the IPRD is located at:
https://flankspeed.sharepoint-mil.us/sites/MyNavyHR_Communities/OPNAV_N13M5.

Note: The IPRD format contains the recommended sections for the document. Table of Content items can be adjusted based on prior analyses or need.

6. Media Selection or MFA

a. Introduction. Media Selection documents the “How” the training will occur for the “To Be” solution. Media Selection can be completed if all the tasks, critical and difficulty and performance criteria are available from the previous analyses or existing data. Media Selection uses the approved task analysis and learning analysis to identify the effective media for teaching each training task to produce the most effective delivery methodology and modality for each learning objective based on the science of learning and cost constraints. Media

Selection maps the learning objectives, instructional strategies and methods identified in the learning analysis to a set of media prioritized alternatives. Each of these alternatives needs to be justified in terms of instructional effectiveness, cost and risk (e.g., AAE). Media Selection includes a complete description of the alternatives (including costing data for development, implementation and sustainment) and a recommended solution with supporting rationale. Additionally, provides specifications for the media selection model used, a description of primary and alternate media requirements and functional requirements for the instructional delivery system. The media data is used to develop a technical assessment of potential training system solution(s) based on the required operational capability and environment using the results of the TRSA, media selection process and an analysis of training alternatives, a complete description of the alternatives (including costing data for development, implementation and sustainment) and a recommended solution with supporting rationale. Each of the media alternatives needs to be justified in terms of instructional effectiveness, cost and risk (e.g., AAE). The IMRD purpose is to serve as the baseline for instructional media performance specifications. The SYSCOM and PMT Advisory Board will validate the Media Selection deliverables before RS approval. The following Exhibit 2-6 shows the Product Roles and Responsibility Chart for Media Selection.

Product	Acquisition Timeline Completion By	CNO N1	RS CNO (N1, N2/N6, N4, N9, N095)	SYSCOMS	P M / P E O	T S A	TSP0/ TSPM	T A	NETC	SYSCOM Tech Authority	USFLTFO RCOM, COMPAC FLT and TYCOM(s)	USMC Combat and Integration and HQ Aviation	CNIC	N A V F A C
Media Selection IMRD / TDCP	MS C	C	A		L	C	C	C	C	V,C	C		C	C

A - Approval C - Concurrence L - Lead V - Validation In Black - Not applicable

Exhibit 2-6: Product Roles and Responsibility Chart for Media Selection

- b. Purpose. To document the Media Selection in the IPRD or TDCP.
- c. Deliverable: Tailored IMRD DID, DI-SESS-81519C (that includes the TDCP) or TDCP.
- d. Media Selection Format: The template for the IMRD (that includes the TDCP) or TDCP is located at: https://flankspeed.sharepoint-mil.us/sites/MyNavyHR_Communities/OPNAV_N13M5.

Note: The IMRD and TDCP formats contain the recommended sections for the document. Table of Content items can be adjusted based on prior analyses or need.

7. MCD

a. Introduction. The MCD documents the training system functional description. The MCD provides the required characteristics of a TD, which define the military functions the TD must be capable of performing or simulating. Military characteristics include physical and operational characteristics, but not technical characteristics. The MCD is developed after the media selection. The MCD discusses how the TDs integrate with the training system to fulfill the training requirements outlined in the TSRA. The MCD provides a functional description of the TD within the respective program's training plan (e.g., TE or NTSP). The development phase must consider known constraints on cost, production, supportability and maintainability. The MCD describes the TD's physical, functional and operational characteristics needed to improve human performance to achieve system readiness. The MCD provides a conceptual description of the TD in terms of the functions and tasks it will support, with additional technical details provided later as the concept is refined and physical characteristics and functional descriptions are defined. The MCD ultimately describes the physical and functional baseline characteristics a TTE or TD must have to fulfill its requirements as one component of an instructional system. The MCD must also include results from a task analysis that identifies requirements to operate and maintain the new or modified warfare system. After the distribution of the baselined MCD to the SYSCOM for validation (and PMT Advisory Board as applicable), the PM will adjudicate any comments and produce a proposed final MCD. The RS will approve the MCD for subsequent development of the training solution and update of all related program documentation, e.g., NTSP, TITA, etc. The following Exhibit 2-7 shows the Product Roles and Responsibility Chart for the Military Characteristics Document.

Product	Acquisition Timeline Completion By	CNO N1	RS CNO (N1, N2/N6, N4, N9, N095)	SYSCOMS	P M / P E O	T S A	TSPO/ TSPM	T A	NETC	SYSCOM Tech Authority	USFLTFO RCOM, COMPAC FLT and TYCOM(s)	USMC Combat and Integration and HQ Aviation	CNIC	N A V F A C
MCD	MSC	C	A		L		C			V,C	C			

A - Approval C - Concurrence L - Lead V - Validation In Black - Not applicable

Exhibit 2-7: Product Roles and Responsibility Chart for Military Characteristics Document

b. Purpose. To document the training system functional description in the MCD.

c. Deliverable: MCD using the template see (2) below.

d. MCD Format: The template for the MCD is located at:
https://flankspeed.sharepoint-mil.us/sites/MyNavyHR_Communities/OPNAV_N13M5.

TRAINING ESTIMATE AND NAVY TRAINING SYSTEM PLAN

1. Introduction. The effectiveness of the TE and TLCE and NTSP depends on timely requirements development, validation and implementation. The objectives support program management requirements by developing an efficient and time-phased training program. The format is a multi-phased process to identify the training system solution and manpower required to operate and maintain ESS. The TE or NTSP documents all training requirements to improve the system and human performance identified through the FEA process. The TE or NTSP enables stakeholders to project PMT requirements and funding across the FYDP. Existing operation and maintenance requirements inform out-year PMT estimates for the new or modified warfare system. Upon approval of system engineering technical data, all PMT estimates for the new or modified warfare system provide a basis for planning and programming. These estimates should continue to be refined before submitting required baselined deliverable to the SYSCOM for validation and to the RS for approval and inclusion in the TE and TLCE or NTSP. The following Exhibit 3-1 shows the Product Roles and Responsibility Chart for the NTSP.

Product	Acquisition Timeline Completion By	CNO N1	RS CNO (N1, N2/N 6, N4, N9, N095)	SYSCOMS	P M / P E O	T S A	TSPO/ TSPM	T A	NETC	SYSCOM Tech Authority	USFLTFO RCOM, COMPAC FLT and TYCOM(s)	USMC Combat and Integration and HQ Aviation	CNIC	N A V F A C
NTSP	MS C	C	A		L, V	C	C	C	C	V,C	V,C	V,C		C

A - Approval C - Concurrence L - Lead V - Validation In Black - Not applicable

Exhibit 3-1: Product Roles and Responsibility Chart for the NTSP

2. Purpose. The purpose of the TE is to provide an estimate of PMT requirements determination, costs and scheduling based on logistics and MTRP data in a standard format. Where applicable, identify Program Budget Information System issues and contract amount with estimated delivery dates. Developers must justify any data elements deemed to be non-required. The purpose of the NTSP is to provide comprehensive PMT requirements in support of logistics products in a standard format. Developers must justify any data elements deemed to be non-applicable.

a. A modification is an update or any alteration (e.g., software, hardware, etc.) to an existing ESS.

b. Modifications that apply to a system, platform, component or any portion of a program will also be included in a TE or NTSP.

3. Process. A TE must be initiated by MS B and completed and approved by MS C for new programs. The completion of the TE is the entrance criteria for the full development of an NTSP, unless a modernized or updated system NTSP is in progress. The TE is submitted for RS approval and completed by MS C. RS's are required to obtain concurrence from CNO N1 for PMT before approving a TE. Once approved by the RS, the TE is the official record of Navy PMT requirements. Once an NTSP is submitted and approved, a TE is no longer required. Urgent or Joint Emergent Needs, modernization and rapid or abbreviated programs, including non-developmental items, COTS, GOTS and rapid deployment capability recommended for the transition into a program of record, will be incorporated into a TE and a follow-on NTSP. The TE enables stakeholders to project PMT requirements and funding across the FYDP. Business Systems will follow procedures in reference (m). The PM documents the PMT requirements for an integrated weapon or warfare system in a TE or NTSP. The completion of the TE is a supporting-supported relationship between the PM and key stakeholders. Each TYCOM will coordinate with FFC and PM before approval by the RS in the Department of Navy Enterprise Task Management Software Solution. Table 3-1 shows the comparison of TE and NTSP Requirements in via the format of each document.

TE	NTSP
PART I Equipment, System, Subsystem Description	PART I Technical Program Data
PART II Equipment, System, Subsystem Logistics, Product Supportability Analysis, Training Requirements	PART II Billet and Personnel Requirements
PART III Rating Continuum	PART III Training Requirements
PART IV Logistics Products	PART IV Training Logistics Support Requirements
PART V Formal School House and Fleet Fielding	PART V PMT Plan of Actions and Milestone (POA&M)
	PART VI Decision Items, Actions Required
	PART VII POCs

Table 3-1: Comparison of TE and NTSP Requirements

a. TE and NTSP Development and Approval Process

(1) PMs Responsibilities: PMs are required to implement the MTRP process using enclosure (2). For TE and NTSP development utilize enclosure (3). The PM will submit preliminary Program Budget Information System issues to the RS for consideration that support cost estimates and any unfunded issues within the POM cycle. PMs will produce prescribed products used in the development of a TE and informed by a CNO N1-approved RDA in

coordination with USFLTFORCOM, COMPACFLT and TYCOM. An NTSP must be informed by a product or logistics support analysis and authoritative data. Development of life cycle training solutions will start when the TYCOM, TSA and TA concur, and the RS approves the MTRP products for incorporation into an NTSP.

- (a) Implement the MTRP process using enclosure (2).
- (b) Develop TE and NTSP utilizing enclosure (3).
- (c) Submit preliminary Program Budget Information System issues to the RS for consideration that support cost estimates and any unfunded issues within the POM cycle.
- (d) Produce prescribed products used in the development of a TE and informed by an N1-approved RDA in coordination with USFLTFORCOM, COMPACFLT and TYCOM.
- (e) Ensure the NTSP is informed by a product or logistics support analysis and authoritative data. Development of life cycle training solutions will start when the TYCOM, TSA and TA concur, and the RS approves the MTRP products for incorporation into an NTSP.

(2) TE Approval Process

(a) The TE provides an estimate of training requirements. These estimated training requirements are derived from the program offices approved ESS, Logistics Support Analysis and Product Support Analysis, task analysis (maintenance, operations tasks, Job Task Analysis or Job Duty Task Analysis) and CNO N1approved RDA with USFLTFORCOM, COMPACFLT and TYCOM coordination. This product is developed using MTRP data. The TE will be aligned with the manpower estimate and RS approved preliminary manpower documents if applicable. The analysis will be updated and transitioned into an NTSP at MS C.

(b) The TE is required 5 years before system IOC for programs requiring MILCON, or by the date indicated for programs meeting the criteria in subparagraphs (2)(b)1. through (2)(b)3.

1. 4 years before system IOC for ACAT I or II programs requiring TD research, development, test and evaluation.

2. For all ACATs, a TE is required before initial operational test and evaluation or low-rate initial production, whichever comes first.

3. The TE is required for programs such as non-developmental items, COTS, GOTS, rapid deployment capability, abbreviated acquisition program and urgent need initiatives and will be approved prior to the final decision (ideally 3 months before IOC) in order to ensure that an appropriate training solution is identified and resourced.

a. Reference (e) defines the DON urgent needs and rapid development process that can be fielded in less than 2 years and are below the cost thresholds of ACAT I and information assurance programs. During the Engineering, Manufacturing and Development phase, the acquiring organization will provide the identified training and produce a TE to support the production and development phase requirements. The PM will develop the NTSP to meet operational and sustainment requirements.

(1) Integrated weapons, warfare systems and functional areas (e.g., habitability, damage control, etc.) require NTSPs.

(2) The RS is responsible for funding warfighter training requirements (e.g., initial, interim and follow-on training). The program office obtains concurrence from key stakeholders and submits a TE to the RS. The critical stakeholders include the RS, USFLTFORCOM, COMPACFLT, TYCOM(s), the TSA, SYSCOM, TSPO, TA, OPNAV N13 and CNO N4. These stakeholders are responsible for coordinating with their subordinate activities as applicable. The RS initiates a 21-calendar day concurrence validation via associated tasker system. Stakeholders validate PMT requirements in the TE and submit comments. Coordination with USFLTFORCOM and TYCOMs must be completed via ETMS2 system. If ETMS2 is replaced by a future DON tasker system, then utilize that system for all coordination, deliverables and approvals. The RS will validate the TE using enclosure (3) and communicate the PMT needs and identify any program gaps prior to TE approval. After establishing adequate planning, programming, budgeting and execution resources for training during the initial or interim phases throughout the FYDP, the RSs will obtain concurrence from CNO N1 prior to TE approval. When approved by the RS, the TE is the estimated training requirement until the requirement is finalized via an approved NTSP to include Total Ownership Cost data.

(3) If the RS determines an PMT Advisory Board is warranted, the RS organizes and chairs the board, supported by the PM, SYSCOM and TSPO. The PM will adjudicate all action items identified during the board and publish the results within 30 days. If a board is not warranted, the TE is updated based on the adjudicated comments.

(4) A 21-calendar day concurrence validation, the program office incorporates TE changes and submits the TE and an adjudicated Comment Resolution Matrix to the RS. OPNAV N13 will provide a formal concurrence letter via ETMS2 to the RS prior to approval. The RS will validate the TE is funded for execution before approval. Where the TE lacks funding, the RS will annotate exceptions and upload a TE approval letter into the approved tracking system.

(5) Programs designated with JROC or Joint Capabilities Board-interest and major defense acquisition programs (ACAT 1D and ACAT 1C) will not receive TE approval by the RS until the accomplishment of additional process steps delineated in reference (f).

(6) The TE approval process for abbreviated acquisition programs and rapid acquisition programs, such as rapid deployment capability, will follow the same procedures

stated in subparagraph. Abbreviated acquisition programs will be integrated with a program of record unless the RS requires a stand-alone TE. 3 months before the abbreviated acquisition program's IOC, an approved TE will be posted. At this point, the requirement for the MTRP ends and the development of the NTSP begins using the format in enclosure (3).

4. The program office is responsible for TE updates until approval of the NTSP. A TE is not required if there is an approved NTSP. Updating the TE continues until the program office develops the NTSP. Until lifecycle training is available to the Fleet, the RS and Program Office continues to fund initial and interim training.

5. The NTSP is the output of MTRP. The MTRP consists of the items in Exhibit 2-1 with the applicable product support analysis data, lifecycle management data, maintenance task analysis, credential transparency description language and PQS. Utilize an NTSP development tool in conjunction with existing NTSP web repositories and import information and data from various sources (e.g., TFMMS), CeTARS, Model-Based Product Support, Naval Operational Business Logistics Enterprise, Navy Enterprise Resource Planning, Aviation Sustainment, Supply Chain Management, Navy Maintenance Repair and Overhaul and Product Lifecycle Management). All NTSPs must specifically state in the Executive Summary section of the NTSP whether the NTSP capability is an ACAT (I through IV) Program of Record, a non-ACAT capability or rapid acquisition program.

(3) NTSP Approval Process

(a) The program office submits the baselined NTSP to the RS. The RS initiates a pass one (45-calendar days) requirements validation review (e.g., on ETMS2). USFLTFORCOM, COMPACFLT, TYCOM(s) and applicable stakeholders submit validated comments from the NTSP in ETMS2. The RS determines if a PMT Advisory Board is warranted and organizes and chairs the board, supported by the PM, SYSCOM and TSPO. The PM will adjudicate all action items identified during the board and publish the results expeditiously. If a board is not warranted, the NTSP is updated based on the adjudicated comments.

(b) Following the 45-calendar day requirements validation, the program office incorporates NTSP changes and submits the NTSP and an adjudicated Comment Resolution Matrix to the RS. The originator will notify the RS of any substantive comments and determine if the NTSP process continues. The RS initiates a pass two (30-calendar day) concurrence validation, coordinating applicable stakeholders via the approved ETMS2 tasker system. Upon satisfactory resolution of all comments, OPNAV N13 will provide formal concurrence to the RS. Where the NTSP requirements lack funding, the RS will annotate exceptions and upload the NTSP approval letter, the NTSP and the adjudicated Comment Resolution Matrix in the approved tracking system.

Note: RS will coordinate ACAT I and DON selected programs at the Flag or Senior Executive Service level during pass two validation to obtain OPNAV N13 concurrence.

(c) Mandatory stakeholders on both the pass one requirements validation and the pass two concurrence validation include the RS, USFFC, COMPACFLT, TSA, TA, SYSCOM and TSPO, TYCOM, OPNAV N13 and CNO N4. These stakeholders will coordinate with subordinate activities, as applicable.

(d) The purpose of the pass two concurrence validation is to address or resolve substantial or critical comments identified during pass one and to validate programmed funding is available for development, execution and support of the training solutions.

(e) Programs designated with JROC or Joint Capabilities Board-interest and major defense acquisition programs (ACAT 1D and ACAT 1C) will not receive NTSP approval by the RS until the accomplishment of additional process steps, delineated in reference (f).

(f) If transitioning to a program of record, the NTSP approval process for abbreviated acquisition programs and rapid acquisition programs, such as rapid deployment capability will follow the same procedures as in subparagraph a(3)a through a(3)b, but with a 30-calendar-day requirements validation period (pass one) and a 14-calendar day concurrence (pass two) validation. Abbreviated acquisition and rapid acquisition programs will be integrated with a program of record unless a separate NTSP is required. The approved NTSP will be developed and posted no later than 3 months before the POM cycle and the training solutions will be delivered before RFT.

(g) The program office must determine if program changes (e.g., equipment change, Doctrine, Organization, Training, Materiel, Leadership and Education, Personnel, Facilities and Policy change request, modernization, AAE results, etc.) require an NTSP update. The program office will also conduct an NTSP review annually and will report their findings to the RS no later than 1 February. The program office is encouraged to make pen and ink changes for minor edits (i.e., administrative changes) in lieu of a comprehensive update. Change pages are recommended and are to be submitted to the PM to determine if an update or pen and ink change is warranted. Based on the program office's recommendation, the RS makes the final determination. Upon completion of either a comprehensive update or pen and ink changes, the revised NTSP will be posted to the appropriate NTSP website and all stakeholders notified. Should pen and ink changes be implemented, the next comprehensive NTSP update will incorporate these changes. The following Exhibit 3-1 displays the NTSP approval process.

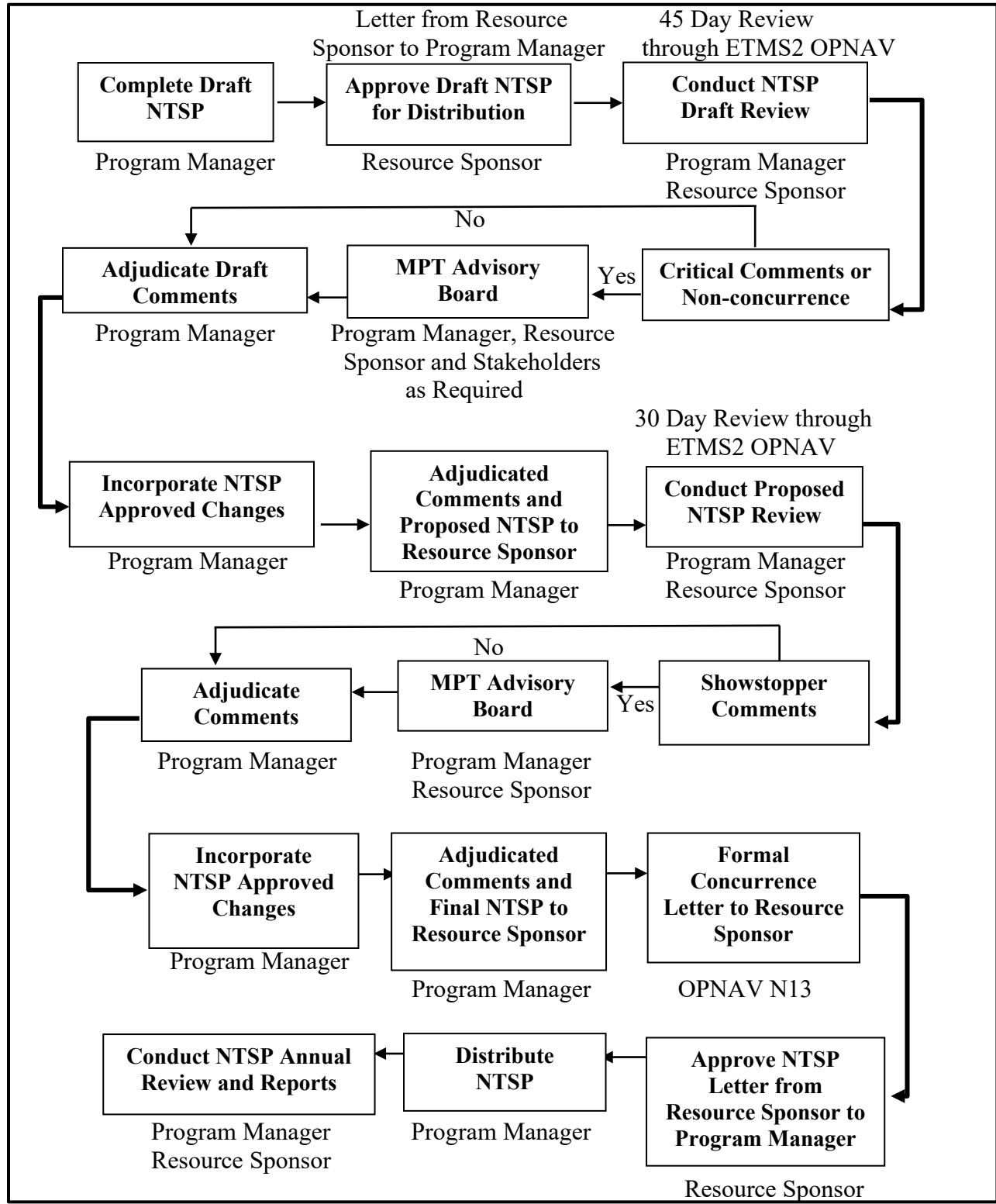


Exhibit 3-1 NTSP Approval Process

(4) Deliverables: TE and NTSP documents

(5) TE Format: The template for the TE is located at: https://flankspeed.sharepoint-mil.us/sites/MyNavyHR_Communities/OPNAV_N13M5.

Note: Ensure TE number, when assigned (same as NTSP), appears on every page of the document on the upper-right hand corner, followed by the date of the TE. The RS will assign a unique identification number before beginning a new TE. The document identification number is alphanumeric, for example:

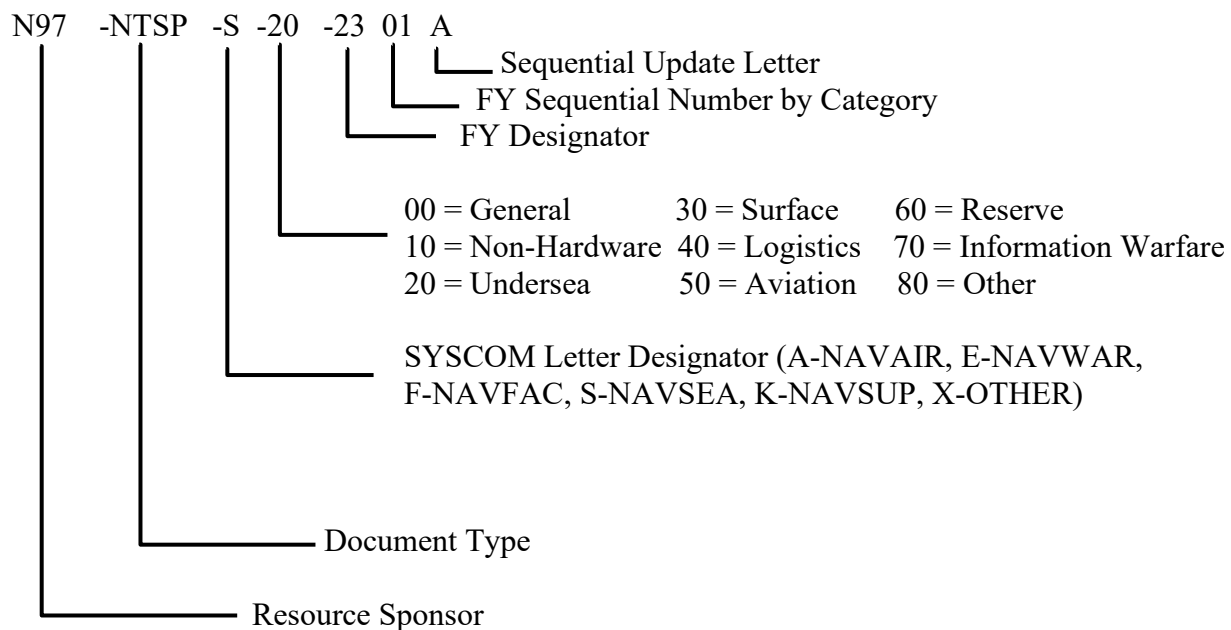


Exhibit 3-2: Sample NTSP Identification Number

(6) NTSP Format: The template for the NTSP is located at: https://flankspeed.sharepoint-mil.us/sites/MyNavyHR_Communities/OPNAV_N13M5.

Note: Use the same document identification number as assigned by the RS for all MTRP documents and the NTSP. Number pages by parts such as I-1, I-2, etc. Ensure NTSP number, when assigned, appears on every page of the document on the upper right-hand corner, followed by the date of the NTSP. Information that does not apply will be marked "Not Applicable" stating why.

ASSESSMENT, ANALYSIS AND EVALUATION

1. Introduction. The AAE strategy describes the process to conduct an assessment, analysis and evaluation using AAE Business Rule 1 or 2. The AAE also identifies methodologies to collect and evaluate performance data of training systems to inform future training requirements or changes that improve readiness (or both). AAE can use data from any of the following: MTRP, Test and Evaluation Master Plan and Electronic Trial Cards to ensure alignment with the OCCSTD and Rating Continuums. Updates will be developed based on MTRP triggers (see Exhibit 2-1) and incorporated into the next scheduled update. An evaluation will be conducted after completion of the pilot training event and when the system and Fleet performance data is available for collection and evaluation. The results of the AAE may drive revisions to MTRP products (FEA, RDA, etc.). The following Exhibit 4-1 shows the Product Roles and Responsibility Chart for the AAE.

Product	Acquisition Timeline Completion By	CNO N1	RS CNO (N1, N2/N6, N4, N9, N095)	SYSCOMS	P M / P E O	T S A	TSPO/ TSPM	T A	NETC	SYS CO M Tech Authority	USFLTFO RCOM, COMPAC FLT and TYCOM(s)	USMC Combat and Integration and HQ Aviation	Commander Naval Installations Command	N A V F A C
TEEP/ TED	MS C	A	C		L C	V C	C	C	C	V,C	V,C			C

A - Approval C - Concurrence L - Lead V - Validation In Black – Not applicable

Exhibit 4-1: Product Roles and Responsibility Chart for the AAE

2. Purpose. To document the strategy, schedule and measures of effectiveness as determined by the PMs, TYCOMs, NETC and other stakeholders.

a. Deliverable: Tailored AAE documented as identified in this enclosure (4).

b. AAE Formats: The Standard Operating Procedure templates for the AAE are located at: https://flankspeed.sharepoint-mil.us/sites/MyNavyHR_Communities/OPNAV_N13M5.

3. Process. The PM is responsible for execution of AAE in coordination with NETC, TYCOM and CNO N1. CNO N1 approves the AAE. The program office must coordinate with the TYCOM and NETC to establish a schedule to develop an AAE. At a minimum, the AAE will be conducted if realignment with the MTRP or Test and Evaluation Master Plan (or both) and Electronic Trial Cards if required resulting from changes to a Rating Continuum. The results of the AAE drive revisions to MTRP products (FEA, TSRA, RDA, etc.). These tasks will be uploaded in RCDC. Updates will be developed based on MTRP triggers and AAE recommended courses and action. Requirement for the AAE will follow the AAE Business Rules, Strategy, Process Maps, and Standard Operating Procedures located at: https://flankspeed.sharepoint-mil.us/sites/MyNavyHR_Communities/OPNAV_N13M5

a. The AAE identifies and describes the plan to conduct an assessment, analysis, and

evaluation of the system's performance levels and proficiency requirements to include training or other solutions as appropriate. Updates will be developed based on MTRP triggers.

b. The AAE, where applicable, provides analysis of training capability based on demonstrated trainee performance improvements directly attributable to the training received.

c. The task analysis (system or rating) data is the authoritative source for the system and tasks (learning objectives) linked to OCCSTD providing the foundational starting point for planning.

d. The PM, in coordination with the RSs or TYCOM, will select an Executing Organization to execute the AAE. The RS(s) must consider the extent and complexity of AAE desired in the selection of the Executing Organization. The Executing Organization will possess or be provided dedicated resources and be able to report results directly to the RS(s). The Executing Organization should have no organizational conflict of interest and be able to report results directly to the RS. The Executing Organization could be the TA, TYCOM or a team of personnel from these organizations (with a designated lead activity).

e. The AAE Business Rule 1 includes a quantitative evaluation strategy that identifies measures of effectiveness, reaction, summative, behavior and result evaluations. Reaction evaluations are assessments by the students receiving the training of their reactions and opinions of their learning experience. Summative evaluations are assessments of the knowledge, or the skills gained during the training. Typically, summative evaluations provide a quantitative grade and make a judgment about a person's knowledge, skills and achievement based on the training. Behavior evaluations provide the measures regarding the degree to which students apply what they learned during training when they are back on the job. Result evaluations provide information regarding the degree to which targeted outcomes occur as a result of the training.

f. A plan for result evaluations will be addressed as early as possible in the training development timeline.

g. An AAE for system training will be conducted after completion of the pilot training event and when the system and Fleet occupational performance data is available from a trigger per Business Rule 2 to collect and evaluate. The results of the AAE may drive revisions to MTRP products (FEA Analyses, etc.).

TRAINING INSTALLATION AND TRANSFER AGREEMENT

1. Introduction. A TITA is required before the RFT date. A TITA is required for complete training systems. The application of SYSCOM procedures for Configuration Data Managers Database Open Architecture, configuration changes and installation execution are authorized. The TITA or IDP pairing ensures that all approved NTSP training resources and capabilities are in place to support the execution of the transfer of responsibility from the PMT RS, Enterprise SYSCOM PM, TSPO and the TSA to the TA. The TITA is designed for the Warfare Enterprise RS, Enterprise SYSCOM's PM, TSPO, TSA and TA to ensure the coordinated, effective training transfer from procurement to sustainment training. The TITA executes the transfer of responsibility for a complete sustainable training system from the SYSCOM's PM, TSPO or the TSA to the TA.

2. Purpose. To document the training products, equipment, facilities and infrastructure are in place to support the implementation of the transfer of responsibility for a complete training system from the TSA to the identified TA. To establish guidelines for development and submission of a TITA for planning, coordinating and to deliver new or modernized training system products identified in a RS approved NTSP and subsequently, the installation and transfer of those training products and their complete logistics outfitting. It also includes supporting technical documents and curriculum per this instruction. The IDP is Part B of the TITA. The following Exhibit 5-1 shows the Product Roles and Responsibilities Chart for the Training Installation and Transfer Agreement.

Product	Acquisition Timeline Completion By	CNO N1	RS CNO (N1, N2/N6, N4, N9, N095)	SYSCOMS	P M / P E O	T S A	TSPO/ TSPM	T A	NETC	SYSCOM Tech Authority	USFLTFOR COM, COMPACF LT and TYCOM(s)	USMC Combat and Integration and HQ Aviation	CNIC	N A V F A C
TITA	Full rate production	A	C		L , V	C	C	C	C	V,C	V,C	V,C		C

A - Approval C - Concurrence L - Lead V - Validation In Black - Not applicable

Exhibit 5-1: Product Roles and Responsibility Chart for the TITA

3. Policy

a. An initial TITA will be developed and approved no later than 180 days after MS C for all ACAT programs and non-ACAT programs. The TITA will be implemented for all new or modified programs of record systems. The content of the TITA template is not meant to be prescriptive. The developer, as concurred with by the Warfare Enterprise Manpower and Training RS and the Warfare Enterprise SYSCOM and TSPO can tailor the TITA. All TITA documents will be updated to reflect changes by the assigned SYSCOM PM, TSPO and TSA and submitted to the SYSCOM for validation, then to the Warfare Enterprise PMT RS for coordination and concurrence with the TA. These updates permit the Warfare Enterprise RS and

CNO N1 in coordination with NETC to identify necessary resources to ensure successful delivery and sustainment of training items identified at the CNO approved RFT date as identified in the NTSP.

b. Training risk levels will be identified via stoplight ratings and mitigations prioritized before RFT, for items in the TITA that are not labeled with a “Green” status. In such cases, the appropriate Warfare Enterprise PMT RS will coordinate with NETC TA and TYCOM to develop a risk mitigation strategy for all items not in a “Green” status.

c. Training will transition at RFT provided all items listed in the TITA are in a “Green” status or have exceptions acknowledged and concurred in by principal stakeholders with the proper execution of the Phase III Final IDP. The TITA does not apply to Naval Nuclear Propulsion or Strategic Weapons Systems training system deliveries.

d. The IDP document development process initiates using a Trainer change proposal (CP) routed through the TSPO or other delegated activities for document type (IDP or IDP Short Form) determination and number assignment. The TSPO and TA should consider the use of the complete IDP for all initial training capability installations and those situations requiring documentation of significant facility impacts to ensure adequate logistics and facility planning is completed, thus limiting the potential for future support conflicts.

e. The TITA will be used to ensure software installation for lifecycle events are achieved at Navy schoolhouses.

f. A TITA identification number will be used to facilitate tracking and identification. The first character identifies the TSA's command, followed by a consecutively assigned number, which is unique to each TITA. An additional four characters may be assigned for further command identification. Each IDP will be assigned the same number as the parent TITA, with sequential letter identifiers on end to identify multiple IDPs designated under the same TITA. An example of the identification system is:

Command X X X X #
IDP Identifier Optional command unique designator Sequential Number

C - Command	A - NAVAIR	E - NAVWAR
F - NAVFAC	N - NAVAIR Training Systems Division	U - NAVSUP
M - BUMED	S - NAVSEA (Surface)	SS - NAVSEA (Submarine)
		X - Other

4. Process

a. The TITA process intent is to ensure successful documentation of training systems and necessary logistics support, as identified in the approved NTSP for the subject system transition into sustainment requirements. This agreement validates the execution status of PMT items identified in the CNO approved NTSP and associated approved SYSCOM training system

installation and logistics outfitting to achieve RFT and begin sustainment training. The TITA is intended to be a coordination and a management tool to facilitate effective planning for training system delivery at the Deputy Chief of Naval Operations level.

b. Status indicators identified in subparagraphs 4b(1) through 4b(3) apply to the training items listed in the TITA template. The Warfare Enterprise RS, SYSCOM's PM, TSPO and TSA have the latitude to jointly tailor the status and justification of any items (e.g., funding or schedule changes), upon coordination with the TA and any other entities as necessary.

(1) Red. Training system and product lifecycle management data requirements are not programmed in the RS POM or are unfunded to meet the NTSP approved RFT date, or the deadline established by TYCOM governing issuance. Risk mitigation OQE response and corrective action from the responsible activity or authority is required.

(2) Yellow. Training system requirement and product lifecycle management data requirements are programmed in the POM, funded, but behind schedule to meet NTSP approved RFT date. A risk mitigation response from the responsible activity or authority is required.

(3) Green. Training system requirements are funded and on schedule to meet RFT.

c. TITA signatories could include the following: Warfare Enterprise RS (Tactical and PMT), Enterprise SYSCOM's PM, TSPO, TSA, TA, OPNAV N13 and should be defined and the attached template modified as necessary as part of initial training solutions planning.

5. Discussion

a. The provisions of this instruction are applicable to all situations where the procurement of or modification to training systems (e.g., TTE) or TD and logistics support) is required to establish or sustain formal training at naval training commands, other naval commands which conduct formal training courses and at other DoD facilities where training is conducted under U.S. Navy auspices. For all such procurements, an IDP must be prepared to:

(1) Define facility requirements, including MILCON and special projects, for installation of the training system.

(2) Identify all associated logistics support elements.

b. The TITA or IDP will be an integral part of the system process and will support the introduction of new training systems and equipment, which are planned following reference (t), training system changes and sustaining requirements.

c. The TSA or Installing Activity will ensure that TITA or IDP development and execution status is reported to the TA and assist in coordinating accomplishment of the schedule leading to

RFT and final acceptance. TSA or Installing Activity will also chair the Pre-Installation and Post-Installation briefs using an agenda, minutes or report and checklists.

d. When submitting Phase I, all data items should be addressed. Include any available Phase II data. This critical document states the training system facility support requirements and all logistics support elements necessary to support the establishment of a training system at a Navy Learning Facility. The complete IDP must reflect the most current information available.

e. The IDP is intended to provide a format to consolidate the relevant data to be provided to the TA in support of a planned installation. TSAs (i.e., PMs) or SYSCOMs may develop the IDPs, which contain additional and clarifying information. The IDP contains the typical considerations necessary to provide either a new training or updated capability to a learning facility.

f. In those cases where a change to an existing training asset does not impact the facility infrastructure. An IDP Short Form, consisting of modified Installation Design Documentation, can be used as a standalone document. The IDP Short Form can be recommended for limited facility infrastructure installations by the TSA in the CP, but only used after a training capability has been initially installed by an IDP.

g. Representative process diagrams are provided as Tables 5-1 through 5-6. Table 5-1 shows the IDPs short 16-week process. These tables present facility circumstances and lead times for the development of TITAs, before the RFT date:

(1) 6 years when associated MILCON Major Construction Project for facilities is required.

(2) 149+ weeks when associated unspecified minor construction MILCON or Section 219 (a) funded project for facilities is required.

(3) 97+ weeks when special installation (demolition paid by TSA funds up to 750K) or associated Special Project for facilities (funded by appropriations available for operations and maintenance) is required.

(4) 56+ weeks for installation only (for equipment installations with minimal facility impacts).

(5) 32+ weeks when installation only (accelerated schedule).

h. When MILCON is required, the schedule allows sufficient time for the MILCON planning, programming and budgeting process. When MILCON is not required, the TSA and TA use the time to identify all requirements (equipment and support) and the host installation PWD to plan for the installation, be it a Special Project or routine installation. Depending on the complexity, additional time may be allocated for project completion, that is, site preparation and

installation of the training system. After project identification, a CP form or facsimile is filled out by the TSA and submitted to the TA for concurrence. Upon TA concurrence, the TSA will assign an IDP number and the project will be reported on the appropriate websites (Aviation Technical Training (AVTECHTRA), COMNAVSEASYS COM Training Acquisition, Human Analysis and Requirements Planning System (HARPS)). Following the document number assignment, an Initial IDP (Phase I) is filled out and forwarded to the TA. The IDP will be released as a complete plan to include all three phases. During development and upon submission, all available data will be included. It is preferred that data submission be made electronically as well, except large or detailed drawing packages which may be in a format (i.e., AutoCAD) not readily usable by the reviewers. On revision, all phases that have not been executed (signed by the TSA and TA) will be updated. A signed IDP and updates are considered a binding agreement and are historical data. In sections where complete and detailed information is not immediately available, a paragraph or statement of intent will be provided in place of "To Be Determined (TBD)" or leaving a section blank. If a particular section is not applicable, so state. Include any drawings and depictions that may serve to amplify the IDP data. Each TSPO will maintain its schedule and notify stakeholders upon changes.

Suppliers	Inputs	Process	Outputs	Customers
OPNAV, NETC, SYSCOMs, PEOs	Funding, ID Requirements, Approved Training Requirements	Short development of POA&M, Work Plan, Work Scope, Draft Installation Design Documentation, Fleet Readiness Certification Board, Accreditation	Draft Work Plan Draft IDP Short 12 Weeks	TA
		TSA Signoff	TSA Signoff Sheet, Pre-install Checklist, Signed IDP Short	
	Enablers, OPNAVINST 1500.76E, OPNAVINST 11010.20J,	Phase 3: Install Meeting	Test Report, OPNAV 4790/CK, Signed IDP Completion Report	

	Training Requirements Document, NTSP	Testing or System Operability Verification Test	4 Weeks	
Total Process Cycle Time: 16 Weeks				

Table 5-1: IDP Short - 16 Weeks Process Time

IDP Process Nominal TimelineInstallation Only (Rapid)				
Suppliers	Inputs	Process	Outputs	Customers
OPNAV, NETC, SYSCOMs, PEOs	Funding, ID Requirements, Approved Training Requirements	Phase 1&2 Combined: POA&M Work Plan, Work Scope development, Fleet Readiness Certification Board, Accreditation	Draft Work Plan 23 Weeks Phase 2 Signoff Sheet, Pre-install Checklist	TA
		TSA Signoff	Test Report, OPNAV 4790/CK, Signed IDP Completion Report	
	Enablers, OPNAVINST 1500.76E, OPNAVINST 11010.20J, Training Requirements Document, NTSP	Phase 3: Install Meeting	10 Weeks	
Total Process Cycle Time: 33 Weeks				

Table 5-2: IDP Process Nominal Timeline Installation Only (Rapid) 33 Week Process Time

Suppliers	Inputs	Process	Outputs	Customers
OPNAV, NETC, SYSCOMs, PEOs	Funding, ID Requirements, Approved Training Requirements, Enablers, OPNAVINST 1500.76E, OPNAVINST 11010.20J, Training Requirements Documents, NTSP	Phase 1: Develop Statement of Requirements, Develop Preliminary Site Survey, POA&M, Funding Approach	Arrangement Drawing Draft WorkPlan DD Form 1391 Description of Training, System Design Elements 12 Weeks	TA, PWD
		TSA, TA Phase 1 Signoff	Signed Phase 1,	
		Phase 2: Define Training System, Engineering Site Survey, Develop Trainer Lifecycle Support Package, Develop Train the Trainer	Draft Drawing Package 25 Weeks Phase 2 Test Report	
		TSA, TA Phase 2 Signoff	Signed Phase 2	
		Phase 3: Pre-Install Meeting, Site Prep, Install Training System Testing, Identify Discrepancies, Post-Install Meeting	Test Report, OPNAV 4790/CK, IDP Completion Report 19 Weeks	
		TSA, TA Phase 3 Signoff	Signed Phase 3	
Total Process Cycle Time: 56 Weeks				

Table 5-3: IDP Process Nominal Timeline Installation Only (Rapid) 56 Weeks Process Time

Suppliers	Inputs	Process	Outputs	Customers
OPNAV, NETC, SYSCOMs, PEOs	Funding, ID Requirements, Approved Training Requirements	Phase 1: Develop Statement of Requirements, Develop Preliminary Site Survey, POA&M, Funding Approach	Arrangement Drawing Draft WorkPlan DD Form 1391 12 Weeks	TA, PWD
	Enablers, OPNAVINST 1500.76E, OPNAVINST 11010.20J Training Requirements Document, NTSP	TSA, TA Phase 1 Signoff	Signed Phase 1	
		Phase 2: Define Training System, Engineering Site Survey, Develop Trainer Lifecycle Support Package, Develop Train the Trainer	Description of Training, System Design Elements Draft IPD 29 Weeks	
		TSA, TA Phase 2 Signoff	Signed Phase 2	
		Phase 3: Pre-Install Meeting, Site Prep, Install Training System	Test Report, OPNAV 4790/CK, IDP Completion Report	

		Testing, Identify Discrepancies, Post-Install Meeting		
		TSA, TA Phase 3 Signoff	Signed Phase 3	
Total Process Cycle Time: 97 Weeks				

Table 5-4: IDP Process Nominal Timeline Special Projects or Facility Modifications 97 Weeks Process Time

Suppliers	Inputs	Process	Outputs	Customers
OPNAV, NETC, SYSCOMs, PEOs	Funding, ID Requirements, Approved Training Requirements	Phase 1: Develop Statement of Requirements, Develop Preliminary Site Survey, POA&M, Funding Approach	Arrangement Drawing Draft WorkPlan DD Form 1391 12 Weeks	TA, PWD
		TSA, TA Phase 1 Signoff	Signed Phase 1	
	Enablers, OPNAVISNT 1500.76E, OPNAVINST 11010.20J Training Requirements Document, NTSP	Phase 2: Define Training System, Engineering Site Survey, Develop Trainer Lifecycle Support Package, Develop Train the Trainer	Description of Training, System Design Elements, Draft IDP 29 Weeks	
		TSA, TA Phase 2 Signoff	Signed Phase 2	
		Phase 3: Pre-Install Meeting, Site Prep, Install Training System	Test Report, OPNAV 4790/CK, IDP Completion Report	

		Testing, Identify Discrepancies, Post-Install Meeting	56 Weeks	
		TSA, TA Phase 3 Signoff	Signed Phase 3	
Total Process Cycle Time: 149 Weeks				

Table 5-5: IDP Process Nominal Timeline: Unspecified Minor Construction MILCON or Section 219(a) Funded 149 Weeks Process Time

Suppliers	Inputs	Process	Outputs	Customers
OPNAV, NETC, SYSCOMs, PEOs	Funding, ID Requirements, Approved Training Requirements	Phase 1: Develop Statement of Requirements, Develop Preliminary Site Survey, POA&M, Funding Approach, Draft POM Budget Request	Arrangement Drawing Draft WorkPlan DD Form 1391 POM Submissions 20 Weeks	TSA, TA, PWD
		RFT -6		
		TSA, TA Phase 1 Signoff	Signed Phase 1	
		Revise Phase 1 Annually	12 Weeks	
		RFT -5		
	Enablers, OPNAVISNT 1500.76E,	Phase 2: Define Training System, Refine Facility Requirements	Description of Training, System Design Elements 60 Weeks	
		RFT -4		
		Engineering Site Survey, Develop Ready to Install Certification,	Draft Drawing Package 24 Weeks	

	OPNAVINST 11010.20J Training Requirements Document, NTSP	Develop Trainer Life-cycle Support Package, Develop Train the Trainer		
		RFT -3		
		Phase 2: Define Training System, Engineering Site Survey, Develop Trainer Lifecycle Support Package, Develop Train the Trainer	Description of Training, System Design Elements, Draft IDP 29 Weeks	
		TSA, TA Phase 2 Signoff	Signed Phase 2	
		Phase 2 Revision and Review Annually, RFT - 2 and RFT -1	12 Weeks	
		RFT -2		
		Phase 3: Pre-Install Meeting, Site Prep, Install Training System Testing, Identify Discrepancies, Post-Install Meeting	Test Report OPNAV 4790/CK, IDP Completion Report 45 Weeks	
		TSA, TA Phase 3 Signoff	Signed Phase 3	
Total Process Cycle Time: 312 Weeks				

Table 5-6: IDP Process Nominal TimelineMILCON Major Construction Project 312 Weeks
Process Time

PART A

TITA Sample

(SYSTEM NAME)

1. Purpose. This agreement is provided to coordinate and ensure the successful transfer of training system products for the subject system. This agreement validates the execution status of PMT items identified in the NTSP and TITA to achieve RFT. The TITA is intended to be a coordination vice management tool to facilitate transition planning.

2. Status Indicators. The status indicators identified in subparagraphs 2a through 2c apply to the training items listed in Table 5-7. The Warfare Enterprise RS, SYSCOM's PM, TSPO and TSA have the latitude to jointly tailor the status and justification of any items (e.g., funding or schedule changes), upon coordination with the TA and any other entities as necessary.

a. Red. Training system and product lifecycle management data requirements are not programmed in the RS POM or are unfunded to meet the NTSP approved RFT date, or the deadline established by TYCOM governing issuance. Risk mitigation OQE response and corrective action from the Responsible Activity or Authority is required.

b. Yellow. Training system and product lifecycle management data requirements are programmed in the POM, funded, but behind schedule to meet NTSP approved RFT date. A risk mitigation response from the Responsible Activity or Authority is required.

c. Green. Training system requirement is funded and on schedule to meet RFT.

3. TITA Signatures

a. RS Signature (Date)

b. OPNAV N13 Signature (Date)

c. PM Signature (Date)

d. TA Signature (Date)

e. TSA Signature (Date)

TRAINING ITEMS DESCRIPTION AND REFERENCES

	Training Item	Description	Reference(s)	Required Deliverable & Date	Responsible Authority	Status (R Y G)
1	Training Requirements Planning FEA Validation	Introduction: The assigned SYSCOMTSPO, TSA, Ship PM or other identified POC can begin developing preliminary training system program products with the approval of a Preliminary Ship Design or an ESS program with an assigned PM. Entrance criteria: Training Situation Analysis – BCS The TE or NTSP for the ship or the ESS based upon the preliminary design and task analysis(es) approved by TYCOMS and key stakeholders. The preliminary cost estimate resource requirements, which will be used later by the designated program office for the Material Solution Analysis and MS B (or other SECNAV or OPNAV directive if not an ACAT program) submittal as part of the Preliminary Lifecycle Cost Estimate. The Facility Gap Analysis and the SYSCOM	Approved NTSP DoDI 5000.02 CH 1 of 8 June 2022 SECNAVINST 5000.2G OPNAVINST 1000.16 Series OPNAVINST 5310.23A SECNAVINST 5400.15D DoDI 5000.95 April 1, 2022	Preliminary TSRA FEA BCS Documented in the TSD: IPRD: IMRD: TDCP: MCD: NTSP: TSA: TA Report: MS B: (Initial) MS C: (Update)	RS TSA CNO N1	

	preliminary IDP to be used by COMNAVFACSYSCOM to establish facility space square foot requirements for MILCON or Special Projects resourcing. The COMNAVFACSYSCOM update of the existing projected facility Baseline Facility Requirements document leading to the development of the DD Form 1391 and future construction or modification of facility space for the emergent training capability. A projected list of FEA and Manpower (MER) products consisting of task analysis, TSRA and TDCP that will be developed from the preliminary BCS list using the BCS list ship or ESS existing Technical Documentation. These preliminary FEA products will be routed through the SYSCOM, Warfare Enterprise RS and training community stakeholders after review sessions are conducted with selected SMEs to receive concurrence and approval from the OPNAV RS to continue development. The Entrance criteria consist of the BCS Analysis, MER, FEA products and the associated cost and facility products generated from them. All entrance criteria products require RS and training community stakeholder approval.		Full Rate Production: (Update)		
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		After final ship or ESS design and upon completion (approval) of the ship or ESS technical data, the FEA products are updated and again routed through the SYSCOM, Warfare Enterprise RS and training community stakeholders for final approval. At this point, the TITA and IDP can be completed and routed for final signature and approval. The exit criteria products consist of the final OPNAV approved NTSP, FEA products and the IDP, all of which have been routed via the training community stakeholders for review and approval and are attached to the TITA cover document for signatures. The			
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		approval of the TITA consummates the successful delivery and transfer of the emergent training capability to the training community and enters the Lifecycle Support phase for the training capability by the SYSCOM (TSA) and NETC (TA) per NAVSO P-1000 as resourced by OPNAV RSs.			
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2a	Facilities Training Space MILCON	Identification of facilities (buildings, spaces, utilities, etc.) required or available (or both). Coordination of facility impacts with the host command and TA requesting MILCON, if required.	NAVSO P-1000 OPNAVINST 11010.20J	MILCON : Start Finish	TSA or TSPO
2b	Facilities Training Space (requirements and funding)	The TSA will provide facility data relative to the applicable facility trainer installation to the TA for the IDP coordination.	NAVSO P-1000 approved NTSP	TD (install): RFT:	TSA

3	IDP (PART B of the TITA)	The IDP, which is Part B of the TITA, is a plan prepared to define facility requirements, including MILCON and special projects, for installation of the training system; identify all associated logistics support elements and transfer training ownership responsibility from the TSA to the TA. The IDP is an integral part of the system procurement process and supports the introduction of new training systems and equipment, training system changes and sustaining requirements. Close liaison between the TSA and TA is key to developing IDPs to meet the established RFT date. In cases where there are multiple training sites, there will be a separate IDP for each training site.	Phase 1: RS Phase 2: Phase 3: CNO N4 MILCON required: TSA RFT: TA
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4	Learning Objectives	Learning objectives describe the knowledge, skills and proficiency level(s) to be, mastered. Terminal and Enabling objectives describe the job behavior, conditions, standards and test criterion. Learning objectives will be SMART: Specific, Measurable, Attainable, Results-Focused and Time-Focused.	Standards as Approved by the Training Agent	IPRD: Training Course Control Document	TA
5	Visual Training Aids (VTA) not embedded in PowerPoint.	Visual training aids include photos, videos with or without sound, models, displays, which the TSA deems necessary and that are called out in the products will be delivered through lifecycle support.	Standards as Approved by the Training Agent	RFT:	TSA

6	Interactive Multimedia Instruction (IMI) i.e. Interactive Courseware or eLearning	IMI i.e. interactive courseware content developed to be used for a schoolhouse and in the Fleet concentration area will be delivered through a Learning Management System. Courseware must be developed using the most recent version of Sharable Content Object Reference Model be compatible with the Learning Management System. Additionally, a compact disk or digital video disk of the courseware must function stand-alone outside of a Learning Management System. All applicable interactive products are delivered and maintained through configuration management by the TSA as required by the TSA.	DoDI 1322.26 of 05 Oct 17 Standards as Approved by the Training Agent	RFT:	RS or TSA
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7	TTE Training Devices Funding	The assigned SYSCOM TSA, TSPO, PM will identify in the NTSP all TTE or TD required for the development and approval of the FEA products per OPNAVINST 1500.76 Policy & Guidance. Once RFT has been achieved and the training capability has been transitioned to the training community principals (TA, etc.) the delivering TSA, TSPO, PM, etc., will retain responsibility for all TTE or TD configuration management tasks to update training materials, operating software, software licenses, hardware and technical data products which are driven by the tactical system modernization program. Configuration management	NAVSO P-1000	RFT:	RS, TSA, TA
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		requirements will be resourced by the cognizant RS to maintain the delivered training capability in sync with Fleet operating systems and equipment.			
8a	TTE or TD Maintenance or Tech Support (Initial Training)	Government or contractor operation and maintenance support are provided for all training equipment in support of initial training. Initial training equipment and support items must take place before the program's IOC date or full rate production decision. Contractor operated maintenance support or of integrated logistics support (or both) is in place. Maintenance and technical support are provided for all training equipment, whether in support of initial or follow-on training.	NAVSO P-1000	The first convening of initial training:	RS, TSA

8b	TTE, TD Maintenance, Tech Support (Follow-on Training)	Government, contractor maintenance and technical support are to be provided for all training equipment in support of follow-on training.	NAVSO P-1000	RFT:	RS, TA
9	Student Guide and Materials	Instructor facilitated interactive training may be offered in one of two formats, Synchronous and Asynchronous. The student guide is organized with all related instructional information grouped within a topic. Each topic is grouped by learning objectives and supported with instruction sheets. Student guides contain all of the content for the course, and they are used as a guide for the trainees. Additional material may include textbooks or tech manuals (or both). Student guides and	Standards as Approved by the Training Agent	RFT: Student Guide	TSA, TA

		materials must be approved by the TA and delivered by the TSA 90 days before the pilot.			
10	Course Requirements	Training course control documents, courseware, curricula and manuals transferred by program management office to the designated Learning Center or TA 90 days before the pilot	Standards as Approved by the Training Agent	RFT: Training Course Control Document:	TSA
11a	Physical Security Certification	A facility certified for delivery, storage and use of classified information and material.	OPNAVINST C5510.93F	Date certified if applicable	TA

11b	Cybersecurity	IS or IT Authority to Operate certification status; Cybersecurity Maintenance requirements defined.	OPNAVINST 5239.1D COMNAVCYBERF ORINST 5239.2A	RFT	TA, TSA
12	Training SustainmentFunding	Military or civilian (including contractor) instructor requirements (Military Personnel Navy or O&MN or OMNR)are programmed before concurrence on a final or updated NTSP where increases ornew NEC producing courses are required for operators and maintainers of new or modernized systems. Provide funds to support and sustain the conduct of training delivered by the TSA are programmedand budgeted.	SECNAVINST 5000.2G NAVSO P-1000	RFT date:	RS, CNO N1, TA

13	Individual's Account Funding	Temporary duty instruction and individual's account funds for training are programmed and budgeted for by RSs and execution by CNO N1. RSs or designated agency ensures funding is programmed	NAVSO P-1000	RFT date:	RS, CNO N1
	Temporary Duty Instruction O&MN Temporary Additional Duty Travel Target Funding(TADTAR)	before approval on a final or updated NTSP. O&MN funds for training are provided to units by the RSs. RSs will approve NTSPs where new, sustained O&MN or O&MNR increases are planned for training. (Note: Must consider the impact on training O&MN TADTAR budget when choosing a training solution. Notify RS via TYCOMs, USFLTFORCOM and COMPACFLT if a new			

		trainingsystem will obligate increases in O&MN TADTAR requirements for units to attendinitial and follow-on training such as “F” schools.)”			
14	Navy Enlisted or Officer Occupation ClassificationSystem Packages	Provides enlisted and officer occupational classification system changes and proposals for implementing changes.	OPNAVINST 1210.2 OPNAVINST 1223.1D	Before the RFT date	RS, TSA, CNO N1
15	RFT	The date of a training system and its associated logistics, maintenance, syllabusand instructors are certified to be availablefor training at the learning facility. This date is predicated on the availability of a new, modified or rehabilitated learning facility for training purposes. All aspects of the facility must be		Primary: Secondary: Iterative:	RS, TSA, TA

		ready, including building completion, completed site preparation, training system installation and testing, trained instructors, furnishings, e.g., non-technical collateralequipment. This is the ultimate planningdate for the new, modified and changed the training system and its readiness for use.			
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Table 5-7: Training Items, Description and References

PART B

SAMPLE CP

FROM: (Code and name of PM originating request)

TO: TSA (with appropriate contact information)

SUBJ: Request for Approval to Develop: IDP ☐ IDP (Short) ☐

Project Title: (Brief Title of training equipment installation)

Executive Summary: (Provide a high-level summary of what this change accomplishes, why this change is needed and provide all references that established the requirement for this installation – (NTSP, OPNAV Requirement, etc.). Identify all or any facility impacts anticipated; if none, so state.)

Applicable Ship Classes: (Identify applicable ship classes)

Learning Facilities (Identify applicable learning Facilities) Building and Room No.: (If known)

Related Changes: (List all related changes and specify if prerequisite or conjunctive with this change)

PM: (SYSCOM Manager responsible for a tactical system)Name:

Code:

Phone (DSN or COM):

E-mail:

TSA: (Activity responsible for training program, if different from PM)

Name:

Code:

Phone (DSN or COM):

E-mail:

Installation Activity and Document Preparer*: (*If different - Please specify)Name:

Code:

Phone (DSN or COM):E-mail:

SAMPLE CP

(Complete either the IDP or IDP Short schedules as applicable)

IDP Development and Installation Schedule:

- (1) Preliminary Site Survey:
- (2) Phase I IDP Submittal to TSA:
- (3) Engineering Site Survey:
- (4) Phase II IDP Submittal to TSA:
- (5) Stage Installation and Integrated Logistics Support:
- (6) Installation Start:

IDP Short Development and Installation Schedule:

- (1) Site Survey (if Required)
- (2) Start IDP Short Development:
- (3) Complete IDP Short Development:
- (4) Draft IDP Short Submittal to TSA:
- (5) Final IDP Short Submittal to TSA:
- (6) Stage Installation and Integrated Logistics Support:
- (7) Installation Start:

Remarks: Attach additional pages as desired to provide amplifying information to assist in understanding the proposed change.

Signature:

Date:

Change Review Board Approval:

CP is: Approved Disapproved

Number is assigned for development of this change.

Copy to:

TSA Internal Codes Appropriate Learning Center Learning Facility
ISEA

Exhibit 5-2: Change Proposal

IDP FOR TRAINING SYSTEM INSTALLATION AND TRANSFER OF [SYSTEM AND
EQUIPMENT NOMENCLATURE] AT

[NAME AND LOCATION OF LEARNING FACILITY TO RECEIVE THE INSTALLATION]

[TA COMMAND LOGO]

DATE [ORIGINATION]
REVISED [LATEST DATE OF REVISION]

TABLE OF CONTENTS

Phase Data or Activity

- I. Initial Agreement
 - A. General Information and Training Program Data
 - B. Facility Data
 - C. Major MSs
- II. Engineering Agreement
 - A. General Information
 - B. Engineering Site Survey Data
 - C. Training Support Package
 - D. Initial Instructor, Support Personnel Training Requirements
 - E. Installation Data
 - F. Actions or Decisions (or both) Required

III. Transfer of Training Responsibility

APPENDIX A Points of Contact

APPENDIX B Installation Design Documentation

IDP Phase I Initial Agreement

Facility Requirements Identification(Preliminary Site Survey)

The initial IDP documentation (preliminary site survey) for facility requirements identification for [System or Equipment Nomenclature] is provided per OPNAVINST 1500.76E.

Facility Requirement Summary

Requirement (check appropriate requirement) Responsible Command

MILCON Required* TSA

Special Project Required* TSA

Installation Only Required TSA

*See attached preliminary Site Survey Comments, Remarks:

Date TSA

(Include: Name, Title, Command and Code)

Date RS

(Include: Name, Title, Command and Code)

The attached initial IDP documentation (preliminary site survey) is accepted as the identification

of facility requirements.

Date TA
Include: Name, Title, Command and Code
If signed by direction, so state

Distribution List:
TA (IDP Coordinator)
Learning Center (IDP Coordinator)

SAMPLE

PRELIMINARY SITE SURVEY DATA

1. General Information and Training System Data.

a. NTSP references or establishing authority. Cite NTSP number, date and status. If there is no NTSP, cite other documentation, which establishes the requirement, etc. to include the date of requirement approval.

(1) Operational Use Summary. (UNCLASSIFIED) Describe the operational use of the equipment as it applies to total force use. Note ship classes on which equipment is installed.

(2) Training Concept Summary. Describe the scope of training to be established, sustained or revised as a result of the installation (i.e., Intermediate or Organizational Level Maintenance, Operator Training, etc.). If known, identify course name, title (CIN) and length. If non-specific, so state and describe.

(3) Functional Description. Provide a brief functional description of the training equipment. Refer to sponsor approved requirements documents as appropriate.

(4) Listing of Training Equipment to be installed. Based on the best available information, list the training equipment to be provided by category: TTE and TD. Provide quantity, procuring activity and delivery status. If applicable, cite any identifying equipment tracking numbers (NTSP, etc.).

(5) Interface or Impacts on Other Training Equipment. Describe the interfaces or impacts on other training equipment that will be relocated and equipment that is presently installed or planned for installation. Consider instances of multiple, phased installations that are covered by other IDPs. Where applicable, reference other IDPs.

(6) Timing and Synchronization. Any issues must be addressed in the IDP to document system requirements such as signal type (i.e., 1 PPS, 5MHz, 100 kHz) number of and type of connections to new and existing equipment and impacts to existing equipment and systems. Specific details of the timing and synchronization system are a required part of the Installation Design Documentation. If new timing and synchronization equipment are required, it must be done in a manner that meets all known requirements and future facilities' expansion plans.

(7) Training Equipment to be replaced. Based on the best information available, list all training equipment that will be replaced or relocated (or both) as a result of this installation. Include, if known, the intended disposition of removed equipment according to regulations outlined in this instruction.

(8) Learning Facility (Command or Location). Provide learning facility's address as listed in the current Standard Navy Distribution List. Include zip code and unit identification code.

2. Facility Data.

a. Training Equipment Location at Learning Facility.

(1) Building and Room No.

(2) Floor Plans and Elevations. Provide floor plans and elevations drawings for each room affected as an attachment-based on the best available information.

(3) Other.

b. Basic Facility Requirements of Training Equipment.

(1) Air Conditioning and Heating.

(2) Water.

(3) Power.

(4) Physical Requirements.

(5) Other.

c. Description of Facility and Infrastructure Requirements. Cite MILCON and any other applicable data.

d. Description of Environmental and Safety Impacts. Describe any National Environmental Policy Act requirements, impacts to environmental permits or other safety or environmental compliance issues.

3. Major MSs.

a. Training System RFT Date. The CNO established RFT date as outlined in a current NTSP or other references.

b. Beneficial Occupancy Date. Date facility will be ready to begin the installation of the necessary personal property equipment.

c. IDP Data Scheduled Updates (Minimum). Update required (month and year) for MILCON programming. Update required (month and year) for special project programming. Phase II Engineering Site Survey Scheduled Date (month and year).

IDP

Phase II Engineering and Training Support Package Elements Agreement
For Installation and Transfer Requirements
(Engineering Site Survey and Training Support Package Elements IDP)

Note: The IDP Phase II installation (engineering site survey) and transfer requirements (training support package elements) required for transfer of training responsibility for the training system are provided per OPNAVINST 1500.76E.

The list of training support package elements is certified as complete unless otherwise indicated in the Comments and Remarks section.

Comments, Remarks:

Add attachments if necessary.

Signatures:

Date Training Agent
Name, Title, Command and Code
If signed by direction, so state

The attached IDP Phase II engineering documentation (engineering site survey and listing of training support package elements) is accepted as the determination of installation and transfer requirements and as the basis for the transfer of training responsibility.

Signatures:

Date Training Agent
Name, Title, Command and Code
If signed by direction, so state

ENGINEERING SITE SURVEY AND TRAINING SUPPORT PACKAGE ELEMENTS

1. General Information.

- a. Date Engineering Site Survey Conducted (day, month and year).
- b. Training Course(s) Identification. Identify the course name, title (CIN) and course length. If nonspecific, so state and describe.
- c. Planned Date of Transfer. Planned date of transfer of responsibility for the training system from the TSA to the TA.
- d. List of Equipment.
 - (1) TTE. In column format, state the precise nomenclature of the TTE, the quantity to be provided, the procuring activity and the delivery status. If applicable, cite any identifying equipment tracking numbers.
 - (2) TD. Identify in the same format as paragraph 1d (1).
- e. System Interface with Other Installed or Future Training Systems or equipment. Update information previously provided in Phase I, paragraph 1g.
- f. For Command, Control, Communications, Computers and Intelligence and other computer network-based Training Systems, provide system changes and operating authority documentation such as Interim Authority to Operate and Certification Assessment and Summary Evaluation Form for the Fleet Readiness Certification Board, according to SECNAVINST 5000.36A, 60 days in advance of the planned installation.
- g. Identify the Installation Design Documentation, including the completed Drawing Package that will be submitted with phase II of the IDP.

2. Major MSs.

- a. IDP Schedule. The IDP Schedule is presented only as an example. Use flexibility with regard to its content and format to meet TSA and TA needs.

SAMPLE INSTALLATION DESIGN MILESTONE PLAN

IDP Number: Requirement Establishment:
Training Agent: (Appropriate Learning Center or Installation Manager:
NETC)
Learning Facility: PM:
Facility POC: IDP Originator:
Facility Manager:

CY	20XX									
FY	20XX									
Action Month	J	F	M	A	M	J	J	A	S	
Initiate Project	X	X	X	X	X	X	X	X	X	X
Preliminary Site Survey	X	X	X	X	X	X	X	X	X	X
Engineering Site Survey	X	X	X	X	X	X	X	X	X	X
Beneficial Occupancy Date	X	X	X	X	X	X	X	X	X	X
Site Prep or Installation Prep	X	X	X	X	X	X	X	X	X	X
TTE, TD or Simulator Delivery	X	X	X	X	X	X	X	X	X	X
Engineering Data Submission (Phase II)	X	X	X	X	X	X	X	X	X	X
Installation	X	X	X	X	X	X	X	X	X	X
Certification	X	X	X	X	X	X	X	X	X	X
Curriculum	X	X	X	X	X	X	X	X	X	X
Software	X	X	X	X	X	X	X	X	X	X
Spares	X	X	X	X	X	X	X	X	X	X
Test Equipment (General or Specification)	X	X	X	X	X	X	X	X	X	X
Tools (General or Specification)	X	X	X	X	X	X	X	X	X	X
Technical Data Support Package	X	X	X	X	X	X	X	X	X	X
Instructor or O&M Training	X	X	X	X	X	X	X	X	X	X
OPNAV 4790 or CK Submission	X	X	X	X	X	X	X	X	X	X
On-Site Testing	X	X	X	X	X	X	X	X	X	X
Installation Complete	X	X	X	X	X	X	X	X	X	X
CMS	X	X	X	X	X	X	X	X	X	X
Ready For Use	X	X	X	X	X	X	X	X	X	X
RFT	X	X	X	X	X	X	X	X	X	X

Legend: 0 - Current Schedule X - Actual Accomplishment
Sample IDP Chart

3. Engineering Site Survey Data

Note: Include summarized results of the engineering site survey to address what the existing conditions or capacities are versus the new requirements and identify who will perform and provide for the difference.

a. Security and safety factors:

(1) Security classification of equipment, courses, etc., include any cybersecurity requirements as outlined in the U.S. Navy cybersecurity program

(2) Environmental safety (federal, state, local)

(3) Pressurized equipment safety, include identification of pressurized equipment safety specifications

(4) Nuclear safety, include identification of nuclear safety requirements

(5) Electromagnetic radiation safety, include any radiofrequency hazards to personnel, fuel and ammunition

(6) Electromagnetic security (TEMPEST survey), if required, a TEMPEST Vulnerability Assessment Request must be prepared according to references, before equipment use

(7) Eye and ear hazards, include identification of eye and ear safety requirements

(8) Fire extinguishment requirements

b. Type of extinguishing agent(s) recommended

c. Special precautions to be observed

d. Do existing facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

(1) Airspace clearance criteria (specify clearance requirements between the equipment and the room to accommodate operation and maintenance tasks, airflow, etc.)

(2) Other (specify any other security and safety factors not previously addressed)

e. Utilities: air conditioning

f. Temperature and heating parameters of the equipment (list high and low temperatures of each unique parameter and summarize total)

g. Heating requirements:

(1) Room (list current BTU capabilities)

(2) Equipment (list equipment BTU requirements)

(3) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

h. Air conditioning or cooling requirements:

(1) Room (list current BTU capabilities)

(2) Equipment (list equipment BTU requirements)

(3) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

i. Ventilation requirements:

(1) Room (specify duct and air handling or flow requirements)

(2) Equipment (specify duct and air handling or flow requirements)

(3) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

j. Humidity factors:

(1) Room (specify dew point or relative humidity parameters)

(2) Equipment (define operating and non-operating humidity parameters)

(3) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

k. Air cleanliness requirements:

(1) Room (specify air particulate requirements)

(2) Equipment (specify air particulate requirements)

(3) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

l. Water requirements: (include flow rate, temperature, purity, etc.)

(1) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

(2) Plumbing

m. Type, volume or capacity, pollution abatement requirements, etc.

(1) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

n. Electrical power requirements:

(1) Room (list current capabilities; include shunt trip and emergency shutdown)

(2) Equipment (list equipment's voltage, hertz, amps and phase requirements)

(3) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

o. Grounding requirements (alternating current, direct current, digital, etc.)

(1) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

p. Requirements for generators, transformers, converters, etc. (rating, capacity, etc.)

(1) Do facilities meet this requirement? (If not, provide corrective action and identify the responsible party)

q. Lighting requirements (identify lumens, color, dimming controls, etc.)

(1) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

r. Uninterrupted power requirements (identify facility and equipment requirements by quantity and rating, etc.)

(1) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

s. Lightning protection requirements (identify facility and equipment requirements rating, etc.)

(1) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

t. Hydraulics, compressed gases, steam, petroleum, oil and liquids and other fluids (identify facility and equipment requirements by quantity, quality, rating, etc.)

(1) Hydraulic power requirements

(2) Compressed gas requirements

(3) Steam requirements

(4) Petroleum, oil and liquids product requirements

(5) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

u. Exhaust requirements: (local, state, federal) (identify facility and equipment requirements by quantity, quality, rating, etc.)

(1) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

v. Noise abatement requirements: (state, federal, local) (identify facility and equipment requirements by quantity, quality, rating, etc.)

(1) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

w. Telecommunication Requirements: (Identify internal, external requirements, including the number of stations, etc.). Consider communication connectivity requirements. Connectivity may be required in support of the installation, testing and normal operations. If existing connectivity is identified and agreed upon during the site survey, the IDP will identify the connectivity and describe which systems or testing requirements this connectivity will support. The IDP will specify what resources such as multiplexer or router ports, cable strands and patch panel ports are reserved for the installation. New terrestrial connectivity is usually ordered via a Feeder Telecommunication Service Request. The IDP will provide sufficient technical circuit

data to enable the originator to prepare the request. The training site is responsible for submitting the document to Defense Information Systems Agency (DISA).

(1) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

x. Other: Specify any other utilities not previously addressed

y. Physical Factors and Requirements:

(1) Limitations

z. Space:

(1) Minimum physical cube requirements into which the equipment will fit

(2) Space for foreseeable growth requirements. Floor loading, false flooring, false overhead and cable runs or waveguide requirements: (identify flooring and floor loading requirements)

(3) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

aa. Windows or special access requirements (or both): (identify window, door and any other special access requirements)

(1) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

ab. Overhead hoist requirements: (identify facility and equipment requirements by quantity, quality, rating, etc.)

(1) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

ac. Storage container and storage area requirements: (identify trainer requirements by type, size or capacity, etc.)

(1) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

ad. Security or safety limitation requirements: (identify trainer requirements and include any classified data in computer memory)

(1) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

ae. Antennas. (identify equipment requirements by type, quantity, rating, mounting, etc.)

(1) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

af. Other Special Considerations. (identify trainer requirements and include any quantitative or qualitative data)

(1) Do facilities meet this requirement? (if not, provide corrective action and identify the responsible party)

4. Trainer Configuration Item Index.

Configuration Before Change				
• Equipment Identification Code	• RIC, APL, AEL	• Part #	• Ref Designator	• Model Type & Noun Name
Configuration After Change				
• Equipment Identification Code	• RIC, APL, AEL	• Part #	• Ref Designator	• Model Type & Noun Name

RIC: repairable identification code APL: allowance parts list AEL: allowance equipage list

Table 5-9: Sample Configuration Item Index

5. Training Support Package.

a. Parts (Spare or Repair)

Delivery Description Identification Procuring Activity Delivery Status

b. Test Equipment

General Purpose:

Nomenclature Model # COG Code FSCM SCAT NSN QTY Procuring Activity FY Delivery
Activity Funded Status

COG: cognizant FSCM: federal supply code for manufacturers SCAT: sub-category NSN:
national stock number

Special Purpose:

Description Identification Procuring Activity Delivery Status

c. Tools

General Purpose:

Nomenclature Quantity Procuring Activity Delivery Status

Special Purpose:

Nomenclature Quantity Procuring Activity Delivery Status

d. Support Equipment

Nomenclature Quantity Procuring Activity Delivery Status

(1) List support equipment such as roll around carts, overhead hoists, workbenches, etc., which allows for training equipment operation and maintenance or otherwise supports the course of instruction but is not part of the curriculum. Distinguish between government furnished equipment and contractor provided equipment.

e. Technical Data Support Package

Procuring Activity Procuring Identifier Title Quantity Delivery Status

(1) Identify all technical manuals, Planned Maintenance System documentation, etc., supporting the equipment, subsystem or system.

f. Curricula Material

Type of Material Quantity Procuring Activity Delivery Status

(1) List all curricula to be provided to the TA, including NTSP identified course materials and any contractor-developed materials to support TDs. The identification of curricula deliverables should include items such as instructor guides, trainee materials, handbooks, etc. Cite the curricula standard or specification used. List all data item descriptors used. List all training aids to be provided. Examples include mock- ups, pre-faulted modules, fault insertion aids and unique audio-visual equipment.

g. Software

<u>Nomenclature</u>	<u>Quantity</u>	<u>Procuring Activity</u>	<u>Software Support</u>	<u>Delivery Status</u>
---------------------	-----------------	---------------------------	-------------------------	------------------------

(1) Operational

(2) Maintenance

(3) Simulation

h. Other Equipment Documentation

(1) List all other equipment documentation to be provided, such as software, software license, software support activity contact information and security documentation, custody and inventory records, data lists, parts lists, etc.

i. Unique Identification Requirements:

(1) Provide unique identification according to Under Secretary of Defense for Acquisition, Technology and Logistics signed policy for unique identification of tangible items - new equipment, major modifications and re-procurements of equipment and spares.

6. Initial Instructor and Support Personnel Training Requirements.

a. Describe all initial training required for assigned instructor personnel and for follow-on maintenance support before the RFT date to include the location of the training, length of the training, any special certifications, qualifications or prerequisites required. Use subheadings as required to show the various types of initial training to be provided, e.g., operation, maintenance, other. Include required training for follow-on maintenance support for training equipment through a TA sponsored contractor O&M service or a technical assistance program.

7. CMS or Technician Assistance Program. Identify whether the installed equipment is expected to be maintained through a TA sponsored contractor O&M service or a technical assistance program.

8. Information Assurance Maintenance and Certification Continuity. Where necessary, identify plans and means for the execution of required information assurance maintenance and certification requirements.

9. Actions or Decision (or both) Required.

Actions or Decisions Required

<u>Action Item</u>	<u>Command Action</u>	<u>Due Date</u>	<u>Status</u>
Phase III			
Transfer of training Responsibility for:			
[System or equipment nomenclature]			
[Learning facility name and location to receive the installation]			

1. IDP

a. Ship's Configuration Change Form (OPNAV 4790/CK): Installing activity will initiate and complete OPNAV 4790/CKs and provide them to the Learning Facility 3M Coordinator upon completion of installation. Learning Facility 3M Coordinator will assign a job sequence number and work center to the OPNAV 4790/CKs, sign and forward to the CDM.

b. Installation activity will note that the Installation Design Documentation completion report submitted to the TSA is now required. Forwarding of signed Installation Design Documentation completion report signifies learning facility 3M coordinator receipt of these completed forms.

c. TSA final acceptance signifies that OPNAV 4790/CKs have been filled out and provided to the Learning Facility 3M Coordinator.

d. TA final acceptance signifies that OPNAV 4790/CKs have been submitted by the learning facility to the CDM.

e. Validation of training support package delivery will be performed by the learning facility in coordination with the TSA.

f. Facility and trainer as built drawings:

(1) The TSA is responsible for the preparation and delivery of trainers and facility as-built drawings. Redlined drawings are to be delivered upon completion of the installation and checkout. Finished as-built drawings will be delivered within 60 days following installation, checkout and inventory before final acceptance.

(2) Drawings or diagrams will be developed in digital format, e.g., AutoCAD (dwg), according to MIL-PRF-5480H, Data, Engineering and Technical-Reproduction, and ASME Y14.100-2017, Engineering Drawing Practices of 2017, and will reference codes, specifications, procedures, equipment drawings and each installation as-built drawing.

(3) Drawing documentation will be maintained as a baseline for the trainers and are therefore considered mandatory for an update, if impacted: Architectural and Engineering,

power distribution, electronic cooling water, heating, ventilation and air conditioning, cableway layouts, facility ground distribution, patch panel layouts, floor plans and elevations, antenna layouts and elevations, cable block diagrams, circuit wiring detail drawings, wiring tables (i.e., cable running sheets and cross- connect records) and distribution frame and junction box layouts.

Installation Design

Ready For Use Agreement

Signature below affirms completion of the [equipment nomenclature] installation at [name and location of learning facility receiving installation] with the known deficiencies or discrepancies and that the ship's configuration change forms (OPNAV 4790/CK) have been filled out and provided to the learning facility:

<u>Deficiency</u>	<u>Code Responsible To Correct</u>	<u>Date Correction Will Be Accomplished</u>
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Further, the named training system (including maintenance of the system or equipment), according to the IDP Phase II Engineering Agreement for Installation and Transfer Requirements, except for those items listed, is hereby transferred and **CONDITIONALLY** accepted by the TA. The final transfer of responsibility for the training system will be affected by the completion and signature of Part A of the respective TITA document.

Signatures:

TSA

Name, Title, Command, Code, Date
If signed by direction, so state

TA

Name, Title, Command, Code, Date
If signed by direction, so state

2. IDPs Deficiencies

a. A well-written deficiency in an IDP clearly and briefly states the problem. These descriptions must specifically identify the action and deficiency number and, where required, include specific nomenclature (i.e., title, part number, control number and quantity). Additionally, the following items must be assigned during the equipment transfer process to identify who has responsibility for the corrective action: (1) responsible command or code; (2) estimated completion date; and (3) remarks, as required, that will provide further assistance for timely completion of the action. These deficiencies should also be prioritized for completion according to the schedule identified in subparagraphs 2a(1) through 2a(5):

(1) Priority 1 – An error that prevents the accomplishment of an operational or mission essential function per official requirements or which jeopardizes personnel and equipment safety.

(2) Priority 2 – An error that adversely degrades the accomplishment of an operational or mission essential function or prevents the accomplishment of an operational or mission essential mode per official requirements and for which no alternative workaround exists.

(3) Priority 3 – An error which adversely degrades the accomplishment of an operational or mission essential mode or prevents the accomplishment of an operational or mission essential sub mode; or an error which adversely degrades the accomplishment of an operational or mission essential function or prevents the accomplishment of an operational or mission essential mode and for which there exists a reasonable workaround solution.

(4) Priority 4 – An error that is an operator inconvenience or annoyance and does not adversely degrade a required operational or mission essential function or mode or sub-mode.

(5) Priority 5 – All other errors, including documentation.

b. These deficiencies, when properly written, will consist of one or two sentences of description and identifiers. If specific spare part lists, etc. are available and space is limited in the engineering documentation of the IDP, use appendices. The lists may be merely referencing materials for a specific deficiency, such as an itemization of courses and curricula or a clause guaranteeing the return of borrowed or residual equipment to the installer or provided of the equipment. An effort should be made to see if such references are available and obtainable for inclusion in the IDP.

3. Submission of OPNAV 4790/CK forms is required at the time of the post installation brief. Failure to submit these forms at the time of the post installation brief will not be accepted as a deficiency. Installation activity must now forward the signed Installation Design Documentation completion report with a deficiency list (as applicable) and post installation briefing checklist to the TSA prior to final transfer of the IDP for processing. Installation Activity and Facility 3M Coordinator signature on the Installation Design Documentation completion report signifies that completed OPNAV 4790/CK forms have been received and are in process.

Installation Design Documentation

Title of Installation

Location (Including Building and Room No.)

Date:
Change or Revision:

SAMPLE

TABLE OF CONTENTS

Items with an asterisk (*) are requirements for submission/approval of the IDP Short for Training System Changes.
IDP Short approval cover sheet is included on the last page of this enclosure.

Section Title

1. Planning Data

- 1.1 Title of Change*
- 1.2 Impact Summary*
- 1.3 Estimated Man-Hours and Items Affected*
- 1.4 Site Applicability*
- 1.5 Security and Safety Requirements*

2. Accomplishment Plan

- 2.1 Site Preparation
- 2.2 Installation
- 2.3 Detailed POA&M Schedule
- 2.4 Installation Parts and Materials
- 2.5 Special Tools and Test Equipment
- 2.6 Documentation Required

3. Installation Requirements

- 3.1 Equipment Configuration Impacts
- 3.2 Power Requirements
- 3.3 Electronic Cooling Water Requirements
- 3.4 Heating, Ventilation and Air Conditioning Requirements
- 3.5 Foundation Requirements
- 3.6 Cable Requirements
- 3.7 Cable Routing or Wireway Requirements
- 3.8 Grounding Requirements

4. Testing Plan

- 4.1 Test Procedure Identification
- 4.2 Pre-Installation Testing
- 4.3 Post Installation Testing

5. Training System Configuration Changes

5.1 Configuration Item Index of Hardware or Software Installed*

5.1.1 Configuration Item Index of all Tactical, Modified Hardware or Software Installed*

5.1.2 Configuration Item Index of all TD, Hardware or Software Installed*

5.2 Classified Material Installed/Removed*

5.3 Documentation*

5.4 Change Marking and Identification Plates*

6. Integrated Logistics Support Requirements

6.1 Onboard Repair Parts*

6.2 Tools and Test Equipment*

6.3 Maintenance Plan and Responsibilities*

6.3.1 Corrective and Preventative Maintenance*

6.3.2 Overhaul Requirements*

7. Training

7.1 Curriculum*

7.2 Instructor or O&M Training*

8. Post Installation Brief

8.1 Installation Design Documentation Completion Report

9. Drawing Package Attachments. Identify necessary attachments, such as the items in subparagraphs 9.1 through 9.17, required to support this IDP. If existing facility and trainer drawings are not impacted, so state.

9.1 Floor Plans and Elevations Drawings

9.2 Power Distribution Drawings

9.3 Electronic Cooling Water Drawings

9.4 Heating, Ventilation and Air Conditioning Drawings

9.5 Cableway Layouts Drawings

9.6 Facility Ground Distribution Drawings

9.7 Patch Panel Layouts Drawings

9.8 System Configuration Detail Drawings

9.9 Antenna Layouts and Elevations Drawings

9.10 Cable Block Diagrams

9.11 Circuit Wiring Detail Drawings

9.12 Wiring Tables (cable running sheets and cross connect records)

9.13 Distribution Frame and Junction Box Layouts Drawings.

- 9.14 Installation Material Lists
- 9.15 Equipment Technical Data Summary Drawings
- 9.16 Installation Detail Drawings (i.e., foundations, mechanical assembly, mounting)
- 9.17 Test Plans or Test Procedures (or both)

Provide the following as attachments as necessary to support IDP Short Template: Installation Procedures, Test Procedures, Equipment Unique Identification Requirements

10. Planning Data

10.1 Title of Change: Enter a complete, descriptive change title for the trainer or change being installed. The title should include the official nomenclature of the configuration item (e.g., unit, equipment, software, system or subsystem). All abbreviations and acronyms in the title will be spelled out.

10.2 Impact Summary: Enter a complete high-level executive summary description of the hardware and software changes included as part of this trainer change. This is to include information on major equipment(s) being added or deleted and their impacts to cabling requirements, foundations, electronic cooling water, heating, ventilation or air conditioning and software changes required as part of the installation. This section should also discuss any problems or deficiencies in the existing system or subsystem corrected by the installation of this change. This section should also provide information concerning any prerequisite, conjunctive or related changes and information on any temporary changes closed out by this installation.

10.3 Estimated man-hours and Items affected: The total estimated man-hours required to install or test the Installation Design Documentation will be provided. The total requirements will be broken out as to pre- installation testing, site preparation testing, check out and grooming testing and certification and acceptance testing (include test of all training system elements). These summary man-hour estimates will correspond to the detailed POA&M provided in Section 2.4 of the Installation Design Documentation.

10.4 Site Applicability: Provide complete identification of the learning facility, including building number, floor, room(s), etc.

Location Building Room(s)

10.5 Security and Safety Requirements: Provide information or references (or both) to security and safety issues that apply to this installation. This includes such items as references to the local command's lock out and tag out procedures, security access requirements or equipment security (classified parts or software or electromagnetic interference handling requirements, etc.).

This section should also identify any related security and safety issues that are the responsibility

of the training command to provide in support of the installation

11. Accomplishment Plan

11.1 Site Preparation: Describe all site preparation work to be performed, including utility requirements such as sewage, hydraulics, compressed gasses, steam, petroleum oil or other fluids, special exhaust requirements, with particular emphasis on areas requiring special considerations or outside assistance by the installing activity. The site preparation description should identify the installation process, what will take place and who will be involved.

11.2 Installation: Provide a general description of the equipment installation work to be performed. The detailed hardware and software installation procedures to be followed are to be provided in Section 3 of the Installation Design Documentation. All contractors or government activities (or both) that are to be involved in the actual equipment installation and their responsibilities, as part of the installation process, should be identified.

11.3 Detailed POA&M Schedule: The POA&Ms will provide a detailed installation schedule reflecting start and stop dates for all major events required as part of the installation process. The POA&M will be jointly developed by the installing activity, the learning facility and testing activities (if any). The installation activity will coordinate the development of the POA&M with the learning facility. This POA&M should normally reflect the data elements shown in the sample Installation Design Documentation Installation Schedule. A sample is provided in Exhibit 5-3.

11.4 Installation Parts and Materials: Separate lists for hardware and software will be provided to identify all Government Furnished Material and Contractor Furnished Material required by the installation. These lists will, at a minimum, include the data elements with the providing activity and the disposition of any excess or unused materials identified. Included in this section are any 2D or 3D mockups required by the installation activity. The listing of Government Furnished Material will be provided as Government Furnished Information to the installing activity and is to identify all items the installing activity is to receive as part of pre-cable kits or other installation materials required by the installation. The Contractor Furnished Equipment list, will reflect all material being procured or supplied by the installation activity. Include the Item Number with description, cage and part number, quantity, provided by and disposition or remarks.

11.5 Special Tools and Test Equipment: These lists will, at a minimum, include the data elements with the providing activity and the disposition of any excess or unused test equipment identified. This would include such items as automatic cable testers or other specialized tools and equipment not normally available to the installing activity. Include the Item Number with description, cage and part number, quantity, provided by and disposition or remarks.

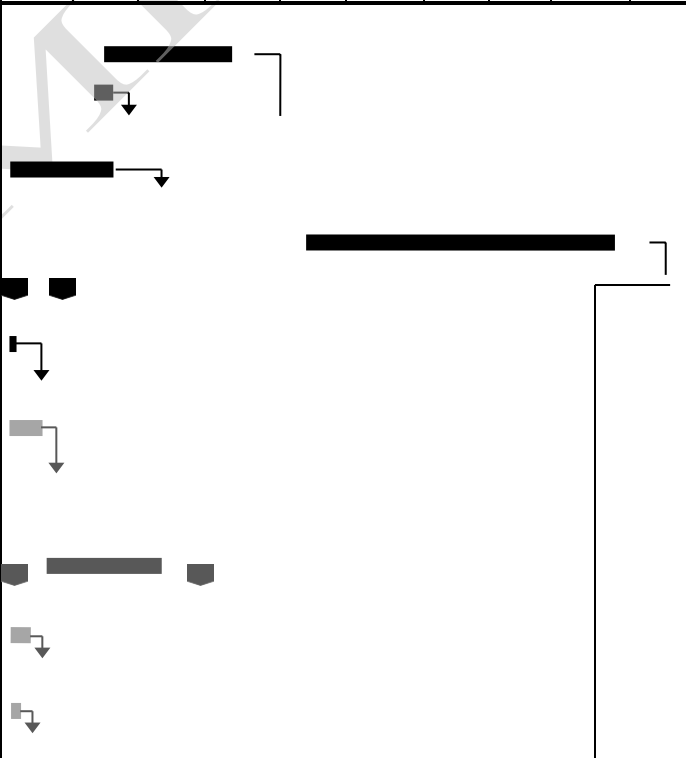
11.6 Documentation Required: These lists will, at a minimum, include the data elements with the providing activity and the disposition of any excess or unused documentation identified.

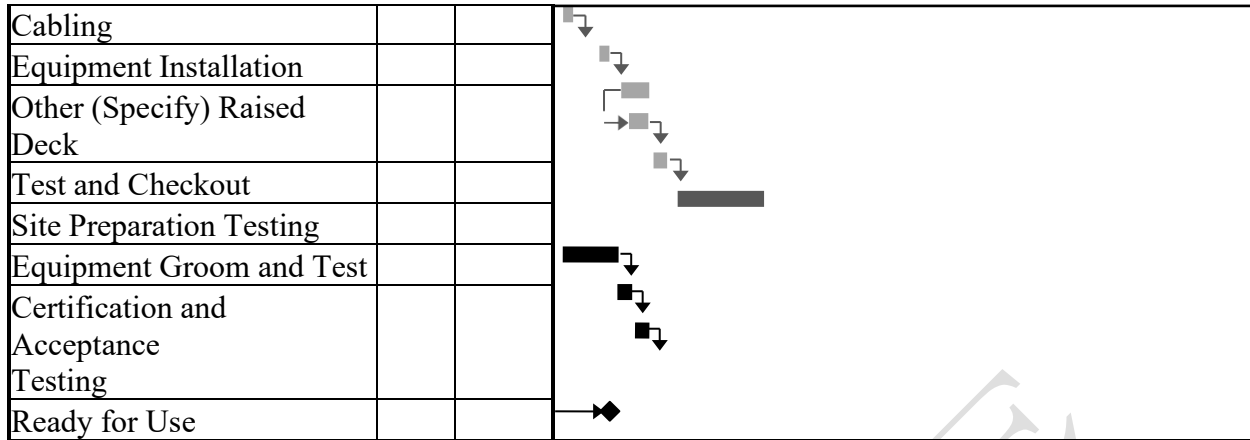
The types of documentation to be identified may include but are not limited to the following: technical manuals, Planned Maintenance System documentation, repairs standards, test procedures, drawings, software documentation or specifications and other related documents. The change or revision level is to be provided where appropriate. Include the title, number, quantity, provided by and remarks.

12. Installation Requirements. The section includes detailed hardware and software installation procedures to be followed. The procedures should be written to a level of detail so that any qualified technician would be able to accomplish the installation solely using the documentation and the drawings provided in the Installation Design Documentation package. Each section should contain all necessary references to the associated drawings provided as attachments to the Installation Design Documentation.

TRAINER INSTALLATION SCHEDULE

IDP No. Learning Facility: (School Name) Trainer:Name or Device Number

Task Name	Start	Finish	FYXX											
			Dec	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep		
Design Development														
Preliminary Lab Arrangement Drawings														
Trainer Change Installation Documentation														
Installation														
Pre-Installation Briefing with Learning Facility														
Trainer Equipment and Integrated Logistics Support Staged on Site (before install)														
Lab Down														
Ripout														
Electronic and Auxiliary Fresh Water														
Heating, Ventilation and Air Conditioning														
Foundations														



Sample Installation POA&M

12.1 Equipment Configuration

a. Provide a summary sheet of all equipment being added or deleted (by nomenclature and part number) and whether its addition or removal will have an impact on power, cooling, air conditioning, cabling, wires, foundations and arrangements. This is to be a quick reference chart; with just an "X" placed under each category to signify, that particular piece of equipment will impact that installation category.

b. Provide a summary sheet of all software required as part of this installation (by nomenclature, part number and revision/version level). If none, so state.

c. Provide detailed pre- and post-installation hardware configuration information. Reporting will be a top-down breakdown to the Lowest Repairable Unit and will reflect impacts on the configuration status accounting from this installation.

12.2 Power Requirements: Provide or provide as an attachment, information showing the power requirements, power panel, breaker number or size and cable designation or type for each equipment that requires an electrical connection. If no impact, so state.

12.3 Electronic Cooling Water Requirements: List or provide an attachment, all equipment requiring cooling water. Specify cooling requirements, manifold, supply and return headers, gauges or valves, flow meters, etc., required for each piece of equipment. If no electronic cooling water requirements, so state.

12.4 Heating, Ventilation and Air Conditioning Requirements: Provide or provides as an attachment, information showing the ventilation changes required to ducting, diffusers, controllers, dampers, etc. If no ventilation changes are necessary, so state.

12.5 Foundation Requirements: Identify or provide as an attachment, information that shows the foundation changes required as part of this installation. If no foundation changes are

required, so state.

12.6 Cable Requirements: List or provide as an attachment, complete wiring table information showing equipment, part numbers, vendor information, cable number, cable class, electromagnetic interference, electromagnetic compatibility, unit, jack number, plug type, backshell, clamp and cable type. If no cable requirements, so state.

12.7 Cable Routing or Wireway Requirements: List or provide as an attachment, information showing all necessary cable routing information, trays, penetrations, wireways and hangers and cable length required for the proper cable installation. If no cable routing requirements, so state.

12.8 Grounding Requirements: Show or provide all necessary equipment grounding information required for proper cable installation as an attachment. This should include equipment identification, terminal number, cable designation, ground termination and hardware required and quantity. If no grounding requirements, so state.

13. Testing

13.1 Test Procedure Identification. Identify all of the test procedures to be conducted as part of the installation. These procedures should describe the type of testing to be accomplished and the test documents to be followed. The procedures should also include the type of tools and test equipment (both general purpose and special purpose) to be used, the safety practices to be followed and identify the persons to be involved in performing and witnessing the testing.

Note: The actual Test Plans or Test Procedures (or both) are to be provided as an attachment or presented as separate documentation to the learning facility with appropriate lead time to permit review and comprehension before testing. A copy of the final redlined and marked-up procedures will be provided to the learning facility. Ensure the inclusion of any necessary information assurance testing required to complete (I) Authority To Operate certifications.

13.2 Pre-Installation Testing. Includes all operability testing necessary to ensure that a trainer or training system, including related data processing hardware, software and interfaces are operating according to trainer system specifications or other documentation before any new equipment or modifications being installed. The pre-installation testing (or baseline testing) may also include system-level operability tests to verify that an entire training system, including subsystems and multi-trainer interfaces, operates and performs as specified in the governing trainer system specifications. Any pre-existing discrepancies noted by this testing are the responsibility of the learning facility to correct and not the responsibility of the installation activity. The installation activity in consultation with the learning facility must determine if

these problems must be corrected before the start of the training system change installation and what re-testing is required once the pre-existing problem has been fixed.

13.3 Post Installation Testing. Verifies that the newly installed equipment is working correctly and has not adversely impacted the operation of the trainer. Testing is conducted in four phases:

13.3.1 Phase 1 Testing - external hardware inspection

13.3.2 Phase 2 Testing - cable continuity and power testing

13.3.3 Phase 3 Testing - equipment powers up and check-out, grooming and post installation system testing

13.3.4 Phase 4 Testing - certification and acceptance testing

13.4 Phase 1 and Phase 2 testing includes facility power verification (voltage and phase), power cable verification, pre-equipment installation tests, hydrostatic pressure tests, cleaning tests, power or water or hydraulic interface validation tests and cable continuity and insulation resistance tests. It also includes the external examination of equipment before any power being applied to the equipment. If the external examination shows the potential for internal damage, the TSA will be notified.

13.5 Phase 3 testing includes equipment check-out, grooming and post-installation testing. It provides for power application to equipment and the performance of all operability testing necessary to ensure that a subsystem or standalone trainer (or both) are operating according to trainer or trainer system specifications. This testing will include system-level operability tests to verify that an entire trainer system, including ESS and multi-trainer interfaces, operates and performs as a complete and composite system within tolerances and parameters specified in the governing trainer system specifications.

13.6 Phase 4 certification and acceptance testing consists of portions of the equipment check out and groom tests that demonstrate trainer performance within the approved design and performance specifications. This testing is a prerequisite for the turnover of the trainer to the learning facility.

14. Training System Configuration Changes

14.1 Configuration item index of hardware and software

14.1.1 Configuration item index of tactical, modified tactical hardware or software

- a. For Tactical Hardware or Software Configuration Before Change include:
 - (1) Host Software Component, Functional Group Code or Repairable Identification Code
 - (2) Model Type and Noun Name
 - (3) Cage or part number
 - (4) Revision

(5) Quantity

- b. For Tactical Hardware or Software Configuration After Change include:
 - (1) Host Software Component, Functional Group Code or Repairable Identification Code
 - (2) Model Type and Noun Name
 - (3) Cage or part number
 - (4) Revision
 - (5) Quantity

14.1.2 Configuration Item Index of all TD Hardware or Software installed

- a. For Tactical Hardware or Software Configuration Before Change include:
 - (1) Host Software Component, Functional Group Code or Repairable Identification Code
 - (2) Model Type and Noun Name
 - (3) Cage or part number
 - (4) Revision
 - (5) Quantity
- b. For Tactical Hardware or Software Configuration After Change include:
 - (1) Host Software Component, Functional Group Code or Repairable Identification Code
 - (2) Model Type and Noun Name
 - (3) Cage or part number
 - (4) Revision
 - (5) Quantity

Note: Provide detailed pre- and post-installation hardware and software Configuration Item Index information for all Tactical and TD. Section 13.1 will be a top-down breakdown to the Lowest Provisioned Unit and will reflect all impacts to the configuration status accounting from this installation. This information will be reported to the CDM for the Learning Facility via the OPNAV 4790/CK as required by reference (ac).

14.2 Classified Material Installed or Removed: List only those materials that are classified and will, at a minimum, include the data elements shown in subparagraphs 14.1.1 and 14.1.2. All classified material will be handled according to DoD and Navy policies.

Classified Material to be Installed or Removed
--

• Model Type and Noun Name	• Classification	• Quantity	• Installed	• Removed	• Remarks
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Sample Classified Materials

14.3 Configuration Documentation

Configuration Documentation							
DocumentType	Title	Class	Number	Config Before	Config After	Provided By	ProvidedY or N
Ship SystemManual	X	X	X	X	X	X	X
Technical Manual	X	X	X	X	X	X	X
MaintenancePlan	X	X	X	X	X	X	X
Maintenance Index Page	X	X	X	X	X	X	X
Maintenance RequirementCard	X	X	X	X	X	X	X
Pre-FaultedModules	X	X	X	X	X	X	X

Sample Documentation

14.3.1 This list includes but is not limited to the data elements shown in subparagraph 14.3 with the providing activity and the disposition of any of the documentation upon completion of the installation identified. The types of documentation to be identified may include but are not limited to the following: ship systems manuals; technical manuals; Planned Maintenance System documentation such as maintenance index page, maintenance requirement card; repair standards; drawings or sketches; software documentation or specifications; pre-faulted modules and other related documents. The change or revision level is to be provided where appropriate.

14.3.2 Drawings or sketches are to be provided as IDP as attachment "A." Room arrangement drawings or sketches are required whenever there are changes in the arrangement of a room or equipment rack. When a third party is installing, cabling drawings or sketches must identify the cable numbers and the jack numbers that identify how the equipment is to be connected within the trainer.

14.4 Change marking and identification plates: The section will provide detailed instructions for the placing of identification markings, nameplates and information plates. MIL-STD-130 (Identification Marking of U.S. Military Property) and MIL-P-15024/10A (Nameplate, ORDAIT Plates and Information Plates) can be used. Nameplates and information plates will be generally mounted on the front of the item. Change labelplates will be located as near as possible to the equipment nameplates. Systems or set nameplates will be mounted on the principal or most prominent item of the major assembly. Plates will be located in easily accessible places during operation. The mounting and location of the plates will be shown on the assembly drawing of the item. Plates will be positioned to not interfere with controls or obscure other required information.

15. Integrated Logistics Support Requirements

15.1 Onboard Repair Parts: Provide complete identification of all parts being provided to the learning facility as spares to support the repair and maintenance of this equipment/change.

• Nomenclature	• Cage Code	• Part #	• NICN or NIIN	• Qty or Site	• Estimated Delivery Date
----------------	-------------	----------	----------------	---------------	---------------------------

NICN: Navy item control number NIIN: national item identification number

15.2 Tools and test equipment: The section should identify all new tools and test equipment required to operate or maintain the equipment as a result of this change. The disposition of the test equipment after the installation must be identified. This would include such items as automatic cable testers or other specialized tools and equipment not usually available at the learning facility.

15.3 Maintenance Plan and Responsibilities

15.3.1 Corrective and planned maintenance: Identify the activity responsible for corrective and planned maintenance, to include information assurance maintenance and certification of this equipment (e.g., local Navy authority, contractor). If it is to be maintained by the contractor, then provide the status of the actions taken to incorporate coverage for this equipment in the TA funded contracts and the effective date of coverage. Learning center approval must be received by the PM for inclusion into the maintenance service contract.

15.3.2 Overhaul requirements: Identify all newly installed equipment that may require

overhaul and provide suggested schedule.

16. Training. Provide information on all training and training materials being provided/updated as a result of this installation.

16.1 Curriculum: Identify the course name, number (CIN) and status of the training materials if not provided as part of the installation kit. If no impact on training materials, enter N/A for not applicable. The identification of curriculum deliverables should include items such as instructor guides, trainee materials, interactive multimedia instruction, handbooks, etc. List all training aids to be provided. Examples include audio/visual material, mock-ups, pre-faulted modules, fault insertion aids and unique audio-visual equipment.

• CIN or Publication #	• Title	• Rev or Change #	• Classification	• Qty or Site	• Estimated Delivery Date
---------------------------	---------	----------------------	------------------	------------------	------------------------------

16.2 Instructor and O&M training: Describe all initial training as provided at the time of installation. Use sub-headings as required to show the various types of initial training to be provided, including the number of hours and days involved. Include required training for assigned instructor personnel and follow-on maintenance support for the training system.

17. Post Installation Brief. Following the installation of a training system change, the installing activity will conduct a post installation brief with the learning facility and contractor support. During this brief, 4790/CKs and any red lines to the training system change documentation or trainer drawings (or both) will be reviewed with the TA. Documents will be provided to the contracting officer or their designated representative for dissemination. During this briefing, the status of all integrated logistics support and training materials to be provided as part of the training system. Change installations will be reviewed (onboard repair parts, tools, test equipment, technical manuals, Planned Maintenance System documentation and training materials) and verified as delivered or noted as an installation discrepancy. Status of all installation discrepancies, resolution responsibility and schedules for resolution will be discussed at this meeting.

17.1 Completion Reporting

17.1.1 The TSA or installing activity will coordinate the review and completion of the attached post installation completion report. The learning facility will complete the post installation completion report. The completion report documents acceptance of the trainer by the learning facility.

17.1.2 The completion of the completion report signifies that the trainer is ready for use by the learning facility for instructor familiarization and to conduct training. Any outstanding discrepancies will be noted on the completion report and tracked until they are satisfactorily resolved. The TSA or installing activity will provide copies of the completed completion report

to the commanding officer of the learning facility and the applicable learning center coordinator following the post installation brief.

18. IDP Deficiencies

18.1 A well-written deficiency in an IDP clearly and briefly states the problem. These descriptions must specifically identify the action and deficiency number and, where required, include specific nomenclature (i.e., title, part number, control number and quantity). Additionally, the items identified in subparagraphs 18.1.1. through 18.1.3 must be assigned during the equipment transfer process to identify who has responsibility for the corrective action:

18.1.1 Responsible command or code

18.1.2 Estimated completion date

18.1.3 Remarks, as required, will provide further assistance for the timely completion of the action. These deficiencies should also be prioritized for completion according to the schedule:

18.1.3.1 Priority 1 - An error that prevents the accomplishment of an operational or mission essential function according to official requirements or which jeopardizes personnel and equipment safety.

18.1.3.2 Priority 2 - An error which adversely degrades the accomplishment of an operational or mission essential function or prevents the accomplishment of an operational or mission essential mode, according to official requirements and for which no alternative work around solution exists.

18.1.3.3 Priority 3 - An error which adversely degrades the accomplishment of an operational or mission essential mode or prevents the accomplishment of an operational or mission essential sub-mode; or an error which adversely degrades the accomplishment of an operational or mission essential function or prevents the accomplishment of an operational or mission essential mode and for which there exists a reasonable workaround solution.

18.1.3.4 Priority 4 - An error that is an operator inconvenience or annoyance and does not adversely degrade a required operational or mission essential function, mode or sub-mode.

18.1.3.5 Priority 5 - All other errors, including documentation.

18.1.3.6 These deficiencies, when properly written, will consist of one or two sentences of description and identifiers, as described in subparagraphs 18.1.3.1 through 18.1.3.5. If specific spare part lists, etc. are available and space is limited in the engineering documentation of the IDP, use appendices. The lists may be merely referencing materials for a

specific deficiency, such as an itemization of courses and curricula or a clause guaranteeing the return of borrowed or residual equipment to the installer or provider of the equipment. An effort should be made to see if such references are available and obtainable for inclusion in the IDP.

18.1.3.7 Submission of OPNAV 4790/CK forms is required at the time of the post-installation brief. Failure to submit these forms at the time of the post-installation brief will not be accepted as a deficiency. Installation activity must now forward the signed Installation Design Documentation completion report with a deficiency list (as applicable) and post-installation briefing checklist to the TSA for IDP final transfer to be processed. Installing activity and facility 3M coordinator signature on the Installation Design Documentation completion report form signifies that completed OPNAV 4790/CK forms have been received and are being processed.

IDP (Short)

Training System Change Cover Sheet

{Change Tracking Number}

{Title of Change} Applicable Ship Classes

Applicable Learning Facilities

Prepared By: {Name, Address, Phone No. of Preparer} Prepared For: {Name, Address, Phone No. of Sponsor}

Reviewed by:
(Date)

Approved by:
(Date)

This cover sheet is used for approval of training system changes to existing capabilities at a learning facility. It is only used in conjunction with the IDP (Short), i.e., the modified Installation Design Documentation.

RATING CONTINUUM

1. Introduction. According to NAVPERS 18068F, a rating is a broad enlisted career field that identifies occupational specialties that encompass related aptitude, training, experience, knowledge and skills for career development and advancement. A rate identifies personnel occupations by paygrades E-1 through E-9 and reflects levels of aptitude, training, experience, knowledge, skill and responsibility. Navy rates are identified in three levels; apprentice, (E-1 through E-4), journeyman (E-5 through E-6) and master (E-7 through E-9). A rating continuum focuses on “what” tasks (system and rating), skills, certifications, qualifications and education outcomes are required of asailor during a career. Rating continuums are multi-phased, sequenced products that align Navyratings in career fields and broad communities providing a foundation for future development and more flexible and sustainable career paths for sailors. Rating continuums include the technical, professional and leadership tasks required of a sailor during a career as depicted in reference (p). The following Exhibit 6-1 shows the Product Roles and Responsibility Chart for the Rating Continuum Career Domain Analysis.

Product	Acquisition Timeline Completi n By	CN O N1	RS CN O (N1, N2/ N6,N 4, N9, N09 5)	SYS COM S	P M / P E O	T S A	TSPO/ TSPM	T A	NET C	SYS CO M Tech Authorit y	USFLT F ORCOM, COMPA CFLT and TYCOM(s)	USMC Combat and Integrati on and HQ Aviation	CNI C	N A V F A C
RDA/CFM	Post MS C	A	C		C	C	L	C	C	C	V,C		C	C

A - Approval C - Concurrence L - Lead V - Validation In Black - Not applicable

Exhibit 6-1: Product Roles and Responsibility Chart for the Rating Continuum Career Domain Analysis

2. Purpose. A Rating continuum is the output of a Career-Long Learning Continuum Development effort and produces an RDA and CFM. All NECs are organized according to their career fields. The foundation for a rating continuum is the technical, professional and leadership training and education needed throughout a career. A rating continuum focuses on “what” tasks (system and rating), skills, certifications, qualifications and education outcomes are required of a sailor during a career. Rating continuums are multi-phased, sequenced products that align Navy ratings in career fields and broad communities, providing a foundation for future development and more flexible and sustainable career paths for sailors. RDA determines “when” work occurs. CNO N1 is responsible for the planning, funding and execution of an RDA. For the listing of ratings and associated stakeholders, refer to the RRL Matrix located at:
https://flankspeed.sharepoint-mil.us/sites/MyNavyHR_Communities/OPNAV_N13M5.

GLOSSARY AND ACRONYMS

1. AAE. The AAE provides the strategy, process, and standard operating procedure for evaluating training Business Rule 1 and performance Business Rule 2. For further guidance refer to AAE Business Rules located at: https://flankspeed.sharepoint-mil.us/sites/MyNavyHR_Communities/OPNAV_N13M5.
2. Acquisition Strategy Document. A business and technical management approach designed to achieve program objectives within the resource constraints imposed. It is the framework for planning, directing contracting for and managing a program.
3. Analysis of alternatives. The evaluation of the performance, operational effectiveness, operational suitability and estimated costs of alternative systems to meet a mission capability. The analysis of alternatives assesses the advantages and disadvantages of alternatives being considered to satisfy capabilities, including the sensitivity of each alternative to possible changes in key assumption or variables. The analysis of alternatives is one of the key inputs to defining the system capabilities in JCIDS.
4. BCS. A current operational system or a composite of current operational subsystems, which most closely represents the design, operational and support characteristics of the new system under development and which possesses mature manpower and training data.
5. Billet Training Profile. Documents-specific training required for each billet, including course and training description, periodicity, course number, reference and training points.
6. Business Case Analysis. The Business Case Analysis provides a cost analysis for the proposed training systems from the MTRP. The Business Case Analysis also captures any risk associated with each option. NETC will propose training systems, as a course of action for each training system, to the requirement sponsor and RS. Collectively, these entities will select one of the training systems courses of action that NETC can then use to develop or revise a course to satisfy the training requirement. It is the output of the business case analysis (decisions by the requirement sponsor and RS) that will determine the and knowledge outcomes that the course will be developed to achieve, and tests built to assess.
7. Career Domain Analysis. Will include the rating master task list and CFM, connecting all training to applicable task requirements (RCDC), contain a priority of implementation with cost order of magnitude data to execute POM.
8. CFM. Graphically illustrates all rating-level requirements year zero across a 30-year career addressing leadership, professional and technical requirements.
9. Configuration. A collection of an item's descriptive and governing characteristics, which can be expressed in functional terms, (e.g., what performance the item is expected to achieve) and in

physical terms (e.g., what the item should look like and consist of when it is built).

10. CONOPS. A verbal or graphic statement, in broad outline, of a commander's assumptions or intent regarding an operation or series of operations. The CONOPS is frequently embodied in campaign plans and operation plans; in the latter case, mainly when the plans cover a series of connected operations to be carried out simultaneously or in succession. CONOPS is designed to give an overall picture of the operation.

11. COTS. A procurement approach for non-developmental items that is available in the commercial marketplace.

12. Crew Scheduling and Phasing Plan. Crew Scheduling and Phasing Plans will include requirements authorized in the Total Force Manpower Management System for each phase vice total requirements in a PSMD or PSQMD (unfunded PSMD or PSQMD requirements will be annotated in the appendix of a Crew Scheduling and Phasing Plan but not included in any of the phases). It is expected that full ship crewing requirements identified in the PSMD or PSQMD will be funded by the RS in the corresponding POM and budget estimate submission to eliminate gaps in the Fleet. A Crew Scheduling and Phasing Plan identifies the pre-commissioning training requirements by billet for each member of the crew. Using the Crew Scheduling and Phasing Plan as a guideline, personnel will be ordered to the ship per manning directives and aligned with the shipbuilder, OEM construction and contractor training schedules. Enlisted personnel will receive all required NEC and initial training before reporting aboard and TYCOM training requirements (including team training) upon arrival. Refer to reference (r) and OPNAVINST 1000.16L for more information.

13. Curricula Materials. Includes all materials required for the delivery of information and the development of knowledge, skills and abilities for a particular task.

14. Distributed Mission Training. Enables system operators to train with each other in realistic, real-time scenarios while being geographically separated.

15. Electronic Trial Card. Utilized to document known noncompliant discrepancies, including items identified in outstanding Quality Deficiency Reports or Corrective Action Reports.

16. Equipment Requirements List. Aviation, surface, expeditionary, subsurface and information warfare TTE requirements in support of naval training is identified by the training activities in equipment requirements list. The installation of initial training equipment and support items must take place before installation at operational units.

17. Executing Organization. The RS's selection as to who or what organization, is to conduct the AAE. The Executing Organization must have no organizational conflict of interest, must possess or be provided dedicated resources and must be able to report results directly to the RS. The Executing Organization could be the RS, TA, PEO, SYSCOM, PM, contractor or a team of personnel from these organizations (with a designated lead activity). Results of an AAE will be

provided in a letter signed by the senior member of the Executing Organization or lead activity.

18. FEA. A structured process used to examine manpower, training requirements planning requirements and identify alternative approaches to training job tasks. The process identifies job tasks to be performed, analyzes the skills and knowledge needed to perform them, assesses the technologies available for training the skills and knowledge, performs a media analysis to recommend the best mix of delivery media and provides cost and lead-time comparisons for the alternatives

19. Follow-on Training. Formal training that has been developed and documented in a formal training plan or curricula, which begins at RFT. (This training is also known as lifecycle support training, pipeline training and sustainment training).

20. Formal training. Training that has been developed and documented in a formal training plan or curricula. Normally provided as factory training or at a formal land-based organic training site and is designed to train replacement personnel.

21. Human Systems Integration. The integrated analysis, design and assessment over the lifecycle of a system and support infrastructure in the domains of manpower, personnel, training, human factors engineering, personnel survivability, habitability, safety and occupational health.

22. IMRD. The IMRD is a DID DI-SESS-81519C that provides specifications for the media selection model used, a description of primary and alternate media requirements and functional requirements for the instructional delivery system. Its purpose is to serve as the baseline for instructional media performance specifications. It is not intended that all the requirements contained within the DID be applied to every program or program phase. Any data requirement contained in this DID is subject to deletion tailoring.

23. Individual's Account. Requirements account for personnel in student, trainee, transient or holdee status and midshipmen on active duty. Holdees include patients, prisoners and personnel in the process of being separated.

24. Informal Training. Training that is not based on formal written curricula. Informal training is often provided by the installation team as part of installation process.

25. Initial Training. Operator and maintenance training resulting from the planned installation of new equipment. It is normally one-time training for personnel performing or involved in developmental and operational testing, for initial cadre maintainers and organic training activity instructors (train the trainers).

26. Inter-service Training Review Organization. Activity for services to voluntarily coordinate inter-service training.

27. Interim Training. Training that is funded and implemented by the PM when follow-on training is required and is not RFT before IOC. Interim training will continue until follow-on training is implemented and RFT is achieved.

28. IOC. The first attainment of the capability to effectively employ a weapon, item of equipment or system of approved specific characteristics and which is manned or operated by a trained, equipped and supported military unit or force.

29. IPRD. The IPRD is a DID DI-SESS-81518D that provides the mission collective and individual task information. This document also provides listings of knowledge, skills, attitudes and learning objectives for the tasks that have been selected for training. The IPRD contains data necessary to support the design of a training program. The Task Analysis and Learning Analysis data are documented in the IPRD. It is not intended that all the requirements contained within the DID be applied to every program or program phase. Any data requirement contained in this DID is subject to deletion tailoring.

30. KPP. Those attributes or characteristics of a system that are considered critical or essential to the development of an effective military capability and those attributes that make a significant contribution to the characteristics of the future joint force as defined in the CONOPS. A KPP must be testable to enable feedback from test and evaluation efforts to the requirements process. KPPs that are designated as joint performance requirement will be validated by the JROC and KPPs that are not designated as JPRs will be validated by the DoD component regardless of the program's joint staffing designator. JCIDS, Capability Development Documents and KPPs are included verbatim in the program baseline.

31. KSA. An attribute or characteristic considered crucial in support of achieving a balanced solution or approach to a KPP or some other key performance attribute deemed necessary by the sponsor. KSA provides decision-makers with an additional level of capability performance characteristics below the KPP level and requires a sponsor, defense agency commander or principal staff assistant to change.

32. Live Virtual Constructive Training. Training environment that connects live, in-person elements with manned virtual simulators and constructive computer-generated forces in conduct of training.

33. Manpower Estimate. An authoritative source for out-year projections of total force requirements and workload. The estimate is prepared by the PM and approved by CNO N1.

34. MCD. Required characteristics of a TD, which define the military functions the TD must be capable of performing or simulating. Military characteristics include physical and operational characteristics, but not technical characteristics. The MCD is developed after the media selection process.

35. Modernization. See SL720-AA-MAN-030, Navy Modernization Process Management and Operational Manual, January 2010.

36. Model-Based Product Support. A COTS application suite used to create, collaborate, manage requirements and specifications, reporting, track and manage programs and analytics.

37. MOS. The duty or related group of duties that a soldier or marine by training, skill and experience is best qualified to perform and that is a basis for the classification, assignment and advancement of enlisted personnel. MOSs are divided into Primary Military Occupational Specialties (PMOS) and non-primary MOSs. PMOS is used to identify the primary skills and knowledge of a marine. Changes to an active component Marine's PMOS without approval from Commandant of the Marine Corps and changes to a reserve component marine's PMOS without approval from Commandant of the Marine Corps resource advocates are not authorized. There are three types of non-PMOSs: 1) Necessary MOS) – A non-PMOS with a prerequisite of one or more PMOSs. This MOS identifies a particular skill or training in addition to a marine's PMOS but can only be filled by a marine with a specific PMOS. When entered as a requirement into the TFSMS, a billet bearing a necessary MOS must identify a single associated PMOS even if several PMOSs are acceptable prerequisites; 2) Free MOS – Non-PMOS that can be filled by any marine regardless of primary MOS. A free MOS requires skillsets unrelated to primary skills; and 3) Exception.

38. MS. Major decision points that separate the phases of a program.

39. MTRP. A collection of discrete requirement determination tasks resulting in the development of the technical program data, part I of the NTSP, the output of which is included in the TE. Part I states the underlying concepts that will govern the operation, maintenance, training, the quantity and quality of personnel required and the necessary training hardware. MTRP is designed to provide PMT analysts with a systematic set of tasks that will determine the best possible PMT profile of an initial acquisition or allow examination of alternative system concepts early in the process. The task-based approach to manpower and training requirements determination also provides PMs with the flexibility to tailor the level of PMT analysis that will be applied to their program.

40. Navy Officer Billet Classification. A classification expressed as a 4-digit code, which identifies officer billet requirements and officer occupational experience, acquired through billet experience or a combination of education and experience.

41. Navy Officer Occupational Classification System Board. The Navy officer occupational classification system is the Navy's method to identify skills, education, training, experience and capabilities related to both officer personnel and manpower requirements. This system forms the basis for officer personnel planning, manpower management, procurement, training, promotion, distribution, career development and mobilization. The Navy officer occupational classification

system board serves as the central point for changes to the officer occupational classification system. The board makes recommendations as appropriate and authorizes administrative actions that change the classification structure.

42. NEC. System of which NEC coding is a part. It supplements the enlisted rating structure in identifying personnel on active or inactive duty and billets in manpower authorizations. NEC codes identify a non-rating wide skill, knowledge, aptitude or qualification that must be documented to identify both people and billets for management purposes. Additionally, an NEC can be used to identify special circumstances or situations with approval via the NEOCS process.

43. NEOCS Board. A method the Navy uses to identify enlisted personnel skills and the requirements associated with these skills. The system forms the basis for actions taken concerning enlisted personnel planning, manpower management, procurement, training, advancement, promotion, distribution, assignment and mobilization. The NEOCS board serves as the central point for changes to the NEOCS. It reviews CPs in terms of overall system direction and makes recommendations as appropriate. The board authorizes administrative actions for NEC changes and makes recommendations to CNO N1 for enlisted rating structure changes.

44. Non-Developmental Item. Any hardware or software item that does not require development, such as commercially available items, items developed by other services or agencies or items developed by foreign governments with which the U.S. Government has a cooperation agreement. Items requiring only minor modification to meet the established requirements of the procuring agency are also considered non-developmental items.

45. NTSP. The primary document for defining PMT requirements and resources for new systems development and modernization of existing systems. The NTSP identifies the resources required to establish and maintain an effective training program throughout the lifecycle of the system. Much like the system it addresses, the NTSP goes through periodic updates to reflect advances in technology to sustain and improve combat effectiveness. The NTSP is a lifecycle document that identifies the resources required to establish and maintain an effective training program throughout the new systems' development and modernization. It controls planning for meeting the training requirements of the system and identifies personnel required to install, operate, maintain or otherwise use the system. It is a report that communicates the product support manager's plan for formulating, implementing and executing a sustainment strategy for training. It should communicate the strategy needed to develop and integrate sustainment requirements into system design, development, testing, fielding and operations or support. NTSPs are required for every Navy and integrated Navy and Marine Corps system. The approved NTSP is the official tasking document for establishing the system's PMT requirements identified within the document.

46. NTSP Tool. Can be any process, program or software, websites or other method of producing NTSPs by a PM or a RS.

47. Participating Resource Manager. An independent systems integrator who develops sub-systems and equipment in parallel with ship construction to meet a defined in-yard date.

48. PEO. A military or civilian official who has primary responsibility for directing multiple ACAT I programs and for assigned ACAT II, III and IV programs. A PEO has no other command or staff responsibilities within the component and only reports to and receives guidance and direction from the DoD component executive. PEOs receive program functional support (engineering, logistics, contracting, accounting, legal, etc.) from SYSCOM commanders. Under the direction of the PEO, assigned PMs may assume the duties and responsibilities of PM, TSA and PM for system development, modernization and configuration management. For PEOs responsible for ship and aircraft procurement, modernization, configuration management and lifecycle management, the PEO will coordinate with all PMs providing systems, subsystems and equipment to the ship and aircraft to ensure that training goals, objectives, responsibilities and schedules are defined.

49. Performance Support. Designed as a “Point of Need” job performance aid in direct support of PQS, Planned Maintenance System documentation and infrequently performed tasks.

50. PM. A military or civilian official who is responsible for managing procurement, modernization and configuration management programs.

51. RCDC. A model that depicts a rating continuum showing the technical, professional (on-the-job training, OCCSTD and PQS) and leadership throughout a career. The Rating Continuum must be completed and approved as identified in enclosure (6).

52. RDA. Aligns all required training tasks to the proper point in a Sailor’s career continuum when they will be needed to perform their job and provides a recommendation for when training should be provided. The analysis identifies and sequences the new or revised work a Sailor is expected to perform, and measures performance based on Fleet operational readiness standards for a given Rating Continuum. The analysis will use a combination of AAE and Fleet performance data to inform the continuum and the timing of performance checks.

53. RFT. The date of a training system and its associated logistics, maintenance, syllabus and instructors are certified to be available for training at the learning facility. This date is predicated on the availability of a new, modified or rehabilitated learning facility for training purposes. All aspects of the facility must be ready, including building completion, completed site preparation, training system installation and testing, trained instructors, furnishings, e.g., non-technical collateral equipment. This is the ultimate planning date for the new or modified training system and its readiness for use.

54. RRL. A pillar of Sailor 2025 that drives fundamental changes into our approach to Sailor training. The goal is to provide the right training at the right time in the right way to improve

Sailor performance and enhance mission readiness. To accomplish this, the Navy will develop a career-long learning continuum for every Sailor, modernize training to maximize impact and relevance and accelerate our processes for delivering new training to the Fleet.

55. RS. The RS determines program objectives and time-phased support requirements and appraises programs, readiness and military worth for a given weapon system function or task in support of the goals and objectives. The RS provides day-to-day OPNAV management of the assigned programs by acting as the central POC for the hardware systems coordinators and the PM. An RS is an OPNAV principal official responsible for an identifiable aggregation of resources that constitute inputs to warfare and supporting warfare tasks.

56. Simulation. The imitation of the operation of a real-world process or system over time. The act of simulating something first requires that a model be developed; this model represents the key characteristics or behaviors of the selected physical or abstract system or process. The model represents the system itself, whereas the simulation represents the operation of the system over time.

57. Software Acquisition Pathway. Is for the timely procurement of custom software capabilities developed for the DoD. Programs executing the software acquisition pathway are not subject to the JCIDS and will be handled as specifically provided for by the Vice Chairman of the Joint Chiefs of Staff, in consultation with Under Secretary of Defense for Acquisition and Sustainment and each service executive. Programs using the software acquisition pathway will demonstrate the viability and effectiveness of capabilities for operational use not later than 1 year after the date on which funds are first obligated to develop the new software capability. New capabilities will be delivered to operations at least annually to iteratively meet requirements, but more frequent updates and deliveries are encouraged where practical. For programs using the embedded software path, this annual update applies after initial operational acceptance of the system in which the software is embedded and should be aligned with the associated system's schedule. Before the operational acceptance of the system in which the software is embedded, software deliveries will be delivered to an operationally representative environment at least annually.

58. Student Profile. The term includes the student being trained, the learning from the interactive courseware, including course and training description, periodicity, course number, reference and training points. Used to construct learning situations to acquire the knowledge and skills required for the accomplishment of specific tasks.

59. SYSCOM. The command responsible for implementing policy and procedures for the execution of assigned procurement and modernization programs. They manage the logistics support plan through their functional and program management organizations.

60. System. A grouping of functionally related subsystems and equipment operating together to support a major function or meet a tactical purpose. The term system includes subsystems, a

system of systems, a family of systems and equipment.

61. TA. An office, bureau, command or headquarters exercising command of and providing support to some major increment of the DON's formalized training effort. Depending upon the level of oversight required, the TA may be NETC, a Learning Center or other designated organizations.

62. Task Analysis. Identifies the tasks required for performance on the job that are analyzed for how a task is accomplished, task and element durations, task frequency, task allocation, task complexity, task criticality, environmental conditions, equipment and any other unique factors involved in or required to perform a task into a Master Task List. Using the Master Task List further task analysis is conducted to identify, using various factors, tasks that require training into a Training Task List.

63. TD. Hardware and software which have been designed or modified exclusively for training purposes involving, to some degree, simulation or stimulation in its construction or operation, to demonstrate or illustrate a concept or simulate an operational environment.

64. TDCP. A document used to identify and evaluate alternative approaches to the design of the training system and to recommend the best approach.

65. TE. An authoritative source for out-year projections of training requirements and assessments. The TE is prepared by the PM and approved by RS(s). A MS B product developed using MTRP data. The TE consists of enclosure (3) parts I, V, VI and VII. The TE will be aligned with the manpower estimate or preliminary manpower documents (or both). The analysis will be updated and transitioned into an NTSP at MS C.

66. TITA. Defines how the Navy coordinates the design, installation and incorporation of a training system and transfer to a schoolhouse. A properly developed agreement: (1) defines facility requirements, including MILCON and special project planning, for installation of the training system; (2) identifies all associated logistics support elements; and (3) transfers training ownership responsibility from the TSA to the TA. The agreement is an integral part of the system process as it supports the introduction of new training systems and equipment, incorporates training system changes and identifies sustainment requirements. Close liaison between the TSA and TA are key to implementing training system installation plans that meet an RFT date.

67. TLCE. Assessment of the estimated cost of developing, producing, deploying, maintaining, operating and disposing of a system over its entire lifespan.

68. Training Materials. See curricula materials.

69. Training System. A systematically developed curriculum includes curriculum, including distributed learning, in-service training, onboard training, self-paced computer-based training,

classroom aids, TDs, operational equipment, embedded training capability and trained personnel to operate, maintain or employ a system. The training system includes all the necessary elements of logistics support.

70. TSA. The office, bureau, command or headquarters responsible for supporting the TA by providing material and other forms of support within the cognizance of the office, bureau or command involved. The TSA provides initial training for the ESS until the TA can acquire the capability for training. The TSA is an activity with fiscal responsibility for supporting the TA by providing material and other forms of support within the cognizance of the office, command or headquarters involved. The TSA will be the PM who is assigned funding responsibility for all investments and expense costs required to turn over a training system end item and allow the TA to sustain the capability for training.

71. TSD. The TSD is a DID DI-SESS-81517C that documents the efficiency of a training system to meet training needs and to survey training programs and technologies for applicability to new training needs. It is not intended that all the requirements contained within the DID be applied to every program or program phase. Any data requirement contained in this DID is subject to deletion tailoring.

72. TSPM. Refer to section 6(g) in main policy for the TSPM definition.

73. TSPO. Refer to section 6(g) in main policy for each TSPO definition.

74. TTE. Investment cost end items of operational equipment, devoted to the training and instruction of naval personnel, for which PMs have responsibility for design, development, modernization, configuration management or selection for service or special use.

75. Urgent Need. An urgent need is an unusual request from a Navy or Marine Corps component commander for an additional warfighting capability critically needed by operating forces conducting combat or contingency operations. Failure to deliver the capability requested is likely to result in the inability of units to accomplish their missions or increases the probability of casualties and loss of life. The DON's urgent need process encompasses Navy urgent operational need, Marine Corps urgent universal need statement and joint urgent operational needs assigned to the DON.

<u>Acronym</u>	<u>Term</u>
AAE	Assessment, Analysis and Evaluation
ACAT	Acquisition Categories
ASN	Assistant Secretary of the Navy
BCS	baseline comparison system
CeTARS	Corporate Enterprise and Training Activity Resource Systems
CFM	Career Flow Map
CIN	Course Identification Number
CMS	contractor maintenance service
CNO	Chief of Naval Operations
CNO N1	Deputy Chief of Naval Operations for Personnel, Manpower and Training
CNO N2/N6	Deputy Chief of Naval Operations for Information Warfare and Director of Naval Intelligence
CNO N4	Deputy Chief of Naval Operations for Fleet Readiness and Logistics
CNO N9	Deputy Chief of Naval Operations for Warfighting Requirements and Capabilities
CNO N095	Chief of Navy Reserve
CNIC	Commander, Navy Installations Command
COMPACFLT	Commander, Pacific Fleet
CONOPS	Concept of Operations
COTS	commercial-off-the-shelf
DID	data item description
DoD	Department of Defense
DON	Department of the Navy
ETMS2	Enterprise Task Management Software Solution
ESS	equipment, systems and subsystems
FEA	front end analysis
FY	fiscal year
FYDP	Future Years Defense Program
GOTS	government-off-the-shelf
HQ	headquarters
IMRD	instructional media requirements document
IPRD	instructional performance requirements document
IOC	initial operational capability
ISEA	In-Service Engineering Agent
JCIDS	Joint Capabilities Integration and Development System
JROC	Joint Requirements Oversight Council
KPP	Key Performance Parameter
KSA	Key System Attribute
LCSP	Life-Cycle Sustainment Plan
MCD	military characteristics document

MFA	media fidelity analysis
MILCON	military construction
MOS	Military Occupational Specialty
PMT	manpower, personnel and training
MS	milestone
MTRP	manpower and training requirements planning
COMNAVAIRSYSCOM	Naval Air Systems Command
COMNAVFACSYSCOM	Naval Facilities Engineering Systems Command
COMNAVFACSYSCOM	Naval Facilities Engineering and Expeditionary Warfare
EXWC	Center
COMNAVIFOR	Naval Information Forces Command
NAVMAC	Navy Manpower Analysis Center
COMNAVSEASYS	Naval Sea Systems Command
COMNAVSEASYS	Naval Supply Systems Command
COMNAVSEASYS	Naval Information Warfare Systems Command
COMNAVSEASYS	Naval Information Warfare Systems Command
NEC	Navy enlisted classification
NEOCS	Navy Enlisted Occupational Classification System
NETC	Naval Education and Training Command
NTP	Navy training process
NTSP	Navy training system plan
O&M	operation and maintenance
O&MN	Operation and Maintenance, Navy
O&MNR	Operation and Maintenance, Navy Reserve
OCCSTD	occupational standards
OEM	original equipment manufacturer
OPNAV	Office of the Chief of Naval Operations
OPTEVFOR	Operational Test and Evaluation Force
OQE	objective quality evidence
PEO	Program Executive Office
PfMO	Portfolio Management Office
PM	program manager
PMOS	Primary Military Occupational Specialty
POA&M	plan of actions and milestones
POC	point of contact
POM	Program Objective Memorandum
PQS	personnel qualification standards
PSMD	Preliminary Ship Manpower Document
PSQMD	Preliminary Squadron Manpower Document
PWD	public works department
RCDC	rating career domain continuum
RD&A	research, development and acquisition

RDA	rating domain analysis
RFT	ready for training
ROC	Required Operational Capability
RRL	Ready Relevant Learning
RS	resource sponsor
SECNAV	Secretary of the Navy
SME	subject matter expert
SQMD	Squadron Manpower Document
STAVE	Surface Training Advanced Virtual Environment
SYSCOM	Systems Command
TA	Training Agency
TADTAR	temporary additional duty travel target funding
TD	training device
TDCP	training decision coordinating paper
TE	Training Estimate
TFMMS	Total Force Manpower Management System
TFSMS	Total Force Structure Management System
TITA	Training Installation and Transfer Agreement
TLCE	total life-cycle cost estimate
TSA	Training Support Agency
TSD	training situation document
TSPM	Training Systems Program Manager
TSPO	Training Systems Program Office
TSRA	training systems requirements analysis
TTE	technical training equipment
TYCOM	Type Commander
USFLTFORCOM	U.S. Fleet Forces Command
USMC	U.S. Marine Corps

Table 7-1: Acronyms

EXAMPLES OF PRODUCTS

1. Introduction. The examples of the PMT documents are available to help answer questions and formatting concerns. However, the examples are by no means the only way to complete the products but serve as a good minimum baseline for a program to utilize for their own versions for their ESS.
2. Purpose. To provide examples of the PMT documents to assist program offices and managers in adhering to the guidelines for development and submission of these products. Examples are located on https://flankspeed.sharepoint-mil.us/sites/MyNavyHR_Communities/OPNAV_N13M5.